MATHEMATICS

METHODS OF DIFFERENTIATION

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METHODS OF DIFFERENTIATION

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METHODS OF DIFFERENTIATION

JEE EXERCISES

Exercise J-I

BASED ON DIFFERENTIATION TECHNIQUES

1. If $y = \log\left(x + \sqrt{x^2 + a^2}\right)$ then dy/dx equals :

यदि $y = \log\left(x + \sqrt{x^2 + a^2}\right)$ तो dy/dx बराबर है :

(A) $\frac{x}{\sqrt{x^2 + a^2}}$ (B) $\frac{1}{\sqrt{x^2 + a^2}}$ (C) $\frac{-1}{\sqrt{x^2 + a^2}}$ (D) $\frac{1}{\sqrt{x^2 - a^2}}$

Ans. B

Sol. $y = \log\left(x + \sqrt{x^2 + a^2}\right)$

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + a^2}} \times \left(1 + \frac{2x}{2\sqrt{x^2 + a^2}}\right)$$

$$\Rightarrow \frac{1}{\sqrt{x^2 + a^2}}$$

2. If $y = 1 + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$ then dy/dx equals :

यदि $y = 1 + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$ तो dy/dx बराबर है :

(A) $y - x - 1$ (B) $y + x - 1$ (C) $y + x$ (D) $y - x$

Ans. B

Sol. $y = 1 + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$

$$\frac{dy}{dx} = \frac{2x}{2!} + \frac{3x^3}{3!} + \frac{4x^3}{4!} + ...$$

$$\frac{dy}{dx} = \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$$

$$\frac{dy}{dx} = 1 - 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$$

$$\frac{dy}{dx} = y + x - 1$$

3. The derivative of $\tan x^o$ is :

$\tan x^o$ का अवकलज है :

(A) $\left(\frac{\pi}{180}\right)\cos^2 x$ (B) $\left(\frac{\pi}{180}\right)\sec^2 x$ (C) $\left(\frac{\pi}{180}\right)\cos^2 x^o$ (D) $\left(\frac{\pi}{180}\right)\sec^2 x^o$

Ans. D

Sol. $y = \tan x°$

$$= \tan\left(\frac{\pi x}{180}\right)$$

$$= \frac{\pi}{180}\sec^2 x°$$

4. If $x^y + y^x = a^b$ then $\frac{dy}{dx}$ equals :

(A) $\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (B) $-\frac{x^y \log x + x.y^{x-1}}{y.x^{y-1} + \log y\ y^x}$ (C) $-\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (D) None of these

यदि $x^y + y^x = a^b$ then $\frac{dy}{dx}$ बराबर है :

(A) $\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (B) $-\frac{x^y \log x + x.y^{x-1}}{y.x^{y-1} + \log y\ y^x}$ (C) $-\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $x^y + y^x = a^b$

$x^y \ln x\ y' + yx^{y-1} + xy^{x-1}y' + y^x \ln y = 0$

$y'(x^y \ln x + xy^{x-1}) = -(yx^{y-1} + y^x \ln y)$

$$\frac{dy}{dx} = \frac{-\left(yx^{y-1} + y^x \ln y\right)}{x^y \ln x + xy^{x-1}}$$

5. If $\sin y = x \sin(a+y)$ then dy/dx is : **[AIEEE-2002]**

(A) $\frac{\sin(a+y)}{\sin^2 a}$ (B) $\frac{\sin^2(a+y)}{\sin a}$ (C) $\frac{\sin(a+y)}{\sin^2 a}$ (D) None of these

यदि $\sin y = x \sin(a+y)$ तो dy/dx होगा : **[AIEEE-2002]**

(A) $\frac{\sin(a+y)}{\sin^2 a}$ (B) $\frac{\sin^2(a+y)}{\sin a}$ (C) $\frac{\sin(a+y)}{\sin^2 a}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $x = \frac{\sin y}{\sin(a+y)}$

$$\frac{dx}{dy} = \frac{\cos y \,.\, \sin(a+y) - \sin y \cos(a+y)}{\sin^2(a+y)}$$

$$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

6. If $y = (1+x^{1/4})\,(1+x^{1/2})\,(1-x^{1/4})$, then dy/dx equals :

यदि $y = (1+x^{1/4})\,(1+x^{1/2})\,(1-x^{1/4})$, तो dy/dx बराबर होगा :

(A) –1 (B) 1 (C) x (D) $\sqrt{x}$

Ans. A

Sol. $y = (1+x^{1/4})\,(1+x^{1/2})\,(1-x^{1/4})$

$$y = \left(1+x^{\frac{1}{2}}\right)\left(1+x^{\frac{1}{4}}\right)\left(1-x^{\frac{1}{4}}\right)$$

$$y = \left(1+x^{\frac{1}{2}}\right)\left(1-x^{\frac{1}{2}}\right)$$

$y = 1 - x$

$$\frac{dy}{dx} = -1$$

7. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx}$ equals :

(A) $\frac{1}{(1+x)^2}$ (B) $-\frac{1}{(1+x)^2}$ (C) $\frac{1}{1+x^2}$ (D) None of these

यदि $x\sqrt{1+y} + y\sqrt{1+x} = 0$, तो $\frac{dy}{dx}$ बराबर है :

(A) $\frac{1}{(1+x)^2}$ (B) $-\frac{1}{(1+x)^2}$ (C) $\frac{1}{1+x^2}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $x^2(1+y) = y^2(1+x) \quad \Rightarrow x^2 - y^2 + xy(x-y) = 0$

$\Rightarrow \quad (x-y)(x+y+xy) = 0$

Now $x \neq y$ [does not satisfy the given equation]

$\therefore \quad x + y + xy = 0 \Rightarrow y = \frac{-x}{1+x}$

$\therefore \quad \frac{dy}{dx} = \frac{-(1+x)+x}{(1+x)^2} = -\frac{1}{(1+x)^2}$

8. If $y = \sin^{-1}\sqrt{\sin x}$, then $\frac{dy}{dx}$ equals :

यदि $y = \sin^{-1}\sqrt{\sin x}$, then $\frac{dy}{dx}$ बराबर है :

(A) $\frac{2\sqrt{\sin x}}{\sqrt{1+\sin x}}$ (B) $\frac{\sqrt{\sin x}}{\sqrt{1-\sin x}}$ (C) $\frac{1}{2}\sqrt{1+\operatorname{cosec} x}$ (D) $\frac{1}{2}\sqrt{1-\operatorname{cosec} x}$

Ans. C

Sol. $y = \sin^{-1}\sqrt{\sin x}$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin x}} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{1}{\sqrt{\frac{(1+\sin x)}{1+\sin x} \times (1-\sin x)}} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{\sqrt{1+\sin x}}{\sqrt{1-\sin^2 x}} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{\sqrt{1+\sin x}}{\cos x} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{1}{2}\left(\sqrt{1+\operatorname{cosec} x}\right)$$

9. If $x^y y^x = 1$, then $\frac{dy}{dx}$ equals :

यदि $x^y y^x = 1$, तो $\frac{dy}{dx}$ बराबर है :

(A) $\frac{x\left(y + x \log\ y\right)}{y\left(x + y \log\ x\right)}$ (B) $-\frac{x\left(x + y \log\ y\right)}{y\left(y + x \log\ x\right)}$ (C) $\frac{y\left(y + x \log\ y\right)}{x\left(x + y \log\ x\right)}$ (D) $-\frac{y\left(y + x \log\ y\right)}{x\left(x + y \log\ x\right)}$

Ans. D

Sol. $x^y y^x = 1$

$\log x^y + \log y^x = \log 1$

$y \log x + x \log y = 0$

$$\frac{y}{x} + \frac{dy}{dx}\log x + \log y + \frac{x}{y}\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}\left(\log x + \frac{x}{y}\right) = -\left(\frac{y}{x} + \log y\right)$$

$$\frac{dy}{dx} = \frac{-y}{x}\left(\frac{y + x\log y}{x + y\log x}\right)$$

10. If $y = \tan^{-1}\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$, then $\frac{dy}{dx}$ equals :

यदि $y = \tan^{-1}\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$, तो $\frac{dy}{dx}$ बरारब है :

(A) $-\frac{1}{2\sqrt{1-x^2}}$ (B) $-\frac{1}{\sqrt{1-x^4}}$ (C) $-\frac{x}{\sqrt{1-x^4}}$ (D) $-\frac{x}{2\sqrt{1-x^4}}$

Ans. C

Sol. $y = \tan^{-1}\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$

$$= \tan^{-1}\left(\frac{1+x^2+1-x^2+2\sqrt{1-x^4}}{1+x^2-1+x^2}\right)$$

$$= \tan^{-1}\left(\frac{1+\sqrt{1-x^4}}{x^2}\right)$$

$x^2 = \sin 2\theta$

$$= \tan^{-1}\left(\frac{1+\cos 2\theta}{\sin 2\theta}\right)$$

$$= \tan^{-1}\left(\cot\theta\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{2}-\theta\right)\right)$$

$$y = \frac{\pi}{2}-\theta$$

$$y = \frac{\pi}{2}-\frac{1}{2}\sin^{-1}(x^2)$$

$$\frac{dy}{dx} = \frac{-1}{2}\frac{1}{\sqrt{1-x^4}} - 2x$$

$$= \frac{-x}{\sqrt{1-x^4}}$$

11. If $f(x) = \log_x(\ln x)$, then at $x = e$, $f'(x)$ equals :

यदि $f(x) = \log_x(\ln x)$, तो $x = e$ पर, $f'(x)$ बराबर है :

(A) 0 (B) 1 (C) e (D) 1/e

Ans. D

Sol. $f(x) = \log_x(\ell nx)$

$$\Rightarrow \quad f(x) = \frac{\ell n(\ell nx)}{\ell nx}$$

$$\Rightarrow \quad f'(x) = \frac{\ell nx\left(\frac{1}{\ell nx}\cdot\frac{1}{x}\right) - \frac{1}{x}\ell n(\ell nx)}{(\ell nx)^2}$$

$$\Rightarrow \quad f'(e) = \frac{\frac{1}{e}-0}{1} = 1/e$$

12. If $y = e^{3x} \sin 4x$, then the value of dy/dx is :

यदि $y = e^{3x} \sin 4x$, तो dy/dx का मान है :

(A) $e^{3x} \sin\left(4x + \tan^{-1}\frac{4}{3}\right)$ (B) $e^{3x} \cos\left(4x + \tan^{-1}\frac{4}{3}\right)$

(C) $5e^{3x} \sin\left(4x + \tan^{-1}\frac{4}{3}\right)$ (D) $5e^{3x} \cos\left(4x + \tan^{-1}\frac{4}{3}\right)$

Ans. C

Sol. $y = e^{3x} \sin 4x$

$$\frac{dy}{dx} = 3e^{3x} \sin 4x + 4e^{3x} \cos 4x$$

$$= e^{3x}(3\sin 4x + 4\cos 4x)$$

$$= 5e^{3x}\left(\frac{3}{5}\sin 4x + \frac{4}{5}\cos 4x\right)$$

$$= 5e^{3x} \sin\left(4x + \tan^{-1}\frac{4}{3}\right)$$

13. If $x + y = x^y$, then dy/dx equals :

(A) $\frac{yx^{y-1} - 1}{1 - x^y \log x}$ (B) $\frac{y\, x^{y-1} - 1}{x^y \log x - 1}$ (C) $\frac{y\, x^{y-1} + 1}{x^y \log x + 1}$ (D) None of these

यदि $x + y = x^y$, तो dy/dx बराबर है :

(A) $\frac{yx^{y-1} - 1}{1 - x^y \log x}$ (B) $\frac{y\, x^{y-1} - 1}{x^y \log x - 1}$ (C) $\frac{y\, x^{y-1} + 1}{x^y \log x + 1}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $x + y = x^y$

$$1 + \frac{dy}{dx} = yx^{y-1} + x^y \ln x \frac{dy}{dx}$$

$$\frac{dy}{dx}\left(1 - x^y \ln x\right) = yx^{y-1} - 1$$

$$\frac{dy}{dx} = \frac{yx^{y-1} - 1}{1 - x^y \ln x}$$

14. The derivative of x^{a^x} is :

(A) $x^{a^x}[a^x x^{-1} + a^x \log a \log x]$ (B) $x^{a^x}[xa^x + a^x \log x]$

(C) $x^{a^x}[a^x + xa^x \log x]$ (D) None of these

x^{a^x} का अवकलज है :

(A) $x^{a^x}[a^x x^{-1} + a^x \log a \log x]$ (B) $x^{a^x}[xa^x + a^x \log x]$

(C) $x^{a^x}[a^x + xa^x \log x]$ (D) None of these

Ans. A

Sol. $y = x^{a^x}$

$\ln y = a^x \ln x$

$$\frac{1}{y}\frac{dy}{dx} = a^x \ln a \ln x + \frac{a^x}{x}$$

$$\frac{dy}{dx} = (x)^{a^x}\left[a^x \ln a \ln x + x^{-1}a^x\right]$$

15. If $(\cos x)^y = (\sin y)^x$, then dy/dx equals :

(A) $\dfrac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$ (B) $\dfrac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$

(C) $\dfrac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$ (D) None of these

यदि $(\cos x)^y = (\sin y)^x$, तो dy/dx बराबर है :

(A) $\dfrac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$ (B) $\dfrac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$

(C) $\dfrac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $(\cos x)^y = (\sin y)^x$

$y \log(\cos x) = x \log(\sin y)$

$$\frac{dy}{dx}\log(\cos x) + y \times \frac{-\sin x}{\cos x} = \log(\sin y) + \frac{x \cos y}{\sin y}\frac{dy}{dx}$$

$$\frac{dy}{dx}\left(\log(\cos x) - x\cot y\right) = y\tan + \log(\sin y)$$

$$\frac{dy}{dx} = \frac{y\tan x + \log(\sin y)}{\log(\cos x) - x\cot y}$$

16. If $y = \sin^{-1}\left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right]$, then dy/dx can be equal to :

(A) $\frac{2}{1+x^2}$ (B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x}\sqrt{1-x}}$ (D) None of these

यदि $y = \sin^{-1}\left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right]$, तो dy/dx हो सकता है :

(A) $\frac{2}{1+x^2}$ (B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x}\sqrt{1-x}}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $y = \sin^{-1}\left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right]$

$\because \sin^{-1}a \pm \sin^{-1}b = \sin^{-1}\left(a\sqrt{1-b^2} \pm b\sqrt{1-a^2}\right)$

$y = \sin^{-1}x - \sin^{-1}\sqrt{x}$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}}$$

17. The differential coefficient $(x > 0)$ of $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ w.r.t $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ can be equal to :

$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ का $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ के सापेक्ष अवकलन गुणांक $(x > 0)$ हो सकता है :

(A) 1/2 (B) –1/2 (C) 1 (D) –1

Ans. A

Sol. $y = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ $\qquad$ $z = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

$x = \tan\theta$ $\qquad$ $x = \tan\phi$

$y = \sin^{-1}\left(\frac{\tan\theta}{\sec\theta}\right)$ $\qquad$ $z = \cos^{-1}\left(\frac{1-\tan^2\phi}{1+\tan^2\phi}\right)$

$y = \sin^{-1}(\sin\theta)$ $\qquad$ $z = \cos^{-1}(\cos 2\phi)$

$y = \theta$ $\qquad$ $z = 2\phi$

$y = \tan^{-1}x$ $\qquad$ $z = 2\tan^{-1}x$

$\frac{dy}{dx} = \frac{1}{1+x^2}$ $\qquad$ $\frac{dz}{dx} = \frac{2}{1+x^2}$

$\frac{dy}{dz} = \frac{1}{2}$

18. If $y = \sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\sqrt{x}.....\infty}}}$, then the value of dy/dx is :

यदि $y = \sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\sqrt{x}.....\infty}}}$, तो dy/dx का मान होगा :

(A) $\frac{xy^2}{2-y\log x}$ (B) $\frac{x^2}{y(2-y\log x)}$ (C) $\frac{y^2}{x(2-y\log x)}$ (D) $\frac{y^2}{x(2+y\log x)}$

Ans. C

Sol. $y = \left(\sqrt{x}\right)^y$

$\ln y = \frac{y}{2}\ln x$

$\frac{1}{y}\frac{dy}{dx} = \frac{\ln x}{2}\frac{dy}{dx} + \frac{y}{2x}$

$\frac{dy}{dx}\left[\frac{1}{y} - \frac{\ln x}{2}\right] = \frac{y}{2x}$

$$\frac{dy}{dx}\left[\frac{2-y\ln x}{2y}\right]=\frac{y}{2x}$$

$$\frac{dy}{dx}=\frac{y^2}{x(2-y\ln x)}$$

19. If $y=(\tan\ x)^{\tan x^{\cdots\cdots\infty}}$, then $\tan x\left(\frac{dy}{dx}\right)$ equals :

(A) $\frac{y^2\sec^2 x}{1-y\log\ \tan x}$ (B) $\frac{y^2\sec^2 x}{1+y\log\tan x}$ (C) $\frac{y\sec^2 x}{1-y\log\tan x}$ (D) None of these

यदि $y=(\tan\ x)^{\tan x^{\cdots\cdots\infty}}$, तो $\tan x\left(\frac{dy}{dx}\right)$ बराबर है :

(A) $\frac{y^2\sec^2 x}{1-y\log\ \tan x}$ (B) $\frac{y^2\sec^2 x}{1+y\log\tan x}$ (C) $\frac{y\sec^2 x}{1-y\log\tan x}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y=(\tan\ x)^y$

$$\log y=y\log(\tan x)$$

$$\frac{1}{y}\frac{dy}{dx}=\frac{dy}{dx}\log(\tan x)+\frac{y}{\tan x}\sec^2 x$$

$$\frac{dy}{dx}\left(\frac{1}{y}-\log(\tan x)\right)=\frac{y\sec^2 x}{\tan x}$$

$$\frac{dy}{dx}=\frac{y^2\sec^2 x}{\tan x(1-y\log\tan x)}$$

$$\tan x\frac{dy}{dx}=\frac{y^2\sec^2 x}{(1-y\log\tan x)}$$

20. If $y=\cfrac{x}{a+\cfrac{x}{b+\cfrac{x}{a+\cfrac{x}{b+\ldots\ldots\ldots\ldots}}}}$, then dy/dx equals :

(A) $\frac{b}{a\ (b+2y)}$ (B) $\frac{b}{b+2y}$ (C) $\frac{a}{b\ (b+2y)}$ (D) None of these

यदि $y = \cfrac{x}{a + \cfrac{x}{b + \cfrac{x}{a + \cfrac{x}{b + ...}}}}$, तो dy/dx बराबर होगा :

(A) $\frac{b}{a\ (b+2y)}$ (B) $\frac{b}{b+2y}$ (C) $\frac{a}{b\ (b+2y)}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y = \cfrac{x}{a + \cfrac{x}{b+y}} \Rightarrow y = \frac{x(b+y)}{ab+ay+x}$

$\Rightarrow \quad aby + ay^2 + xy = xb + xy$

$\Rightarrow \quad ab\frac{dy}{dx} + 2ay\frac{dy}{dx} = b \qquad \Rightarrow \frac{dy}{dx} = \frac{b}{ab+2ay}$

21. If $y = \log_y x$, then $\frac{dy}{dx} =$

यदि $y = \log_y x$, तो $\frac{dy}{dx} =$

(A) $\frac{1}{x+\log y}$ (B) $\frac{1}{\log x\,(1+y)}$ (C) $\frac{1}{x\,(1+\log y)}$ (D) $\frac{1}{y+\log x}$

Ans. C

Sol. $y = \log_y x \qquad y = \frac{\log x}{\log y}$

$y \times \log y = \log x$

$\frac{dy}{dx}\log y + y \times \frac{1}{y}\ \frac{dy}{dx} = \frac{1}{x}$

$\frac{dy}{dx}(\log y + 1) = \frac{1}{x} \qquad \boxed{\frac{dy}{dx} = \frac{1}{x(1+\log y)}}$

22. If $x = 3\cos\theta - 2\cos^3\theta$ and $y = 3\sin\theta - 2\sin^3\theta$, then $\frac{dy}{dx} =$

यदि $x = 3\cos\theta - 2\cos^3\theta$ तथा $y = 3\sin\theta - 2\sin^3\theta$, तो $\frac{dy}{dx} =$

(A) $\sin\theta$ (B) $\cos\theta$ (C) $\tan\theta$ (D) $\cot\theta$

Ans. D

Sol. $x = 3\cos\theta - 2\cos^3\theta \quad y = 3\sin\theta - 2\sin^3\theta$

$$\frac{dx}{d\theta} = -3\sin\theta + 6\cos^2\theta \times \sin\theta$$

$$\frac{dy}{d\theta} = 3\cos\theta - 6\sin^2\theta\cos\theta$$

$$\frac{dy}{dx} = \frac{3\cos\theta - 6\sin^2\theta\cos\theta}{-3\sin\theta + 6\cos^2\theta\sin\theta}$$

$$= \frac{\cos\theta - 2\sin^2\theta\cos\theta}{-\sin\theta + 2\cos^2\theta\sin\theta} = \cot\theta$$

23. Let f(x) be a polynomial function of second degree. If f(1) = f(– 1) and a, b, c are in A.P. then f′ (a), f′(b) and f′(c) are in : **[AIEEE-2003]**

(A) Arithmetic - Geometric Progression (B) A.P.

(C) G.P. (D) H.P.

माना कि f(x) द्विघात का बहुपदीय फलन है यदि f(1) = f(– 1) तथा a, b, c स. श्रे. में हो तो f′ (a), f′(b) एवं f′(c) होंगे : **[AIEEE-2003]**

(A) समान्तरीय गुणोत्तर श्रेणी में (B) समान्तर श्रेणी में

(C) गुणोत्तर श्रेणी में (D) हरात्मक श्रेणी में

Ans. B

Sol. Let $f(a) = a_1x^2 + a_2x + a_3$

$\because$ $f(1) = f(-1)$

$\Rightarrow$ $a_1 + a_2 + a_3 = a_1 - a_2 + a_3 \quad \Rightarrow \quad a_2 = 0 \quad \Rightarrow \quad f(x) = a_1x^2 + a_3$

$f'(x) = 2a_1x$

$f'(a) = 2a_1a$

$f'(b) = 2a_1b$

$f'(c) = 2a_1c$

$\because$ a, b, c are in AP

$\Rightarrow$ f′(a), f′(b), f′(c) are also in AP

24. If $x = e^{y+e^{y+.....to\ \infty}}$, $x > 0$, then $\frac{dy}{dx}$ is equal to : **[AIEEE-2004]**

यदि $x = e^{y+e^{y+.....to\ \infty}}$, $x > 0$, तो $\frac{dy}{dx}$ बराबर होगा : **[AIEEE-2004]**

(A) $\frac{x}{1+x}$ (B) $\frac{1}{x}$ (C) $\frac{1-x}{x}$ (D) $\frac{1+x}{x}$

Ans. C

Sol. $x = e^{y+x}$

$x + y = \ell n\, x$

$1 + \frac{dy}{dx} = \frac{1}{x}$

$\frac{dy}{dx} = \frac{1-x}{x}$

25. If $x^m . y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is equal to : **[AIEEE-2006]**

यदि $x^m . y^n = (x+y)^{m+n}$, तो $\frac{dy}{dx}$ बराबर होगा : **[AIEEE-2006]**

(A) $\frac{x+y}{xy}$ (B) xy (C) $\frac{x}{y}$ (D) $\frac{y}{x}$

Ans. D

Sol. $x^m y^n = (x+y)^{m+n}$

taking log both sides

$m\, \ell n\, x + n\, \ell n\, y = (m+n)\, .\, \ell n\, (x+y)$

$$\frac{m}{x} + \frac{n}{y}\frac{dy}{dx} = \frac{(m+n)}{x+y}\left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \quad \frac{dy}{dx}\left(\frac{n}{y} - \frac{(m+n)}{x+y}\right) = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{x}$$

26. $y=\frac{1}{1+(\tan\theta)^{\sin\theta-\cos\theta}+(\cot\theta)^{\cos\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\sin\theta}+(\cot\theta)^{\sin\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\cot\theta}+(\cot\theta)^{\cot\theta-\sin\theta}}$

then $\frac{dy}{d\theta}$ at $\theta=\pi/3$ is :

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) None of these

$y=\frac{1}{1+(\tan\theta)^{\sin\theta-\cos\theta}+(\cot\theta)^{\cos\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\sin\theta}+(\cot\theta)^{\sin\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\cot\theta}+(\cot\theta)^{\cot\theta-\sin\theta}}$

है तो $\frac{dy}{d\theta}$ का मान $\theta=\pi/3$ पर होगा :

(A) 0 (B) 1 (C) $\sqrt{3}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y=\frac{1}{1+(\tan\theta)^{\sin\theta-\cos\theta}+(\cot\theta)^{\cos\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\sin\theta}+(\cot\theta)^{\sin\theta-\cot\theta}}+\frac{1}{1+(\tan\theta)^{\cos\theta-\cot\theta}+(\cot\theta)^{\cot\theta-\sin\theta}}$

$=\frac{1}{1+\frac{(\tan\theta)^{\sin\theta}}{(\tan\theta)^{\cos\theta}}+\frac{(\cot\theta)^{\cos\theta}}{(\cot\theta)^{\cot\theta}}}+\frac{1}{1+\frac{(\tan\theta)^{\cos\theta}}{(\tan\theta)^{\sin\theta}}+\frac{(\cot\theta)^{\sin\theta}}{(\cot\theta)^{\cot\theta}}}+\frac{1}{1+\frac{(\tan\theta)^{\cos\theta}}{(\tan\theta)^{\cot\theta}}+\frac{(\cot\theta)^{\cot\theta}}{(\cot\theta)^{\sin\theta}}}$

$\Rightarrow\frac{1}{1+\frac{(\tan\theta)^{\sin\theta}}{(\tan\theta)^{\cos\theta}}+\frac{(\tan\theta)^{\cot\theta\theta}}{(\tan\theta)^{\cos\theta}}}+\frac{1}{1+\frac{(\tan\theta)^{\cos\theta}}{(\tan\theta)^{\sin\theta}}+\frac{(\tan\theta)^{\cot\theta}}{(\tan\theta)^{\sin\theta}}}+\frac{1}{1+\frac{(\tan\theta)^{\cos\theta}}{(\tan\theta)^{\cot\theta}}+\frac{(\tan\theta)^{\sin\theta}}{(\tan\theta)^{\cot\theta}}}$

$\Rightarrow\frac{(\tan\theta)^{\cos\theta}}{(\tan\theta)^{\sin\theta}+(\tan\theta)^{\cos\theta}+(\tan\theta)^{\cot\theta}}+\frac{(\tan\theta)^{\sin\theta}}{(\tan\theta)^{\sin\theta}+(\tan\theta)^{\cos\theta}+(\tan\theta)^{\cot\theta}}+$

$\frac{(\tan\theta)^{\cot\theta}}{(\tan\theta)^{\sin\theta}+(\tan\theta)^{\cos\theta}+(\tan\theta)^{\cot\theta}}$

$y=\frac{(\tan\theta)^{\sin\theta}+(\tan\theta)^{\cos\theta}+(\tan\theta)^{\cot\theta}}{(\tan\theta)^{\sin\theta}+(\tan\theta)^{\cos\theta}+(\tan\theta)^{\cot\theta}}$

$y=1$

$\frac{dy}{dx}=0$

27. Let $f'(x) = \sin(x^2)$ and $y = f(x^2 + 1)$ then $\frac{dy}{dx}$ at $x = 1$ is :

माना $f'(x) = \sin(x^2)$ तथा $y = f(x^2 + 1)$ है, तो $\frac{dy}{dx}$ का मान $x = 1$ पर होगा :

(A) $2\sin 2$ (B) $2\cos 2$ (C) $2\sin 4$ (D) $\cos 2$

Ans. C

Sol. $f'(x) = \sin(x^2)$ $y = f(x^2 + 1)$

$y = f(x^2 + 1)$

$$\frac{dy}{dx} = f'(x^2 + 1).2x$$

$$\left.\frac{dy}{dx}\right|_{x=1} = f'(1+1).2(1)$$

$$= 2f'(2)$$

$$f'(2) = \sin(2^2) = \sin 4$$

$$\left.\frac{dy}{dx}\right|_{x=1} = 2\sin 4$$

28. If $f(x) = |\sin x - |\cos x||$, then $f'\left(\frac{7\pi}{6}\right) =$

यदि $f(x) = |\sin x - |\cos x||$ है, तो $f'\left(\frac{7\pi}{6}\right) =$

(A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{1-\sqrt{3}}{2}$ (C) $\frac{\sqrt{3}-1}{2}$ (D) $\frac{-1-\sqrt{3}}{2}$

Ans. C

Sol. $f(x) = |\sin x - |\cos x||$

$$x = \frac{7\pi}{6}$$

$$= |\sin x + \cos x|$$

$$f(x) = -\sin x - \cos x$$

$f'(x) = -\cos x + \sin x$

$x = \frac{7\pi}{6}$

$f'(x) = -\cos\frac{7\pi}{6} + \sin\frac{7\pi}{6}$

$f'\left(\frac{7\pi}{6}\right) = \frac{\sqrt{3}-1}{2}$

29.. If $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$; then f'(0) is equal to :

यदि $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$ है तो f'(0) बराबर होगा :

(A) 4 (B) 3 (C) 2 (D) 1

Ans. D

Sol. $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$

$$f'(x) = \frac{1}{2\sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}} \times \frac{\frac{1}{\sqrt{1-x^2}} \times \left(1-\tan^{-1}x\right) + \frac{1}{1+x^2}(1+\sin^{-1}x)}{(1-\tan^{-1}x)^2}$$

$x = 0$

$f'(0) = 1$

30. If $f(x) = 2 + |x| - |x-1| - |x+1|$, then $f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$ is equal to :

यदि $f(x) = 2 + |x| - |x-1| - |x+1|$ है, तो $f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$ का मान होगा :

(A) 1 (B) –1 (C) 2 (D) –2

Ans. D

Sol. $f(x) = 2 + |x| - |x-1| - |x+1|$

$$x = -\frac{1}{2}$$

$$f(x) = 2 - x - x + x - x - 1$$

$$f(x) = 1 - 2x$$

$$f'(x) = -2 = f'\left(-\frac{1}{2}\right) = -2$$

$$x = \frac{1}{2}$$

$$f(x) = 2 + \frac{1}{2} - x + 1 - x - 1$$

$$f(x) = \frac{5}{2} - 2x = f'(x) = -2$$

$$f'\left(\frac{1}{2}\right) = 2$$

$$x = \frac{3}{2}, \frac{5}{2}$$

$$f(x) = 2 + x - x + 1 - x - 1$$

$$f(x) = 2 - x$$

$$f'(x) = -1$$

$$f'\left(\frac{3}{2}\right) = f'\left(\frac{5}{2}\right) = -1$$

$$f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$$

$$= -2$$

31. Let g (x) = *l*n *f* (x) where f (x) is a twice differentiable positive function on (0, ∞) such that f (x + 1) = x f (x). Then for N = 1, 2, 3..... $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$ **[JEE-2008]**

माना g (x) = ln f (x) है जहां f (x) अन्तराल (0, ∞) में दो बार अवकलनीय धनात्मक फलन इस प्रकार है कि f(x + 1) = x f(x) तो N = 1, 2, 3..... के लिये $g''\left(N+\frac{1}{2}\right)-g''\left(\frac{1}{2}\right)=$ **[JEE-2008]**

(A) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right\}$ (B) $4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N-1)^2}\right\}$

(C) $-4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N+1)^2}\right\}$ (D) $4\left\{1+\frac{1}{9}+\frac{1}{25}+.....+\frac{1}{(2N+1)^2}\right\}$

Ans. A

Sol. $g(x+1)=\log[f(x+1)]=\log[xf(x)]=\log x+\log[f(x)]=\log x+g(x)$

$\Rightarrow \quad g(x+1)-g(x)=\log x$

$\Rightarrow \quad g''(x+1)-g''(x)=-\frac{1}{x^2}$

$\Rightarrow \quad g''\left(1+\frac{1}{2}\right)-g''\left(\frac{1}{2}\right)=-4$

$g''\left(1+\frac{3}{2}\right)-g''\left(\frac{3}{2}\right)=-\frac{4}{9}$

.....

$g''\left(N+\frac{1}{2}\right)-g''\left(N-\frac{1}{2}\right)=-\frac{4}{(2N-1)^2}$

Adding all, $g''\left(N+\frac{1}{2}\right)-g''\left(\frac{1}{2}\right)=-4\left\{1+\frac{1}{9}+.....+\frac{1}{(2N-1)^2}\right\}$

32. Let f : (–1, 1) → R be a differentiable function with f(0) = –1, f'(0) = 1. Let g(x) = |f(2f(x) + 2)|^2.] Then g'(0) : **[AIEEE-2010]**

माना f : (–1, 1) → R एक ऐसा अवकलनीय फलन है जिसमें f(0) = –1, f'(0) = 1 है। माना g(x) = |f(2f(x) + 2)|^2. तो g'(0) बराबर है : **[AIEEE-2010]**

(A) 4 (B) –4 (C) 0 (D) –2

Ans. B

Sol. g′(x) = 2f(2f(x) + 2) . f′ (2f(x) + 2) . 2f′(x)

g′(0) = 2f (2f(0) + 2) f′ (2f(0) + 2) . 2f′(0)

$= 2f(0)\, f'(0)\, 2f'(0) = (2)(-1)(1)(2)(1) = -4$

Hence correct option is (1)

अतः सही विकल्प (1) है।

33. For $x \in R$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then : **[JEE(Main)-2016]**

(A) $g'(0) = -\cos(\log 2)$

(B) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$

(C) g is not differentiable at $x = 0$

(D) $g'(0) = \cos(\log 2)$

$x \in R$ के लिए, $f(x) = |\log 2 - \sin x|$ तथा $g(x) = f(f(x))$ हैं, तो : **[JEE(Main)-2016]**

(A) $g'(0) = -\cos(\log 2)$

(B) $x = 0$ पर g अवकलनीय है तथा $g'(0) = -\sin(\log 2)$ है

(C) $x = 0$ पर g अवकलनीय नहीं है

(D) $g'(0) = \cos(\log 2)$

Ans. D

Sol. $g(x) = |\log 2 - \sin|\log 2 - \sin x||$

gx near by of $x = 0$

$g(x) = \log 2 - \sin(\log 2 - \sin x)$

$g'(x) = -(\cos(\log 2 - \sin x))(-\cos x)$

$g'(0) = \cos(\log 2)$

34. If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x}.g(x)$, then g(x) equals : **[JEE(Main)-2017]**

यदि $x \in \left(0, \frac{1}{4}\right)$ के लिए $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ का अवकलन $\sqrt{x}.g(x)$ है, तो g(x) बराबर है : **[JEE(Main)-2017]**

(A) $\frac{3x}{1-9x^3}$ (B) $\frac{3}{1+9x^3}$ (C) $\frac{9}{1+9x^3}$ (D) $\frac{3x\sqrt{x}}{1-9x^3}$

Ans. C

Sol. $f(x) = \tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right) = 2\tan^{-1}\left(3x\sqrt{x}\right)$

$\Rightarrow \quad f'(x) = \frac{9\sqrt{x}}{1+9x^3} = \sqrt{x}.g(x)$

$\Rightarrow \quad g(x) = \frac{9}{1+9x^3}$

DERIVATIVE OF INVERSE OF A FUNCTION

35. Let g(x) be the inverse of f(x) such that $f'(x) = \frac{1}{1+x^5}$, then $\frac{d^2(g(x))}{dx^2}$ is equal to :

माना f(x) का प्रतिलोम फलन g(x) इस प्रकार है कि $f'(x) = \frac{1}{1+x^5}$ है तो $\frac{d^2(g(x))}{dx^2}$ का मान होगा :

(A) $\frac{1}{1+(g(x))^5}$ (B) $\frac{g'(x)}{1+(g(x))^5}$ (C) $5(g(x))^4(1+(g(x))^5)$ (D) $1+(g(x))^5$

Ans. C

Sol. $f'(x) = \frac{1}{1+x^5}$, $\frac{d^2(g(x))}{dx^2} = ?$

$f^{-1}(x) = g(x)$

$f(g(x)) = x$

$f'(g(x)) \times g'(x) = 1$

$g'(x) = \frac{1}{f'(g(x))}$

$= \frac{1}{\frac{1}{1+[g(x)]^2}} = 1+(g(x))^5$

$g''(x) = 0 + 5[g(x)]^4 \times g'(x)$

$= 5[g(x)]^4[1+(g(x))^5]$

36. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f. The value of $g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)$ has the value equal to:

माना $f(x) = x + \sin x$. यदि g फलन f का प्रतिलोम है, तो $g'\left(\frac{\pi}{4}+\frac{1}{\sqrt{2}}\right)$ का मान है :

(A) $\sqrt{2}-1$ (B) $\frac{\sqrt{2}+1}{\sqrt{2}}$ (C) $2-\sqrt{2}$ (D) $\sqrt{2}+1$

Ans. C

Sol. $f(x) = y = x + \sin x$

$$\frac{dy}{dx} = 1 + \cos x$$

$$g'(y) = \frac{dx}{dy} = \frac{1}{1+\cos x}$$

where $y = \frac{\pi}{4} + \frac{1}{\sqrt{2}} = x + \sin x \quad \Rightarrow \quad x = \frac{\pi}{4}$

$$\therefore \quad g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right) = \frac{1}{1+(1/\sqrt{2})} = \frac{\sqrt{2}}{\sqrt{2}+1} = \sqrt{2}(\sqrt{2}-1) = 2-\sqrt{2}\ ]$$

37. Let g is the inverse function of f and $f'(x) = \frac{x^{10}}{\left(1+x^2\right)}$. If $g(2) = a$ then $g'(2)$ is equal to :

यदि g (x), f का प्रतिलोम फलन है और $f'(x) = \frac{x^{10}}{\left(1+x^2\right)}$. यदि $g(2) = a$ तब $g'(2)$ बराबर :

(A) $\frac{5}{2^{10}}$ (B) $\frac{1+a^2}{a^{10}}$ (C) $\frac{a^{10}}{1+a^2}$ (D) $\frac{1+a^{10}}{a^2}$

Ans. B

Sol. $f[g(x)] = x \Rightarrow f'[g(x)] \cdot [g'(x)] = 1 \Rightarrow f'[g(2)]\, g'(2) = 12 \Rightarrow f'(a)\, g'(2)$ [putting $x = 2$]

given, $f'(a) = \frac{a^{10}}{1+a^2} \Rightarrow \quad g'(2) = \frac{1+a^2}{a^{10}}$]

Alternative : $g[f(x)] = x$

$g'[f(x)] \cdot f'(x) = 1$

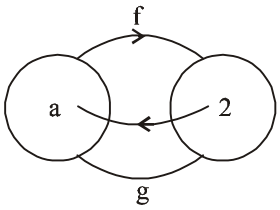
now $g(2) = a \Rightarrow f(a) = 2$

$\therefore$ g and f are inverse of each other

now $f(x) = 2 \Rightarrow g(2) = x = a$

$\therefore \quad g'(2) \cdot f'(a) = 1$

$$g'(2) = \frac{1}{f'(a)} = \frac{1+a^2}{a^{10}}$$



HIGHER ORDER DERIVATIVE

38. If $y = \frac{1}{x^2 - a^2}$, then $\frac{d^2y}{dx^2}$ equals :

यदि $y = \frac{1}{x^2 - a^2}$, तो $\frac{d^2y}{dx^2}$ बराबर है :

(A) $\frac{3x^2 + a^2}{\left(x^2 - a^2\right)^3}$ (B) $\frac{3x^2 + a^2}{\left(x^2 - a^2\right)^4}$ (C) $\frac{2(3x^2 + a^2)}{\left(x^2 - a^2\right)^3}$ (D) $\frac{2(3x^2 + a^2)}{\left(x^2 - a^2\right)^4}$

Ans. C

Sol. $y = \frac{1}{x^2 - a^2}$

$$\frac{dy}{dx} = \frac{-1(2x)}{(x^2 - a^2)^2}$$

$$\frac{d^2y}{dx^2} = \frac{-2(x^2 - a^2)^2 + 2x \times 2(x^2 - a^2)}{(x^2 - a^2)^4} \times 2x$$

$$= \frac{2(x^2 - a^2)[4x^2 - x^2 + a^2]}{(x^2 - a^2)^4}$$

$$= \frac{2(3x^2 + a^2)}{(x^2 - a^2)^3}$$

39. If $y = ae^{\alpha x} - be^{-\alpha x}$, then the value of $\frac{d^2y}{dx^2}$ equals :

यदि $y = ae^{\alpha x} - be^{-\alpha x}$, तब $\frac{d^2y}{dx^2}$ का मान बराबर है :

(A) αy (B) $-\alpha y$ (C) $\alpha^2 y$ (D) $-\alpha^2 y$

Ans. C

Sol. $y = ae^{\alpha x} - be^{-\alpha x}$

$$\frac{dy}{dx} = a\alpha e^{\alpha x} + b\alpha e^{-\alpha x}$$

$$\frac{d^2y}{dx^2} = a\alpha^2 e^{\alpha x} - b\alpha^2 e^{-\alpha x}$$

$$\frac{d^2y}{dx^2} = \alpha^2 y$$

40. If $y^{1/m} = x + \sqrt{1+x^2}$, then $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ equals :

(A) my^2 (B) m^2y (C) m^2y^2 (D) None of these

यदि $y^{1/m} = x + \sqrt{1+x^2}$, तब $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ बराबर है :

(A) my^2 (B) m^2y (C) m^2y^2 (D) इनमें से कोई नहीं

Ans. B

Sol. $y^{\frac{1}{m}} = x + \sqrt{1+x^2}$

$$\frac{1}{m}\ln y = \ln\left(x + \sqrt{1+x^2}\right)$$

$$\frac{1}{my}\frac{dy}{dx} = \frac{1 + \frac{2x}{2\sqrt{1+x^2}}}{x + \sqrt{1+x^2}}$$

$$\frac{dy}{dx} = \frac{my}{\sqrt{1+x^2}}$$

$$\frac{d^2y}{dx^2} = \frac{m\frac{dy}{dx}\sqrt{1+x^2} - my\frac{2x}{2\sqrt{1+x^2}}}{1+x^2}$$

$$(1+x^2)\frac{d^2y}{dx^2} = m\sqrt{1+x^2}\frac{dy}{dx} - \frac{mxy}{\sqrt{1+x^2}}$$

$$(1+x^2)\frac{d^2y}{dx^2} = m\sqrt{1+x^2}\frac{dy}{dx} - x\frac{dy}{dx}$$

$$(1+x^2)+x\frac{dy}{dx}=m\sqrt{1+x^2}\left(\frac{my}{\sqrt{+x^2}}\right)$$

$=m^2y$

41. If $y = a\cos(\log x) + b\sin(\log x)$ then $(x^2y_2 + xy_1)$ equals to :

यदि $y = a\cos(\log x) + b\sin(\log x)$ तब $(x^2y_2 + xy_1)$ बराबर है :

(A) y (B) ay (C) by (D) –y

Ans. D

Sol. $y = a\cos(\log x) + b\sin(\log x)$

$$y_1=\frac{-a\sin(\log x)}{x}+\frac{b\cos(\log x)}{x}$$

$$y_1=\frac{-a\sin(\log x)+b\cos(\log x)}{x}$$

$$y_2=\frac{x\left(\frac{-a\cos(\log x)}{x}-\frac{b\sin(\log x)}{x}\right)-\left(b\cos(\log x)-a\sin(\log x)\right)}{x^2}$$

$$x^2y_2=a\cos(\log x)-b\sin(\log x)-b\cos(\log x)+a\sin(\log x)$$

$xy_1 = -a\sin(\log x) + b\cos(\log x)$

$x^2y_2 + xy_1 = -a\cos(\log x) - b\sin(\log x)$

$= -y$

42. If $y=\sin 2x\cos 6x$, then the value of $\frac{d^2y}{dx^2}$ is :

(A) $8\sin 4x + 32\sin 8x$ (B) $32\sin 8x - 8\sin 4x$ (C) $8\sin 4x - 32\sin 8x$ (D) None of these

यदि $y=\sin 2x\cos 6x$, तब $\frac{d^2y}{dx^2}$ का मान होगा :

(A) $8\sin 4x + 32\sin 8x$ (B) $32\sin 8x - 8\sin 4x$ (C) $8\sin 4x - 32\sin 8x$ (D) इनमें से कोई नहीं

Ans. C

Sol. $y=\frac{2\sin 2x\cos 6x}{2}$

$$y = \frac{\sin 8x - \sin 4x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}[8\cos 8x - 4\cos 4x]$$

$$= 4\cos 8x - 2\cos 4x$$

$$\frac{d^2y}{dx^2} = -32\sin 8x + 8\sin 4x$$

43. If $y = \sqrt{2x - x^2}$, then $yy_2 + y_1^2$ is equal to :

(A) 0 (B) –1 (C) 1 (D) Does not exist

यदि $y = \sqrt{2x - x^2}$ है तो $yy_2 + y_1^2$ बराबर होगा :

(A) 0 (B) –1 (C) 1 (D) विद्यमान नहीं

Ans. B

Sol. $y = \sqrt{2x - x^2}$

$$y_1 = \frac{2 - 2x}{2\sqrt{2x - x^2}} = \frac{1 - x}{\sqrt{2x - x^2}}$$

$$y_2 = \frac{(-1)\sqrt{2x - x^2} - (1 - x)\dfrac{2 - 2x}{2\sqrt{2x - x^2}}}{(2x - x^2)}$$

$$y_2 = \frac{-\sqrt{2x - x^2} - \dfrac{(1 - x)^2}{\sqrt{2x - x^2}}}{2x - x^2}$$

$$y_2 = \frac{-2x + x^2 - 1 - x^2 + 2x}{(2x - x^2)\sqrt{2x - x^2}}$$

$$y_2 = \frac{1}{(x^2 - 2x)\sqrt{2x - x^2}}$$

$$yy_2 + y_1^2 = \frac{\sqrt{2x - x^2}}{(x^2 - 2x)\sqrt{2x - x^2}} + \frac{(1 - x)^2}{2x - x^2}$$

$$= \frac{1-(1-x)^2}{x^2-2x} = -1$$

44. If $f(x) = x^n$, then the value of $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + ... + \frac{(-1)^n f^n(1)}{n!}$ is :

यदि $f(x) = x^n$ तो $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + ... + \frac{(-1)^n f^n(1)}{n!}$ का मान है :

(A) 1 (B) 2^n (C) 2^{n-1} (D) 0

Ans. D

Sol. $f(x) = x^n$

$\therefore$ $f'(x) = nx^{n-1}$, $f''(x) = n(n-1)x^{n-2}$

.....

$f^{\text{''''.....n times}}(x) = n!$

Now $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - ..... \frac{f^{\text{''''.....n times}}(1)}{n!}$

$= 1 - \frac{n}{1!} + \frac{n(n-1)}{2!} - ..... \frac{n!}{n!} = (1-1)^n = 0$

45. Let 'f' be a differentiable real valued function satisfying $f(x+2y) = f(x) + f(2y) + 6xy(x+2y)\ \forall\ x, y \in R$. Then $f''(0), f''(1), f''(2)$ .... are in :

(A) AP (B) GP (C) HP (D) None of these

माना 'f' एक अवकलनीय वास्तविक मान फलन इस प्रकार है कि $f(x+2y) = f(x) + f(2y) + 6xy(x+2y)\ \forall\ x, y \in R$ है तो $f''(0), f''(1), f''(2)$ .... में होगें :

(A) AP (B) GP (C) HP (D) इनमें से कोई नहीं

Ans. A

Sol. $x = y = 0$

$f(0) = 0$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\left[f(x) + f(h) + 3xh(x+h) - f(x)\right]}{h}$$

$$f'(x)=\lim_{h\to 0}\left[\frac{f(h)}{h}+3x(x+h)\right]$$

$$f'(x)=\lim_{h\to 0}\left[\frac{f'(h)}{1}\right]+3x^2$$

$f'(x)=f'(0)+3x^2$

$f''(x)=0+6x$

$f''(0), f''(1), f''(2)$

$0,6,,12 \rightarrow$ A.P.

46. $\frac{d^2x}{dy^2}$ equals : **[JEE-2007]**

$\frac{d^2x}{dy^2}$ बराबर होगा : **[JEE-2007]**

(A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$ (C) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (D) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

Ans. D

Sol. $\frac{d}{dy}\left(\frac{dx}{dy}\right)=\frac{d}{dy}\left(\frac{1}{dy/dx}\right)$

$$=-\left(\frac{dy}{dx}\right)^{-2}\frac{d^2y}{dx^2}\cdot\frac{dx}{dy}=-\frac{d^2y}{dx^2}\left(\frac{dy}{dx}\right)^{-3}$$

L'HOSPITAL RULE

47. Let $f(x)=\frac{e^{\tan x}-e^x+\ln(\sec x+\tan x)-x}{\tan x-x}$ be a continuous function at $x=0$. The value of $f(0)$ equals :

माना $f(x)=\frac{e^{\tan x}-e^x+\ln(\sec x+\tan x)-x}{\tan x-x}$, $x=0$ पर सतत् फलन है तब $f(0)$ बराबर :

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 2

Ans. C

Sol. For continuity of f at $x = 0$, we have

$$k = f(0) = \lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{e^{\tan x} - e^{x}}{\tan x - x} + \lim_{x\to 0} \frac{ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3}\right)x^3}$$

$$= \lim_{x\to 0} \frac{e^{x}(e^{\tan x - x} - 1)}{\tan x - x} + 3\lim_{x\to 0} \frac{ln(\sec x - \tan x) - x}{x^3} = 1 + 3\lim_{x\to 0} \frac{\sec x - 1}{3x^2} \quad \text{(Using L.H. Rule)}$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$ **Ans.**

48. If the normal of $y = f(x)$ at $(0, 0)$ is given by $y - x = 0$, then $\lim\limits_{x\to 0} \frac{x^2}{f(x^2) - 20\, f(9x^2) + 2\, f(99x^2)}$ is equal to :

(A) $\frac{1}{19}$ (B) $\frac{-1}{19}$ (C) $\frac{1}{2}$ (D) non-existent

यदि $y = f(x)$ का $(0, 0)$ अभिलम्ब समीकरण $y - x = 0$ है, तो $\lim\limits_{x\to 0} \frac{x^2}{f(x^2) - 20\, f(9x^2) + 2\, f(99x^2)}$ बराबर होगी :

(A) $\frac{1}{19}$ (B) $\frac{-1}{19}$ (C) $\frac{1}{2}$ (D) विद्यमान नहीं

Ans. B

Sol. The form being $\frac{0}{0}$, so the required limit by L'Hospital rule is

$$\lim_{x\to 0} \frac{2x}{2x\, f'(x^2) - 360x\ f'(9x^2) + 396x\ f'(99x^2)} = \lim_{x\to 0} \frac{1}{f'\left(x^2\right) - 180 f'(9x^2) + 198 f'(99x^2)}$$

For $f(x) = 0$, the slope of tangent is $f'(x)$ and slope of normal at $x = 0$ is $\frac{-1}{f'(0)} = 1$ (Given)

$\Rightarrow f'(0) = -1$

$\therefore$ Limit $= \frac{1}{f'(0) - 180 f'(0) + 198 f'(0)} = \frac{-1}{(1 - 180 + 198)} = \frac{-1}{19}$ **Ans.**

49. $\lim\limits_{x\to\infty} \left(\frac{x^{100}}{e^{x}} + \left(\cos\frac{2}{x}\right)^{x^2} \right)$ equals :

$\lim\limits_{x\to\infty}\left(\frac{x^{100}}{e^x}+\left(\cos\frac{2}{x}\right)^{x^2}\right)$ बराबर होगा :

(A) e^{-1} (B) e^{-4} (C) $(1+e^{-2})$ (D) e^{-2}

Ans. D

Sol. $\lim\limits_{x\to\infty}\frac{x^{100}}{e^x}=0$ (using L'Hospital's Rule)

$\lim\limits_{x\to\infty}\left(\cos\frac{2}{x}\right)^{x^2}$ (1^∞)

$= e^{\lim\limits_{x\to\infty} x^2\left(\cos\frac{2}{x}-1\right)} = e^{-\lim\limits_{x\to0}\left(\frac{1-\cos t}{t^2}\right)\cdot 4}$; put $\frac{2}{x}=t \quad\Rightarrow\quad x=\frac{2}{t}$

$= e^{-2}$ **Ans.**

50. Let $f(x)$ be diffrentiable at $x=h$ then $\lim\limits_{x\to h}\frac{(x+h)f(x)-2hf(h)}{x-h}$ is equal to :

माना f (x); x = h पर अवकलनीय है, तब $\lim\limits_{x\to h}\frac{(x+h)f(x)-2hf(h)}{x-h}$ बराबर :

(A) f(h) + 2hf'(h) (B) 2 f(h) + hf'(h) (C) hf(h) + 2f'(h) (D) hf(h) – 2f'(h)

Ans. A

Sol. Use L' Hospital's rule

51. For $x>0$, $\lim\limits_{x\to0}\left((\sin x)^{1/x}+(1/x)^{\sin x}\right)$ is :

$x>0$ के लिये $\lim\limits_{x\to0}\left((\sin x)^{1/x}+(1/x)^{\sin x}\right)$ है :

(A) 0 (B) –1 (C) 1 (D) 2

Ans. C

Sol. for $x>0$

$\lim\limits_{x\to0}(\sin x)^{\frac{1}{x}}+\left(\frac{1}{x}\right)^{\sin x}$

$$\lim_{x\to 0}(\sin x)^{\frac{1}{x}} + \lim_{x\to 0}\left(\frac{1}{x}\right)^{\sin x} = 0 + \lim_{x\to 0}\left(\frac{1}{x}\right)^{\sin x} = e^{\lim_{x\to 0}(\sin x).\ell n\left(\frac{1}{x}\right)} = e^{\lim_{x\to 0}\frac{\ell n(x)}{-\operatorname{cosec} x}}\left(\frac{\infty}{\infty}\text{ form}\right)$$

$$= e^{\lim_{x\to 0}\frac{\frac{1}{x}}{(\operatorname{cosec} x\cot x)}} = e^{\lim_{x\to 0}\frac{\sin^2 x}{x\cos x}} = e^{o} = 1$$

Exercise J-II

BASED ON DIFFERENTIATION TECHNIQUES

1. Let f, g and h are differentiable functions. If $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are $(fg)'(0) = 6$; $(gh)'(0) = 4$ and $(hf)'(0) = 5$ then compute the value of $(fgh)'(0)$.

माना f, g तथा h अवकलनीय फलन है। यदि $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ तथा इनके युग्मों के गुणनों के $x = 0$ पर अवकलज इस प्रकार है। $(fg)'(0) = 6$; $(gh)'(0) = 4$ और $(hf)'(0) = 5$ तो $(fgh)'(0)$ का मान निकालिए।

Ans. 16

Sol. $(fgh)' = \dfrac{(fg)'h + f(gh)' + g(fh)'}{2}$

$(fgh)'(0) = \dfrac{6\times3 + 1\times4 + 2\times5}{2}$

$= 16$

2. (a) If $y = (\cos x)^{\ln x} + (\ln x)^x$ find $\dfrac{dy}{dx}$ (b) If $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$. Find $\dfrac{dy}{dx}$.

(a) यदि $y = (\cos x)^{\ln x} + (\ln x)^x$ तो $\dfrac{dy}{dx}$ ज्ञात कीजिए (b) यदि $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$ तो $\dfrac{dy}{dx}$ ज्ञात कीजिए।

Ans. (a) $\dfrac{dy}{dx} = (\cos x)^{\ln x}\left[\dfrac{\ln(\cos x)}{x} - \tan x \ln x\right] + (\ln x)^x\left[\dfrac{1}{\ln x} + \ln(\ln x)\right]$

(b) $\dfrac{dy}{dx} = e^{x^{e^x}} \cdot x^{e^x}\left[\dfrac{e^x}{x} + e^x \ln x\right] + e^{x^{x^e}}\ x^{e-1}\ x^{x^e}[1 + e\ln x] + x^{e^{e^x}}\ e^{e^x}\left[\dfrac{1}{x} + e^x \ln x\right]$

Sol. (a) $y = (\cos x)^{\ln x} + (\ln x)^x$

$y_1 = (\cos x)^{\ln x}$

$\log y_1 = \ln x \ln(\cos x)$

$\dfrac{1}{y_1}\dfrac{dy_1}{dx} = \dfrac{\ln(\cos x)}{x} - \ln x \dfrac{\sin x}{\cos x}$

$\dfrac{dy_1}{dx} = (\cos x)^{\ln x}\left[\dfrac{\ln(\cos x)}{x} - \tan x \ln x\right]$

$y_2 = (\ln x)^x$

$\log y_2 = x \ln(\ln x)$

$$\frac{1}{y_2}\frac{dy_2}{dx} = \ln(\ln x) + \frac{1}{\ln x}$$

$$\frac{dy_2}{dx} = (\ln x)^x\left[\ln(\ln x) + \frac{1}{\ln x}\right]$$

$$\frac{dy}{dx} = \frac{dy_1}{dx} + \frac{dy_2}{dx}$$

(b) $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$

$$y_1 = e^{x^{e^x}}$$

$$\log y_1 = x^{e^x}$$

$$\log(\log y_1) = e^x \ln x$$

$$\frac{1}{y_1}\frac{1}{\log y_1}\frac{dy_1}{dx} = e^x \ln x + \frac{e^x}{x}$$

$$\frac{dy_1}{dx} = e^{x^{e^x}}\, x^{e^x}\left[e^x \ln x + \frac{e^x}{x}\right]$$

$$y_2 = e^{x^{x^e}}$$

$$\log y_2 = x^{x^e}$$

$$\log(\log y_2) = x^e \ln x$$

$$\frac{1}{y_2 \log y_2}\frac{dy_2}{dx} = ex^{e-1}\ln x + \frac{x^e}{x}$$

$$\frac{dy_2}{dx} = e^{x^{x^e}}.x^{x^e}\left[ex^{e-1}\ln x + \frac{x^e}{x}\right]$$

$$\frac{dy_2}{dx} = e^{x^{x^e}}.x^{x^e}.x^{e-1}\left[e\ln x + 1\right]$$

$$y_3 = x^{e^{e^x}}$$

$$\log y_3 = e^{e^x} \log x$$

$$\log(\log y_3) = e^x + \log(\log x)$$

$$\frac{1}{y_3 \log y_3}\frac{dy_3}{dx} = e^x + \frac{1}{x \ln x}$$

$$\frac{dy_3}{dx} = x^{e^{e^x}} . e^{e^x} \log x\left[e^x + \frac{1}{x \ln x}\right]$$

$$\frac{dy_3}{dx} = x^{e^{e^x}} . e^{e^x}\left[\ln x\, e^x + \frac{1}{x}\right]$$

$$\frac{dy}{dx} = e^{x^{e^x}} . x^{e^x}\left[e^x \ln x + \frac{e^x}{x}\right] + e^{x^{x^e}} . x^{x^e} . x^{e-1}\left[e \ln x + 1\right] + x^{e^{e^x}} . e^{e^x}\left[\ln x\, e^x + \frac{1}{x}\right]$$

3. If $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ell n\sqrt{x + \sqrt{x^2+1}}$ prove that $2y = xy' + \ln y'$. where ' denotes the derivative.

यदि $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ell n\sqrt{x + \sqrt{x^2+1}}$ तो सिद्ध कीजिये $2y = xy' + \ln y'$ जहाँ y ', y का अवकलज है।

Ans.

Sol. $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ell n\sqrt{x + \sqrt{x^2+1}}$

$$2y = x^2 + x\sqrt{x^2+1} + \ell n\left(x + \sqrt{x^2+1}\right) \quad .....(i)$$

$$2y' = 2x + \sqrt{x^2+1} + \frac{2x^2}{2\sqrt{x^2+1}} + \frac{1 + \frac{2x}{2\sqrt{x^2+1}}}{x + \sqrt{x^2+1}}$$

$$2y' = 2x + \sqrt{x^2+1} + \frac{x^2}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}$$

$$= 2x + \sqrt{x^2+1} + \sqrt{x^2+1}$$

$$y' = \left(x + \sqrt{x^2+1}\right)$$

Put the value of $x + \sqrt{x^2+1}$ in (i)

$2y = xy' + \ell n\, y'$

4. If $y = ln\left(x^{e^x \cdot a^y}\right)^{y^x}$ find $\frac{dy}{dx}$.

यदि $y = ln\left(x^{e^x \cdot a^y}\right)^{y^x}$ तो $\frac{dy}{dx}$ ज्ञात कीजिए।

Ans. $\frac{y}{x} \cdot \frac{x\ \ell n\, x + x\ \ell n\, x\ .\ell n\, y\ + 1}{\ell n\, x\ (1 - x - y\, \ell n\, a)}$

Sol. $y = ln\left(x^{e^x \cdot a^y}\right)^{y^x}$

$y = e^x\, a^y\, y^x\, \ln x$

$\ln y = x + y\ln a + x\ln y + \ln(\ln x)$

$\frac{1}{y}y' = 1 + y'\ln a + \ln y + \frac{x}{y}y' + \frac{1}{x\ln x}$

$\frac{y'}{y} - y'(\ln a) - \frac{x}{y}y' = 1 + \ln y + \frac{1}{x\ln x}$

$y'\left(\frac{1 - y\ln a - x}{y}\right) = \frac{x\ln x + x\ln x\ln y + 1}{x\ln x}$

$\frac{dy}{dx} = \frac{y}{x} \cdot \frac{x\ln x + x\ln x\ln y + 1}{\ln x(1 - y\ln a - x)}$

5. If $y = 1 + \frac{x_1}{x - x_1} + \frac{x_2 \cdot x}{(x - x_1)(x - x_2)} + \frac{x_3 \cdot x^2}{(x - x_1)(x - x_2)(x - x_3)}$ +..... upto (n + 1) terms then prove that

$\frac{dy}{dx} = \frac{y}{x}\left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + ... + \frac{x_n}{x_n - x}\right]$

यदि $y = 1 + \frac{x_1}{x - x_1} + \frac{x_2 \cdot x}{(x - x_1)(x - x_2)} + \frac{x_3 \cdot x^2}{(x - x_1)(x - x_2)(x - x_3)}$ +..... (n + 1) पदों तक तो सिद्ध कीजिये

$\frac{dy}{dx} = \frac{y}{x}\left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + ... + \frac{x_n}{x_n - x}\right]$

Ans.

Sol. $\because y = 1 + \frac{x_1}{x - x_1} + \frac{x_2 x}{(x - x_1)(x - x_2)} + \frac{x_3 x}{(x - x_1)(x - x_2)(x - x_3)} + ...(n+1)$ terms

$= \frac{x}{(x - x_1)} + \frac{x_2 x}{(x - x_1)(x - x_2)} + \frac{x_3 x}{(x - x_1)(x - x_2)(x - x_3)} + ...(n+1)$ terms

$= \frac{x^2}{(x - x_1)(x - a_2)} + \frac{x_3 x}{(x - x_1)(x - x_2)(x - x_3)} + ...(n+1)$ terms

finally we get

$$y = \frac{x^n}{(x - x_1)(x - x_2)(x - x_3)....(x - x_n)}$$

Taking $\log_e$ of both sides, we have

$\ell ny = n\,\ell nx - \ell n(x - x_1) - \ell n(x - x_2) - \ell n(x - x_3) - .....- \ell n(x - x_n)$

Differentiating both sides w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = \left[\frac{n}{x} - \frac{1}{x - x_1} - \frac{1}{x - x_2} - \frac{1}{x - x_3} - ..... - \frac{1}{x - x_n}\right]$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{x}\left[n - \frac{x}{x - x_1} - \frac{x}{x - x_2} - \frac{x}{x - x_3} - ..... - \frac{x}{x - x_n}\right]$$

$$= \frac{y}{x}\left[(1+1+1+1+.....n \text{ times}) - \frac{x}{x - x_1} - \frac{x}{x - x_2} - \frac{x}{x - x_3} - ..... - \frac{x}{x - x_n}\right]$$

$$= \frac{y}{x}\left[\left(1 - \frac{x}{x - x_1}\right) + \left(1 - \frac{x}{x - x_2}\right) + \left(1 - \frac{x}{x - x_3}\right) + ... + \left(1 - \frac{x}{x - x_n}\right)\right]$$

$$\therefore \quad \frac{dy}{dx} = \frac{y}{x}\left[\left(\frac{-x_1}{x - x_1}\right) + \left(\frac{-x_2}{x - x_2}\right) + \left(\frac{-x_3}{x - x_3}\right) + ... + \left(\frac{-x_n}{x - x_n}\right)\right]$$

$$= \frac{y}{x}\left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + ... + \frac{x_n}{x_n - x}\right]$$

6. If $x = \text{cosec}\,\theta - \sin\theta$; $y = \text{cosec}^n\theta - \sin^n\theta$, then show that $(x^2 + 4)\left(\frac{dy}{dx}\right)^2 - n^2(y^2 + 4) = 0$.

यदि $x = \text{cosec}\,\theta - \sin\theta$; $y = \text{cosec}^n\theta - \sin^n\theta$, तो दर्शाइए $(x^2 + 4)\left(\frac{dy}{dx}\right)^2 - n^2(y^2 + 4) = 0$

Ans.

Sol. $x = \operatorname{cosec}\theta - \sin\theta$

$$\frac{dx}{d\theta} = -\operatorname{cosec}\theta\cot\theta - \cos\theta \times \frac{\sin\theta}{\sin\theta}$$

$$\frac{dx}{d\theta} = -\operatorname{cosec}\theta\cot\theta - \cot\theta\sin\theta$$

$$\frac{dx}{d\theta} = -\cot\theta\left(\operatorname{cosec}\theta + \sin\theta\right)$$

$y = \operatorname{cosec}^n\theta - \sin^n\theta$

$$\frac{dy}{d\theta} = x\operatorname{cosec}^{n-1}\theta\left(-\operatorname{cosec}\theta\cot\theta\right) - n\sin^{n-1}\theta \times \cos\theta$$

$$\frac{dy}{d\theta} = -n\operatorname{cosec}^n\theta\cot\theta - n\sin^{n-1}\theta\cos\theta \times \frac{\sin\theta}{\sin\theta}$$

$$\frac{dy}{d\theta} = -n\operatorname{cosec}^n\theta\cot\theta - n\sin^n\theta\cot\theta$$

$$\frac{dy}{d\theta} = -n\cot\theta\left(\operatorname{cosec}^n\theta + \sin^n\theta\right)$$

$$\frac{dy}{dx} = \frac{-n\cot\theta(\operatorname{cosec}^n\theta + \sin^n\theta)}{-\cot\theta(\operatorname{cosec}\theta + \sin\theta)}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{n^2(\operatorname{cosec}^n\theta + \sin^n\theta)^2}{(\operatorname{cosec}\theta + \sin\theta)^2}$$

$x = \operatorname{cosec}\theta - \sin\theta$

$x^2 = \operatorname{cosec}^2\theta + \sin^2\theta - 2 + 2 - 2$

$x^2 = (\operatorname{cosec}\theta + \sin\theta)^2 - 4$

$x^2 + 4 = (\operatorname{cosec}\theta + \sin\theta)^2$

$y = \operatorname{cosec}^n\theta - \sin^n\theta$

$y^2 = \operatorname{cosec}^{2n}\theta + \sin^{2n}\theta - 2 + 2 - 2$

$y^2 + 4 = (\operatorname{cosec}^n\theta + \sin^n\theta)^2$

$$\left(\frac{dy}{dx}\right)^2 = n^2\frac{(y^2+4)}{(x^2+4)}$$

$$(x^2+4)\left(\frac{dy}{dx}\right)^2 - n^2(y^2+4) = 0$$

7. If a curve is represented parametrically by the equations

$x = \sin\left(t+\frac{7\pi}{12}\right)+\sin\left(t-\frac{\pi}{12}\right)+\sin\left(t+\frac{3\pi}{12}\right)$, $y = \cos\left(t+\frac{7\pi}{12}\right)+\cos\left(t-\frac{\pi}{12}\right)+\cos\left(t+\frac{3\pi}{12}\right)$ then find the value of $\frac{d}{dt}\left(\frac{x}{y}-\frac{y}{x}\right)$ at $t=\frac{\pi}{8}$.

यदि एक वक्र निम्न प्राचलिक समीकरणों से दर्शाया जाता है

$x = \sin\left(t+\frac{7\pi}{12}\right)+\sin\left(t-\frac{\pi}{12}\right)+\sin\left(t+\frac{3\pi}{12}\right)$, $y = \cos\left(t+\frac{7\pi}{12}\right)+\cos\left(t-\frac{\pi}{12}\right)+\cos\left(t+\frac{3\pi}{12}\right)$ तो $\frac{d}{dt}\left(\frac{x}{y}-\frac{y}{x}\right)$ का $t=\frac{\pi}{8}$ पर मान ज्ञात कीजिए।

Ans. 8

Sol. $x = \sin\left(t+\frac{7\pi}{12}\right)+\sin\left(t-\frac{\pi}{12}\right)+\sin\left(t+\frac{3\pi}{12}\right)$

$$x = 2\sin\left(\frac{t+\frac{7\pi}{12}+t-\frac{\pi}{12}}{2}\right)\cos\left(\frac{t+\frac{7\pi}{12}-t+\frac{\pi}{12}}{2}\right)+\sin\left(t+\frac{\pi}{4}\right)$$

$$x = 2\sin\left(t+\frac{\pi}{4}\right)\cos\left(\frac{\pi}{3}\right)+\sin\left(t+\frac{\pi}{4}\right)$$

$$x = 2\sin\left(t+\frac{\pi}{4}\right)$$

$$y = \cos\left(t+\frac{7\pi}{12}\right)+\cos\left(t-\frac{\pi}{12}\right)+\cos\left(t+\frac{3\pi}{12}\right)$$

$$y = 2\cos\left(\frac{t+\frac{7\pi}{12}+t-\frac{\pi}{12}}{2}\right)\cos\left(\frac{t+\frac{7\pi}{12}-t+\frac{\pi}{12}}{2}\right)+\cos\left(t+\frac{\pi}{4}\right)$$

$$y = 2\cos\left(t + \frac{\pi}{4}\right)$$

$$\frac{x}{y} - \frac{y}{x} = \frac{x^2 - y^2}{xy}$$

$$= \frac{4\left[\sin^2\left(t + \frac{\pi}{4}\right) - \cos^2\left(t + \frac{\pi}{4}\right)\right]}{4\sin\left(t + \frac{\pi}{4}\right)\cos\left(t + \frac{\pi}{4}\right)}$$

$$= \frac{-4\cos\left(\frac{\pi}{2} + 2t\right)}{2\sin\left(\frac{\pi}{2} + 2t\right)}$$

$$= +2\frac{\sin 2t}{\cos 2t}$$

$$\left(\frac{x}{y} - \frac{y}{x}\right) = +2\tan 2t$$

$$\frac{d}{dt}\left(\frac{x}{y} - \frac{y}{x}\right) = 2\sec^2(2t) \times 2$$

$$= 4\sec^2(2t)$$

$$= 4\sec^2\left(2 \times \frac{\pi}{8}\right)$$

$$= 8$$

8. If $x = \frac{1 + \ell nt}{t^2}$ and $y = \frac{3 + 2\ell nt}{t}$. Show that $y\frac{dy}{dx} = 2x\left(\frac{dy}{dx}\right)^2 + 1$.

यदि $x = \frac{1 + \ell nt}{t^2}$ एवं $y = \frac{3 + 2\ell nt}{t}$ तो दर्शाइये $y\frac{dy}{dx} = 2x\left(\frac{dy}{dx}\right)^2 + 1$

Ans.

Sol. $x = \frac{1+\ln t}{t^2}$

$$\frac{dx}{dt} = \frac{\frac{1}{t}\times t^2 - (1+\ln t)(2t)}{t^4}$$

$$= \frac{t - 2t - 2t\ln t}{t^4}$$

$$\frac{dx}{dt} = \frac{-(1+2\ln t)}{t^3}$$

$$y = \frac{3+2\ell n t}{t}$$

$$\frac{dy}{dt} = \frac{\frac{2}{t}.t - (3+2\ln t)}{t^2}$$

$$\frac{dy}{dt} = \frac{-(1+2\ln t)}{t^2}$$

$$\frac{dy}{dx} = \frac{-(1+2\ln t)}{t^2}\times\frac{t^3}{-(1+2\ln t)}$$

$$\frac{dy}{dx} = t$$

$$\left(\frac{dy}{dx}\right)^2 = t^2$$

$$t^2 = \frac{1+\ln t}{x} \qquad t = \frac{3+2\ln t}{y}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{1+\ln t}{x}$$

$$1 + 2x\left(\frac{dy}{dx}\right)^2 = 2 + 2\ln t + 1$$

$$1+2x\left(\frac{dy}{dx}\right)^2 = 3+2\ln t \Rightarrow 1+2x\left(\frac{dy}{dx}\right)^2 = ty$$

$$1+2x\left(\frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)y$$

9. Differentiate $\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}$ w. r. t. $\sqrt{1-x^4}$.

$\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}$ का $\sqrt{1-x^4}$ के सापेक्ष अवकलन कीजिये।

Ans. $\frac{1+\sqrt{1-x^4}}{x^6}$

Sol. $y = \frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}} \times \frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}$

$$y = \frac{1+x^2+1-x^2+2\sqrt{1-x^4}}{1+x^2-1+x^2}$$

$$y = \frac{2+2\sqrt{1-x^4}}{2x^2} = \frac{1+\sqrt{1-x^4}}{x^2}$$

$A = \sqrt{1-x^4}$

$A^2 = 1-x^4$

$x^4 = 1-A^2$

$x^2 = \sqrt{1-A^2}$

$$y = \frac{1+A}{\sqrt{1-A^2}}$$

$$\frac{dy}{dA} = \frac{1\sqrt{1-A^2}-(1+A)\frac{-2A}{2\sqrt{1-A^2}}}{(1-A^2)}$$

$$= \frac{1-A^2+A+A^2}{\sqrt{1-A^2}(1-A^2)} = \frac{1+A}{(1-A^2)^{3/2}}$$

$$\frac{dy}{dA} = \frac{1+\sqrt{1-x^4}}{(x^4)^{3/2}} = \frac{1+\sqrt{1-x^4}}{x^6}$$

10. Let g(x) be a polynomial, of degree one and $f(x)$ be defined by $f(x) = \begin{cases} g(x), & x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$. Find the continuous function f(x) satisfying $f'(1) = f(-1)$

माना g(x) ईकाई घात का एक बहुपद है तथा $f(x)$ परिभाषित है $f(x) = \begin{cases} g(x), & x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$ सतत फलन f(x) निकालिए जो $f'(1) = f(-1)$ को संतुष्ट करता हैं।

Ans. $f(x) = \begin{cases} -\frac{2}{3}\left[\frac{1}{6} + \ln\frac{3}{2}\right]x & \text{if } x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x} & \text{if } x > 0 \end{cases}$

Sol. $f(x) = \begin{cases} g(x), & x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$

$g(x) = ax + b$

$$f(x) = \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} \qquad f(0^-) = f(0^+)$$

$$b = \left(\frac{1}{2}\right)^{\frac{1}{0}} \Rightarrow b = 0$$

$$\ln(f(x)) = \frac{1}{x}\log\left(\frac{1+x}{2+x}\right)$$

$$\frac{1}{f(x)}f'(x)=\frac{-1}{x^2}\log\left(\frac{x+1}{x+2}\right)+\frac{1}{x}\left(\frac{x+2}{x+1}\right)\left(\frac{x+2-x-1}{(x+2)^2}\right)$$

$$f'(x)=\left(\frac{x+1}{x+2}\right)^{\frac{1}{x}}\left[\frac{-1}{x^2}\log\left(\frac{x+1}{x+2}\right)+\frac{1}{x(x+1)(x+2)}\right]$$

$$f'(1)=\left(\frac{2}{3}\right)\left[-\log\frac{2}{3}+\frac{1}{1.2.3}\right]$$

$$f'(1)=\frac{2}{3}\left[\frac{1}{6}-\log\frac{2}{3}\right]$$

$$f(-1)=g(-1)=-a+b$$

$$\frac{2}{3}\left[\frac{1}{6}+\log\frac{3}{2}\right]=-a+0$$

$$a=\frac{-2}{3}\left[\frac{1}{6}+\log\frac{3}{2}\right]$$

11. If $\sqrt{1-x^6}+\sqrt{1-y^6}=a^3.(x^3-y^3)$, prove that $\frac{dy}{dx}=\frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$.

यदि $\sqrt{1-x^6}+\sqrt{1-y^6}=a^3.(x^3-y^3)$ तो सिद्ध कीजिए $\frac{dy}{dx}=\frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$.

Ans.

Sol. $\sqrt{1-x^6}+\sqrt{1-y^6}=a^3\left(x^3-y^3\right)$

$$\frac{-6x^5}{2\sqrt{1-x^6}}+\frac{-6y^5}{2\sqrt{1-y^6}}y'=3a^3x^2-3a^3y^2y'$$

$$y'\left(\frac{y^5}{\sqrt{1-y^6}}-a^3y^2\right)=-\left(\frac{x^5}{\sqrt{1-x^6}}+a^3x^2\right)$$

$$y'=\frac{\sqrt{1-y^6}}{\sqrt{1-x^6}}\left(\frac{-x^5-a^3x^2\sqrt{1-x^6}}{y^5-a^3y^2\sqrt{1-y^6}}\right)$$

$$y' = \frac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \frac{-x^2}{y^2}\left(\frac{x^3 + a^3\sqrt{1-x^6}}{y^3 - a^3\sqrt{1-y^6}}\right)$$

$$y' = \frac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \frac{x^2}{y^2}\left(\frac{a^3\sqrt{1-x^6} + x^3}{a^3\sqrt{1-y^6} - y^3}\right)$$

$$\sqrt{1-x^6} + \sqrt{1-y^6} = a^3x^3 - a^3y^3$$

12. If $y = x + \cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x + \ldots\ldots\ldots\ldots}}}$, prove that $\frac{dy}{dx} = \cfrac{1}{2 - \cfrac{x}{x + \cfrac{1}{x + \cfrac{1}{x + \ldots\ldots\ldots\ldots}}}}$.

यदि $y = x + \cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x + \ldots\ldots\ldots\ldots}}}$, तो सिद्ध कीजिए $\frac{dy}{dx} = \cfrac{1}{2 - \cfrac{x}{x + \cfrac{1}{x + \cfrac{1}{x + \ldots\ldots\ldots\ldots}}}}$.

Ans.

Sol. $y = x + \cfrac{1}{x + \cfrac{1}{x + \cfrac{1}{x + \ldots\ldots\ldots\ldots}}}$

$$y = x + \frac{1}{y}$$

$$y^2 = xy + 1$$

$$2yy' = y + xy'$$

$$y'(2y - x) = y$$

$$y' = \frac{y}{2y - x}$$

$$y' = \frac{1}{2 - \frac{x}{y}}$$

$$\frac{dy}{dx} = \frac{1}{2 - \dfrac{x}{x + \dfrac{1}{x + \dfrac{1}{x + ...}}}}$$

13. Let $f(x) = x + \dfrac{1}{2x + \dfrac{1}{2x + \dfrac{1}{2x + ...\infty}}}$. Compute the value of $f(100) \cdot f'(100)$.

माना $f(x) = x + \dfrac{1}{2x + \dfrac{1}{2x + \dfrac{1}{2x + ...\infty}}}$ तो $f(100) \cdot f'(100)$ का मान ज्ञात कीजिए।

Ans. 100

Sol. $y = x + \dfrac{1}{2x + \dfrac{1}{2x + \dfrac{1}{2x + ...\infty}}}$

$$y - x = \frac{1}{2x + \dfrac{1}{2x + \dfrac{1}{2x + ...\infty}}}$$

$$y - x = \frac{1}{2x + y - x}$$

$$y - x = \frac{1}{x + y}$$

$(x + y)(y - x) = 1$

$y^2 - x^2 = 1$

$2yy' - 2x = 0$

$\frac{dy}{dx} = \frac{x}{y}$

$y\frac{dy}{dx} = x$

$f(x)f'(x) = x$

$f(100)f'(100) = 100$

14. Find the derivative with respect to x of the function: $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \arcsin\frac{2x}{1+x^2}$ at $x = \frac{\pi}{4}$.

निम्नलिखित फलन का x के सापेक्ष अवकलज निकाले : $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \arcsin\frac{2x}{1+x^2}$, $x = \frac{\pi}{4}$ पर

Ans. $\frac{32}{16+\pi^2} - \frac{8}{\ln 2}$

Sol. $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \arcsin\frac{2x}{1+x^2} = y$

$\frac{\log \sin x}{\log \cos x} \times \frac{\log \sin x}{\log \cos x} + \sin^{-1}\left(\frac{2x}{1+x^2}\right) = y$

$y = \left(\frac{\ln \sin x}{\ln \cos x}\right)^2 + \tan^{-1} x$

$\frac{dy}{dx} = 2\left(\frac{\ln \sin x}{\ln \cos x}\right)\left(\frac{\cot x \ln \cos x + \tan x \ln \sin x}{(\ln \cos x)^2}\right) + 2 \times \frac{1}{1+x^2}$

$x = \frac{\pi}{4}$

$\frac{dy}{dx} = \frac{32}{16+\pi^2} - \frac{8}{\ln 2}$

15. Suppose $f(x) = \tan(\sin^{-1}(2x))$

(a) Find the domain and range of f.

(b) Express $f(x)$ as an algebaric function of x.

(c) Find $f'(1/4)$.

माना कि $f(x) = \tan(\sin^{-1}(2x))$

(a) फलन f का प्रांत तथा परिसर बताइए।

(b) $f(x)$ को x के बीजगणितीय फलन के रूप में प्रदर्शित करे।

(c) $f'(1/4)$ ज्ञात कीजिए।

Ans. (a) $\left(-\frac{1}{2}, \frac{1}{2}\right), (-\infty, \infty)$ (b) $f(x) = \frac{2x}{\sqrt{1-4x^2}}$ (c) $\frac{16\sqrt{3}}{9}$

Sol. $f(x) = \tan(\sin^{-1}(2x))$

(a) $-1 \le 2x \le 1$

$$\frac{-1}{2} \le x \le \frac{1}{2}$$

$$x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$x = \frac{-1}{2} \Rightarrow \tan\left(\frac{-\pi}{2}\right) \quad \text{Rejected}$$

$$x = \frac{1}{2} \Rightarrow \tan\left(\frac{\pi}{2}\right) \quad \text{Rejected}$$

Domain : $x \in \left(\frac{-1}{2}, \frac{1}{2}\right)$

$$f(x) = \tan\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

$$= (-\infty, \infty)$$

(b) $f(x) = \tan(\sin^{-1}(2x))$

$$= \tan\left(\tan^{-1}\frac{2x}{\sqrt{1-4x^2}}\right)$$

$$f(x)=\frac{2x}{\sqrt{1-4x^2}}$$

(c) $$f'(x)=\frac{2\sqrt{1-4x^2}-2x\left(\frac{-8x}{2\sqrt{1-4x^2}}\right)}{1-4x^2}$$

$$\Rightarrow \frac{2(1-4x^2)+8x^2}{(1-4x^2)^{\frac{3}{2}}}$$

$$f'(x)=\frac{2}{(1-4x^2)^{\frac{3}{2}}}$$

$$x=\frac{1}{4}$$

$$f'\left(\frac{1}{4}\right)=\frac{16\sqrt{3}}{9}$$

16. If $y=\tan^{-1}\frac{1}{x^2+x+1}+\tan^{-1}\frac{1}{x^2+3x+3}+\tan^{-1}\frac{1}{x^2+5x+7}+\tan^{-1}\frac{1}{x^2+7x+13}+\ldots\ldots$ to n terms. Find $\frac{dy}{dx}$, expressing your answer in 2 terms.

यदि $y=\tan^{-1}\frac{1}{x^2+x+1}+\tan^{-1}\frac{1}{x^2+3x+3}+\tan^{-1}\frac{1}{x^2+5x+7}+\tan^{-1}\frac{1}{x^2+7x+13}+\ldots\ldots$ पदों तक $\frac{dy}{dx}$ निकालिए, अपना उत्तर दो पदों में दीजिये।

Ans. $\frac{1}{1+(x+n)^2}-\frac{1}{1+x^2}$

Sol. $y=\tan^{-1}\frac{1}{x^2+x+1}+\tan^{-1}\frac{1}{x^2+3x+3}+\tan^{-1}\frac{1}{x^2+5x+7}+\tan^{-1}\frac{1}{x^2+7x+13}+\ldots\ldots$to n terms

$$T_r=\tan^{-1}\left(\frac{1}{x^2+(2r-1)x+(r^2-r+1)}\right)$$

$$T_r=\tan^{-1}\left(\frac{1}{1+x^2+2rx+r^2-x-r}\right)$$

$$= \tan^{-1}\left(\frac{1}{1+(x+r)^2-(x+r)}\right)$$

$$= \tan^{-1}\left(\frac{1}{1+(x+r)(x+r-1)}\right)$$

$T_r = \tan^{-1}(x+r) - \tan^{-1}(x+r-1)$

$T_1 = \tan^{-1}(x+1) - \tan^{-1}(x)$

$T_2 = \tan^{-1}(x+2) - \tan^{-1}(x+1)$

$T_3 = \tan^{-1}(x+3) - \tan^{-1}(x+2)$

. .
. .
. .

$T_n = \tan^{-1}(x+n) - \tan^{-1}(x+n-1)$

$S_n = y = \tan^{-1}(x+n) - \tan^{-1} x$

$$\frac{dy}{dx} = \frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$$

17. If $y = \tan^{-1}\frac{u}{\sqrt{1-u^2}}$ and $x = \sec^{-1}\frac{1}{2u^2-1}$, $u \in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ prove that $2\frac{dy}{dx} + 1 = 0$.

यदि $y = \tan^{-1}\frac{u}{\sqrt{1-u^2}}$ तथा $x = \sec^{-1}\frac{1}{2u^2-1}$, $u \in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ सिद्ध कीजिए $2\frac{dy}{dx} + 1 = 0$

Ans.

Sol. $y = \tan^{-1}\frac{u}{\sqrt{1-u^2}}$ $\qquad$ $x = \sec^{-1}\frac{1}{2u^2-1}$

$u = \sin\theta$ $\qquad$ $u = \cos\theta$

$y = \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)$ $\qquad$ $x = \sec^{-1}\left(\frac{1}{2\cos^2\theta - 1}\right)$

$x = \sec^{-1}\left(\frac{1}{\cos 2\theta}\right)$

$x = \sec^{-1}(\sec 2\theta)$

$y = \theta$ $\qquad$ $x = 2\theta$

$y = \sin^{-1} u$ $\qquad$ $x = 2\cos^{-1} u$

$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$ $\qquad$ $\frac{dx}{du} = \frac{-2}{\sqrt{1-u^2}}$

$\frac{dy}{dx} = \frac{-1}{2}$

$\frac{2dy}{dx} + 1 = 0$

18. If $y = \cot^{-1}\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$, find $\frac{dy}{dx}$ if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$.

यदि $y = \cot^{-1}\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$, $\frac{dy}{dx}$ निकालिये जब $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$ हो।

Ans. $\frac{1}{2}$ or $-\frac{1}{2}$

Sol. $y = \cot^{-1}\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$

$y = \cot^{-1}\left(\frac{1+\sin x+1-\sin x+2\sqrt{1-\sin^2 x}}{1+\sin x-1+\sin x}\right)$

$y = \cot^{-1}\left(\frac{1+|\cos x|}{\sin x}\right)$

$x \in \left(0, \frac{\pi}{2}\right)$

$y = \cot^{-1}\left(\frac{1+\cos x}{\sin x}\right)$

$y = \cot^{-1}\left(\cot\frac{x}{2}\right)$

$$y = \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}$$

$$x \in \left(\frac{\pi}{2}, \pi\right)$$

$$y = \cot^{-1}\left(\frac{1-\cos x}{\sin x}\right)$$

$$y = \cot^{-1}\left(\tan\frac{x}{2}\right)$$

$$y = \frac{\pi}{2} - \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{-1}{2}$$

19. If $y = \tan^{-1}\frac{x}{1+\sqrt{1-x^2}} + \sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right)$, then find $\frac{dy}{dx}$ for $x \in (-1, 1)$.

यदि $y = \tan^{-1}\frac{x}{1+\sqrt{1-x^2}} + \sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right)$, तो $x \in (-1, 1)$ के लिये $\frac{dy}{dx}$ बताइये।

Ans. $\frac{1-2x}{2\sqrt{1-x^2}}$

Sol. $y = \tan^{-1}\frac{x}{1+\sqrt{1-x^2}} + \sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right)$

$x = \sin\theta \qquad x = \cos\phi$

$$y = \tan^{-1}\left(\frac{\sin\theta}{1+\cos\theta}\right) + \sin\left(2\tan^{-1}\sqrt{\frac{1-\cos\phi}{1+\sin\phi}}\right)$$

$$y = \tan^{-1}\left(\tan\frac{\theta}{2}\right) + \sin\left(2\tan^{-1}\left(\tan\frac{\phi}{2}\right)\right)$$

$$y = \frac{\theta}{2} + \sin\left(2\times\frac{\phi}{2}\right)$$

$$y = \frac{\theta}{2} + \sin(\phi)$$

$$y = \frac{\sin^{-1}x}{2} + \frac{\sqrt{1-x^2}}{1}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = \frac{1-2x}{2\sqrt{1-x^2}}$$

20. Let P(x) be a polynomial of degree 4 such that P(1) = P(3) = P(5) = P'(7) = 0. If the real number $x \neq 1, 3, 5$ is such that P(x) = 0 can be expressed as x = p/q where 'p' and 'q' are relatively prime, then (p + q) equals.

माना P(x) एक चार घात का बहुपद है जिसके लिये P(1) = P(3) = P(5) = P'(7) = 0 हो। यदि वास्तविक संख्या $x \neq 1, 3, 5$ इस प्रकार हो कि P(x) = 0 को x = p/q के रूप में प्रदर्शित किया जा सके, जहाँ 'p' तथा 'q' सापेक्षतः अभाज्य संख्याऐं (relatively prime) है, तो (p + q) किसके बराबर होगा?

Ans. 100

Sol. P(1) = P(3) = P(5) = P'(7) = 0

P(x) = (x – 1) (x – 3) (x – 5) (ax + b)

log (p (x)) = log (x – 1) + log (x – 3) + log (x – 5) + log (ax + b)

$$\frac{1}{P(x)}P'(x) = \frac{1}{x-1} + \frac{1}{x-3} + \frac{1}{x-5} + \frac{a}{ax+b}$$

x = 7

$$0 = \frac{1}{6} + \frac{1}{4} + \frac{1}{2} + \frac{a}{7a+b}$$

$$\frac{a}{7a+b} = \frac{-11}{12}$$

89 a = – 11b

$$\frac{-b}{a}=\frac{89}{11}$$

$$x=\frac{89}{11}=\frac{P}{q}$$

$$P+q=100$$

21. $$f(x)=\begin{cases} b\sin^{-1}\left(\dfrac{x+c}{2}\right), & -\dfrac{1}{2}<x<0 \\ \dfrac{1}{2} & \text{at } x=0 \\ \dfrac{e^{ax/2}-1}{x}, & 0<x<\dfrac{1}{2} \end{cases}$$

If f(x) is differentiable at x = 0 and |c| < 1/2 then find the value of 'a' and prove that $64b^2 = 4 - c^2$.

$$f(x)=\begin{cases} b\sin^{-1}\left(\dfrac{x+c}{2}\right), & -\dfrac{1}{2}<x<0 \\ \dfrac{1}{2} & \text{at } x=0 \\ \dfrac{e^{ax/2}-1}{x}, & 0<x<\dfrac{1}{2} \end{cases}$$

यदि f (x), x = 0 पर अवकलनीय है तथा | c | < 1/2 हो तो 'a' का मान बताइए एवं सिद्ध कीजिये कि $64b^2 = 4 - c^2$

Ans. a = 1

Sol. $$f(x)=\begin{cases} b\sin^{-1}\left(\dfrac{x+c}{2}\right), & -\dfrac{1}{2}<x<0 \\ \dfrac{1}{2} & \text{at } x=0 \\ \dfrac{e^{ax/2}-1}{x}, & 0<x<\dfrac{1}{2} \end{cases}$$

$$f(0^-)=f(0)=f(0^+)$$

$$b\sin^{-1}\left(\frac{c}{2}\right)=\frac{1}{2}=\lim_{x\to 0^+}\frac{e^{\frac{ax}{2}}-1}{2}$$

$$= \lim_{x\to 0^+} \frac{\frac{a}{2}e^{\frac{ax}{2}}}{1} = \frac{a}{2}$$

$a = 1$

$$f'(x) = \begin{cases} \dfrac{b\left(\frac{1}{2}\right)}{\sqrt{1-\left(\frac{x+c}{2}\right)^2}} & ; \quad -\dfrac{1}{2} < x < 0 \\ 0 & ; \quad x = 0 \\ \dfrac{\frac{e^{\frac{x}{2}} \times x}{2} - \left(e^{\frac{x}{2}} - 1\right)}{x^2} & ; \quad 0 < x < \dfrac{1}{2} \end{cases}$$

LHD = RHD at $x = 0$

$$LHD = \frac{b}{2\sqrt{1-\frac{c^2}{4}}} = \frac{b}{\sqrt{4-c^2}}$$

$$RHD = \lim_{x\to 0} \frac{\frac{xe^{\frac{x}{2}}}{2} - e^{\frac{x}{2}} + 1}{x^2} = \lim_{x\to 0} \frac{e^{\frac{x}{2}}(x-2)+2}{2x^2}$$

$$= \lim_{x\to 0} \frac{e^{\frac{x}{2}} + \frac{e^{\frac{x}{2}}(x-2)}{2}}{4x}$$

$$= \lim_{x\to 0} \frac{\frac{e^{\frac{x}{2}}}{2} + \frac{e^{\frac{x}{2}}(x-2)}{4} + \frac{e^{\frac{x}{2}}}{2}}{4}$$

$$= \frac{\frac{1}{2} + \frac{-2}{4} + \frac{1}{2}}{4}$$

$= \frac{1}{8}$

RHD = LHD

$\frac{1}{8} = \frac{b}{\sqrt{4-c^2}}$

$\frac{1}{64} = \frac{b^2}{4-c^2}$

$64b^2 = 4 - c^2$

22. If $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ and $g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y)$ for all $x, y \in R$. If right hand derivative at $x = 0$ exists for $f(x)$. Find derivative of $g(x)$ at $x = 0$.

यदि $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ तथा $g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y)$ सभी $x, y \in R$ है। यदि $x = 0$ पर $f(x)$ का दायां अवकलज अस्तित्व में है। $x = 0$ पर $g(x)$ का अवकलज बताइये।

Ans. $g'(0) = 0$

Sol. $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ .....(1)

$g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y)$.....(2)

in (1) $x = y$

$f(0) = f(x)\, g(x) - f(x)\, g(x)$

$f(0) = 0$

$y = 0$ in (1)

$f(x) = f(x)\, g(0) - f(0)\, g(x)$

$f(x) = f(x)\, g(0) - (0)\, g(x)$

$g(0) = 1$

$f'(0^+) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$

$= \lim_{h \to 0} \frac{f(0).g(-h) - f(-h).g(0) - 0}{h}$

$= \lim_{h \to 0} \frac{f(-h)}{-h} = f(0^-)$

$x = y$ in (2)

$g(0) = g^2(x) + f^2(x)$

$0 = 2g(x)\, g'(x) + 2f(x)\, f'(x)$

$g'(0)\, g(0) + f(0)\, f'(0) = 0$

$g'(0) \times 1 + 0 = 0$

23. Let $f(\theta) = \sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan\theta)}(f(\theta))$ **[JEE-2011]**

माना $f(\theta) = \sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, जहाँ $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. तो $\frac{d}{d(\tan\theta)}(f(\theta))$ का मान होगा। **[JEE-2011]**

Ans. 1

Sol. Let $f(\theta) = \sin\alpha$ where $\alpha = \tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)$

$\Rightarrow \quad \tan\alpha = \frac{\sin\theta}{\sqrt{\cos 2\theta}}$

$\Rightarrow \quad \sin\alpha = \frac{\sin\theta}{\cos\theta} = \tan\theta \quad \left(\because \theta \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)\right)$

$\Rightarrow \quad f(\theta) = \tan\theta$

$\Rightarrow \quad \frac{d(f(\theta))}{d(\tan\theta)} = 1$

DERIVATIVE OF INVERSE OF A FUNCTION

24. (a) Let $f(x) = x^2 - 4x - 3$, $x > 2$ and let g be the inverse of f. Find the value of g' where $f(x) = 2$.

(b) Let $f: R \to R$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and $g(x) = f^{-1}(x)$, then find $g'(-1)$.

(c) Suppose f^{-1} is the inverse function of a differentiable function f and let $G(x) = \frac{1}{f^{-1}(x)}$.

If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.

(a) माना कि $f(x) = x^2 - 4x - 3$, $x > 2$ तथा g, f का प्रतिलोम फलन है। g' का मान जबकि $f(x) = 2$ है क्या होगा?

(b) माना $f: R \to R$ में परिभाषित है $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ तथा $g(x) = f^{-1}(x)$, हो तो $g'(-1)$ निकालिये।

(c) माना f^{-1} एक अवकलनीय फलन f का प्रतिलोम फलन है तथा $G(x) = \frac{1}{f^{-1}(x)}$ है।

यदि $f(3) = 2$ एवं $f'(3) = \frac{1}{9}$ तो $G'(2)$ का मान ज्ञात कीजिये।

Ans. (a) $\frac{1}{6}$ (b) $\frac{1}{14}$ (c) -1

Sol.

(a) $f(x) = x^2 - 4x - 3$ and $f(x) = 2$

For $x = 5, -1$

$\Rightarrow \quad x = 5 \qquad (x > 2)$

Now $y = f(x) \Rightarrow f^{-1}(y) = x$

$\Rightarrow \quad g(y) = x$

$$\Rightarrow \quad g'(y) = \frac{dx}{dy} = \frac{1}{2x-4} = \frac{1}{6}$$

(b) $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$

$x = 0$

$f(0) = -1$

$g(f(x)) = x$

$g'(f(x)).f'(x) = 1$

$$g'(f(x)) = \frac{1}{f'(x)}$$

$x = 0$

$$g'(f(0)) = \frac{1}{f'(0)}$$

$f'(x) = 3x^2 + 6x + 6 + 8e^{2x}$

$f'(0) = 14$

$$g'(-1) = \frac{1}{14}$$

(c) $G(x) = \frac{1}{f^{-1}(x)}$ $f(3) = 2$ $f'(3) = \frac{1}{9}$

$h(x) = f^{-1}(x)$

$$G(x) = \frac{1}{h(x)}$$

$$G'(x) = \frac{-h'(x)}{h^2(x)}$$

$h(f(x)) = x$

$x = 3$

$h(2) = 3$

$h'(f(x))f'(x) = 1$

$$h'(f(x)) = \frac{1}{f'(x)}$$

$x = 3$

$$h'(f(3)) = \frac{1}{f'(3)} \Rightarrow h'(2) = 9$$

$$G'(2) = \frac{-h'(2)}{[h(2)]^2} = \frac{-9}{(3)^2} = -1$$

25. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is. **[JEE-2009]**

यदि फलन $f(x) = x^3 + e^{\frac{x}{2}}$ तथा $g(x) = f^{-1}(x)$ है, तो $g'(1)$ का मान ज्ञात कीजिए। **[JEE-2009]**

Ans. 2

Sol. $f(x) = x^3 + e^{x/2}$, $g(x) = f^{-1}(x)$

$\Rightarrow g'(f(x)).f'(x) = 1$

Put $f(x) = 1 \Rightarrow x^3 + e^{x/2} = 1$

$\Rightarrow x = 0$

$$\Rightarrow g'(1).f'(0) = 1,\ f'(x) = 3x^2 + e^{x/2}.\frac{1}{2}$$

$\Rightarrow g'(1) = 2$

HIGHER ORDER DERIVATIVE

26. (a) If $y = y(x)$ and it follows the relation $e^{xy} + y\cos x = 2$, then find (i) $y'(0)$ and (ii) $y''(0)$.

(b) A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions $f(0) = 2$; $f'(0) = -5$ and $f''(0) = 3$.

The function $g(x)$ is defined by $g(x) = e^{ax} + f(x)\ \forall\ x \in R$, where 'a' is any constant. If $g'(0) + g''(0) = 0$. Find the value(s) of 'a'.

(a) यदि $y = y(x)$ है एवं यह सम्बन्ध $e^{xy} + y\cos x = 2$ के अनुगामी है तो (i) $y'(0)$ तथा (ii) $y''(0)$ निकालिये।

(b) यदि $f(x)$ एक दो बार अवकलनीय फलन है तथा वो निम्न शर्तो को संतुष्ट कर रहा है

$f(0) = 2$; $f'(0) = -5$ तथा $f''(0) = 3$

फलन $g(x)$ परिभाषित है $g(x) = e^{ax} + f(x)\ \forall\ x \in R$, जहाँ 'a' कोई नियतांक है। यदि $g'(0) + g''(0) = 0$ हो तो 'a' के मान बताइयें।

Ans. (a) (i) $y'(0) = -1$; (ii) $y''(0) = 2$ (b) $a = 1, -2$

Sol. (a) $y = y(x)$ $e^{xy} + y\cos x = 2$

$x = 0$

$e^{0(y)} + y\cos(0) = 2$

$1 + y = 2$

$y(0) = 1$

$e^{xy} + y\cos x = 2$

$e^{xy}(y + xy') + y'\cos x - y\sin x = 0$

$x = 0$

$e^0(y(0) + 0 \times y') + y'(0)\cos 0 - y\sin 0 = 0$

$y'(0) = -1$

$e^{xy}(y + xy')^2 + e^{xy}(y' + y' + xy'') + y''\cos x - y'\sin x - y'\sin x - y\cos x = 0$

$x = 0$

$y''(0) = 2$

$f(0) = 2$ $f'(0) = -5$ $f''(0) = 3$

$g(x) = e^{ax} + f(x)$

$g'(x) = ae^{ax} + f'(x)$

$g'(0) = a + f'(0)$

$= a - 5$

$g''(x) = a^2e^{ax} + f''(x)$

$g''(0) = a^2e^0 + f''(0)$

$g''(0) = a^2 + 3$

$g'(0) + g''(0) = 0$

$a - 5 + a^2 + 3 = 0$

$a^2 + a - 2 = 0$

$a = 1, -2$

27. If $x = 2\cos t - \cos 2t$ and $y = 2\sin t - \sin 2t$, find the value of (d^2y/dx^2) when $t = (\pi/2)$.

यदि $x = 2\cos t - \cos 2t$ तथा $y = 2\sin t - \sin 2t$ तो (d^2y/dx^2) का $t = (\pi/2)$ पर मान बताइए।

Ans. $\frac{-3}{2}$

Sol. $x = 2\cos t - \cos 2t$

$$\frac{dx}{dt} = -2\sin t + 2\sin 2t$$

$y = 2\sin t - \sin 2t$

$$\frac{dy}{dt} = 2\cos t - 2\cos 2t$$

$$\frac{dy}{dx} = \frac{2\cos t - 2\cos 2t}{-2\sin t + 2\sin 2t} = \frac{\cos t - \cos 2t}{\sin 2t - \sin t}$$

$$\frac{dy}{dx} = \tan\left(\frac{3t}{2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{3}{2}\sec^2\left(\frac{3t}{2}\right) \times \frac{dt}{dx}$$

$$= \frac{3}{2}\sec^2\left(\frac{3t}{2}\right) \times \frac{1}{2(\sin 2t - \sin t)}$$

$$t = \frac{\pi}{2}$$

$$\frac{d^2y}{dx^2} = \frac{-3}{2}$$

28. Find the value of the expression $y^3 \frac{d^2y}{dx^2}$ on the ellipse $3x^2 + 4y^2 = 12$.

दीर्घवृत्त $3x^2 + 4y^2 = 12$ पर व्यंजक $y^3 \frac{d^2y}{dx^2}$ का मान बताइए।

Ans. $\frac{-9}{4}$

Sol. $3x^2 + 4y^2 = 12$

$6x + 8yy' = 0$

$8yy' = -6x$

$4yy' = -3x$

$4(y')^2 + 4yy^2 = -3$

$$\frac{d^2y}{dx^2} = \frac{-3 - 4\left(\frac{dy}{dx}\right)^2}{4y}$$

$$y^3\frac{d^2y}{dx^2}=\frac{-\left[3+4\left(\frac{dy}{dx}\right)^2\right]y^3}{4y}$$

$$=\frac{-\left[3+4\left(\frac{-3x}{4y}\right)^2\right]y^2}{4}$$

$$=-\left[\frac{12y^2+9x^2}{16y^2}\right]\times y^2$$

$$=-\left[\frac{3(12-3x^2)+9x^2}{16}\right]$$

$$=-\frac{9}{4}$$

29. If $f:R\to R$ is a function such that $f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3)$ for all $x\in R$, then prove that $f(2)=f(1)-f(0)$.

यदि $f:R\to R$ एक फलन इस प्रकार है कि $f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3)$ सभी $x\in R$ के लिये हो तो सिद्ध कीजिये $f(2)=f(1)-f(0)$

Ans.

Sol. $f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3)$

$f'(1)=a \qquad f''(2)=b$

$f'''(3)=c$

$f(x)=x^3+ax^2+bx+c$

$f'(x)=3x^2+2ax+b$

$x=1$

$a=3+2a+b$

$a+b=-3$ ..... (1)

$f''(x)=6x+2a$

$x=2$

$f''(2)=12+2a$

$b = 12 + 2a$ ..... (2)

from (1) and (2)

$a = -5$

$b = 2$

$f'''(x) = 6$

$x = 3$

$c = 6$

$f(x) = x^3 - 5x^2 + 2x + 6$

$f(2) = 8 - 20 + 4 + 6 = -2$

$f(1) = 4$

$f(0) = 6$

$f(2) = f(1) - f(0)$

$-2 = 4 - 6$

$-2 = -2$

DIFFERENTIATION OF DETERMINANT

30. If $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ then $f'(x) = \lambda \cdot \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$. Find the value of λ.

यदि $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ तो $f'(x) = \lambda \cdot \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$ है। λ का मान ज्ञात कीजिए।

Ans. 3

Sol. $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$

$$f'(x)=\begin{vmatrix}4(x-a)^3 & (x-a)^3 & 1\\ 4(x-b)^3 & (x-b)^3 & 1\\ 4(x-c)^3 & (x-c)^3 & 1\end{vmatrix}+\begin{vmatrix}(x-a)^4 & 3(x-a)^2 & 1\\ (x-b)^4 & 3(x-b)^2 & 1\\ (x-c)^4 & 3(x-c)^2 & 1\end{vmatrix}+\begin{vmatrix}(x-a)^4 & (x-a)^3 & 0\\ (x-b)^4 & (x-b)^3 & 0\\ (x-c)^4 & (x-c)^3 & 0\end{vmatrix}$$

$$f'(x)=0+3\begin{vmatrix}(x-a)^4 & (x-a)^2 & 1\\ (x-b)^4 & (x-b)^2 & 1\\ (x-c)^4 & (x-c)^2 & 1\end{vmatrix}+0$$

$\lambda = 3$

31. If $f(x)=\begin{vmatrix}\cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2)\\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2)\\ \sin 2x & 0 & \sin 2x^2\end{vmatrix}$ then find f'(x).

यदि $f(x)=\begin{vmatrix}\cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2)\\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2)\\ \sin 2x & 0 & \sin 2x^2\end{vmatrix}$ तो f'(x) ज्ञात कीजिए।

Ans. $2(1 + 2x) \cdot \cos 2(x + x^2)$

Sol. $f(x)=\begin{vmatrix}\cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2)\\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2)\\ \sin 2x & 0 & \sin 2x^2\end{vmatrix}$

$$f(x)=\sin 2x\left[\cos(x^2+x-x+x^2)\right]+\sin 2x^2\left[\cos(x+x^2+x-x^2)\right]$$

$= \sin 2x \cos 2x^2 + \sin 2x^2 \cos 2x$

$f(x) = \sin (2x^2 + 2x)$

$f'(x) = (4x + 2) \cos (2x^2 + 2x)$

$= 2(2x + 1) \cos 2(x + x^2)$

32. If α be a repeated root of a quadratic equation $f(x) = 0$ and $A(x)$, $B(x)$, $C(x)$ be the polynomials of degree 3, 4 amd 5 respectively, then show that $\begin{vmatrix}A(x) & B(x) & C(x)\\ A(\alpha) & B(\alpha) & C(\alpha)\\ A'(\alpha) & B'(\alpha) & C'(\alpha)\end{vmatrix}$ is divisible by f(x), where dash denotes the derivative.

यदि α द्विघात समीकरण $f(x) = 0$ का एक दोहराया गया मूल है तथा $A(x)$, $B(x)$, $C(x)$ क्रमशः 3, 4 तथा 5 घात के बहुपद हो तो

सिद्ध कीजिये $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ फलन $f(x)$ द्वारा भाज्य है जहाँ y' का तात्पर्य अवकलज है।

Ans.

Sol. $f(x) = c(x-\alpha)^2$

$$F(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

Now $F(\alpha) = 0$ $\quad\therefore\quad (x-\alpha)$ is factor of $F(x)$.

$$F'(x) = \begin{vmatrix} A'(x) & B'(x) & C'(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

Now $F'(\alpha) = 0 \Rightarrow x-\alpha$ is a factor of $F'(x)$. So $(x-\alpha)$ must be repeated at least two times in $F(x)$.

$\Rightarrow$ $F(x)$ is divisible by $f(x)$.

L'HOSPITAL RULE

Evalute the following limits using L'Hospital's Rule or otherwise :

निम्नलिखित सीमाओं को L' Hospital के नियम से या अन्य विधि से हल कीजिये :

33. $\lim\limits_{x\to 0}\left[\dfrac{1}{x\sin^{-1}x} - \dfrac{1-x^2}{x^2}\right]$

Ans. $\dfrac{5}{6}$

Sol. $\lim\limits_{x\to 0}\left[\dfrac{1}{x\sin^{-1}x} - \dfrac{1-x^2}{x^2}\right]$

$x = \sin\theta$

$$= \lim_{\theta\to 0}\left(\frac{1}{\theta\sin\theta} - \frac{1-\sin^2\theta}{\sin^2\theta}\right)$$

$$= \lim_{\theta\to 0}\left(\frac{\sin\theta - \theta + \theta\sin^2\theta}{\theta\sin^2\theta}\right)$$

$$= \lim_{\theta \to 0} \left(\frac{\theta - \frac{\theta^3}{3!} + ... - \theta + \theta^3 \left(\frac{\sin\theta}{\theta} \right)^2}{\theta^3 \left(\frac{\sin\theta}{\theta} \right)^2} \right)$$

$$= \lim_{\theta \to 0} \left(\frac{-\frac{1}{3!} + 1}{1} \right)$$

$$= \frac{5}{6}$$

34. $\lim\limits_{x \to 0} \dfrac{x + \ln\left(\sqrt{x^2+1} - x\right)}{x^3}$

Ans. $\dfrac{1}{6}$

Sol. $\lim\limits_{x \to 0} \dfrac{x + \ln\left(\sqrt{x^2+1} - x\right)}{x^3}$

$$\lim_{x \to 0} \frac{1 + \frac{1}{\sqrt{x^2+1} - x} \left(\frac{2x}{2\sqrt{x^2+1}} - 1 \right)}{3x^2}$$

$$\lim_{x \to 0} \frac{1 + \frac{1}{\sqrt{x^2+1} - x} \left(\frac{x - \sqrt{x^2+1}}{\sqrt{x^2+1}} \right)}{3x^2}$$

$$= \lim_{x \to 0} \frac{1 - \frac{1}{\sqrt{x^2+1}}}{3x^2}$$

$$= \lim_{x \to 0} \frac{\frac{1}{2}\left(x^2+1\right)^{-\frac{3}{2}}(2x)}{6x}$$

$$=\frac{1}{6}(1)^{-\frac{3}{2}}=\frac{1}{6}$$

35. $\lim\limits_{x\to 0}\left[\frac{1}{x^2}-\frac{1}{\sin^2 x}\right]$

Ans. $-\frac{1}{3}$

Sol. $\lim\limits_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{\sin^2 x}\right)$

$$\lim_{x\to 0}\frac{\sin^2 x-x^2}{x^2\sin^2 x}$$

$$\lim_{x\to 0}\frac{(\sin x-x)(\sin x+x)}{x^2\sin^2 x}$$

$$\lim_{x\to 0}\frac{\left(x-\frac{x^3}{\underline{|3}}+.....-x\right)(x+\sin x)}{x^2\sin^2 x}$$

$$\lim_{x\to 0}\frac{x^3\left(-\frac{1}{6}+....\right)\times 2\left(1+\frac{\sin x}{x}\right)}{x^4\left(\frac{\sin x}{x}\right)^2}$$

$$\frac{-\frac{1}{6}\times(1+1)}{1}=-\frac{1}{3}$$

36. (a) $\lim\limits_{x\to 0^+} x^{(x^x-1)}$

(b) $\lim\limits_{x\to 0^+}\left(\cot x-\ell n^2 x\right)$

Ans. (a) 1 (b) does not exist

Sol. (a) $\lim\limits_{x\to 0^+}(x)^{(x^x-1)}$

$= (e)^{\lim\limits_{x\to 0^+}(x^x-1)\ln x} = (e)^2$

$L = \lim\limits_{x\to 0^+}(x^x-1)\ln x$

$= \lim\limits_{x\to 0^+}\dfrac{x^x-1}{\frac{1}{\ln x}}$

Apply L′ Hospital Rule

$= \lim\limits_{x\to 0^+}\dfrac{x^x(1+\ln x)}{-\frac{1}{(\ln x)^2}\left(\frac{1}{x}\right)}$

$\lim\limits_{x\to 0^+} -x(\ln x)^2(1+\ln x) \qquad \left(\because \lim\limits_{x\to 0} x^x = 1\right)$

put $\ln x = t$

$\lim\limits_{x\to -\infty} -e^t t^2(1+t)$

put $t = -u$

$\lim\limits_{u\to\infty} -e^{-u}u^2(1-u)$

$L = \lim\limits_{u\to\infty}\dfrac{-u^2(1-u)}{e^u} = 0$

$\Rightarrow e^u = e^0 = 1$

(b) $\lim\limits_{x\to 0^+}(\cot x - \ln^2 x)$

$\lim\limits_{x\to 0^+}\dfrac{\cos x - \sin x \ln^2 x}{\sin x}$

DNE

37. $\lim\limits_{x\to 0}\dfrac{1+\sin x-\cos x+\ell n(1-x)}{x\cdot\tan^2 x}$

Ans. $-\frac{1}{2}$

Sol. $\lim\limits_{x\to 0}\dfrac{1+\sin x-\cos x+\ln(1-x)}{x\cdot\tan^2 x}$

$$\lim_{x\to 0}\frac{1+x-\frac{x^3}{\underline{|3}}+.....-1+\frac{x^2}{\underline{|2}}-\frac{x^4}{\underline{|4}}.....-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}....}{x^3\left(\frac{\tan x}{x}\right)^2}$$

$$\lim_{x\to 0}\frac{x^3\left(-\frac{1}{6}+.....-\frac{x}{24}-\frac{1}{3}.....\right)}{(1)}$$

$$=-\frac{1}{6}-\frac{1}{3}=-\frac{1}{2}$$

38. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\sin x-(\sin x)^{\sin x}}{1-\sin x+\ln(\sin x)}$

Ans. 2

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\sin x-(\sin x)^{\sin x}}{1-\sin x+\ln(\sin x)}$

$$\lim_{x\to\frac{\pi}{2}}\frac{\cos x-(\sin x)^{\sin x}\left[\cos x\log\sin x+\cos x\right]}{-\cos x+\frac{\cos x}{\sin x}}$$

$$\lim_{x\to\frac{\pi}{2}}\frac{1-(\sin x)^{\sin x}\left[\log\sin x+1\right]}{-1+\frac{1}{\sin x}}$$

$$\lim_{x\to\frac{\pi}{2}}\frac{1-(\sin x)^{\sin x}\left[\log(\sin x)+1\right]}{\operatorname{cosec} x-1}$$

$$\lim_{x\to\frac{\pi}{2}} \frac{-(\sin x)^{\sin x}(\cos x \log \sin x + \cos x) - (\sin x)^{\sin x}\frac{\cos x}{\sin x}}{-\text{cosec } x \cot x}$$

$$\lim_{x\to\frac{\pi}{2}} \frac{-(\sin x)^{\sin x}\left[\log \sin x + 1 + \frac{1}{\sin x}\right]}{-\frac{1}{\sin^2 x}}$$

$$+(1)^1 \frac{(0+1+1)}{+\frac{1}{1}} = 2$$

39. If $\lim\limits_{x\to 0} \frac{a\sin x - bx + cx^2 + x^3}{2x^2.\ell n(1+x) - 2x^3 + x^4}$ exists and is finite, find the values of a, b, c and the limit.

यदि सीमा $\lim\limits_{x\to 0} \frac{a\sin x - bx + cx^2 + x^3}{2x^2.\ell n(1+x) - 2x^3 + x^4}$ अस्तित्व में है एवं नियत हो तो a, b, c तथा सीमा के मान बताइये।

Ans. $a = 6, b = 6, c = 0; \frac{3}{40}$

Sol. $\lim\limits_{x\to 0} \frac{a\sin x - bx + cx^2 + x^3}{2x^2 \ln(1+x) - 2x^3 + x^4}$

$$\lim_{x\to 0} \frac{ax - \frac{ax^3}{6} + \frac{ax^5}{120} + ... - bx + cx^2 + x^3}{2x^2\left(\frac{x}{1} - \frac{x^2}{2} + \frac{2x^5}{3} + ...\right) - 2x^3 + x^4}$$

$a - b = 0$

$a = b$ $\qquad c = 0$ $\qquad -\frac{a}{6} + 1 = 0$

$a = 6$

$b = 6$

$$\text{limit} = \frac{a}{120 \times \frac{2}{3}} = \frac{6}{80} = \frac{3}{40}$$

40. Evaluate : $\lim\limits_{x\to 0} \dfrac{x^{6000}-(\sin x)^{6000}}{x^2\cdot(\sin x)^{6000}}$

मान लिखिये : $\lim\limits_{x\to 0} \dfrac{x^{6000}-(\sin x)^{6000}}{x^2\cdot(\sin x)^{6000}}$

Ans. 1000

Sol. $\lim\limits_{x\to 0} \dfrac{x^{6000}-(\sin x)^{6000}}{x^2(\sin x)^{6000}}$

$$\lim_{x\to 0} \frac{\left(\frac{x}{\sin x}\right)^{6000}-1}{x^2}$$

$$\lim_{x\to 0} \frac{6000\left(\frac{x}{\sin x}\right)^{5999}\left[\frac{\sin x - x\cos x}{\sin^2 x}\right]}{2x}$$

$$\lim_{x\to 0} \frac{6000\left(\frac{x}{\sin x}\right)^{5999}\left[x-\frac{x^3}{\lfloor 3}+.....-x+\frac{x^3}{\lfloor 2}\right]}{2x\frac{\sin^2 x}{x^2}\times x^2}$$

$$\lim_{x\to 0} 6000\left(\frac{x}{\sin x}\right)^{5999}\frac{\left(x^3\left(\frac{1}{2}-\frac{1}{6}\right)\right)}{2x^3\left(\frac{\sin x}{x}\right)^2}$$

$$\Rightarrow \frac{6000\times(1)\left(\frac{2}{6}\right)}{2}=1000$$

41. (a) If $\lim\limits_{x\to 0} \dfrac{1-\cos x\cdot\cos 2x\cdot\cos 3x.....\cos nx}{x^2}$ has the value equal to 253, find the value of n (where $n \in N$).

(b) Find the value of $\lim_{x\to 0}\frac{1-\cos^5 x\cos^3 2x\cos^3 3x}{x^2}$.

(c) If $\lim_{x\to 0}\frac{1-\cos 3x\cdot\cos 9x\cdot\cos 27x.....\cos 3^n x}{1-\cos\frac{1}{3}x\cdot\cos\frac{1}{9}x\cdot\cos\frac{1}{27}x.....\cos\frac{1}{3^n}x}=3^{10}$, find the value of n.

(a) यदि $\lim_{x\to 0}\frac{1-\cos x\cdot\cos 2x\cdot\cos 3x.....\cos nx}{x^2}$ का मान 253 के बराबर है, तो n का मान बताइए (जहां $n\in N$)

(b) $\lim_{x\to 0}\frac{1-\cos^5 x\cos^3 2x\cos^3 3x}{x^2}$ का मान बताइयें।

(c) यदि $\lim_{x\to 0}\frac{1-\cos 3x\cdot\cos 9x\cdot\cos 27x.....\cos 3^n x}{1-\cos\frac{1}{3}x\cdot\cos\frac{1}{9}x\cdot\cos\frac{1}{27}x.....\cos\frac{1}{3^n}x}=3^{10}$ तो n का मान बताइए।

Ans. (a) 11, (b) 22, (c) 4

Sol. (a) $\lim_{x\to 0}\frac{1-\cos x\cos 2x\cos 3x.....\cos nx}{x^2}$

$\lim_{x\to 0}\frac{1-f(x)}{x^2}$

$\lim_{x\to 0}\frac{-f'(x)}{2x}$

$f(x)=\cos x\cos 2x\cos 3x....\cos nx$

$\log[f'(x)]=\log(\cos x)+\log(\cos 2x)+\log(\cos 3x)+....+\log(\cos nx)$

$\frac{1}{f(x)}f'(x)=-\frac{\sin x}{\cos x}-\frac{2\sin 2x}{\cos 2x}-\frac{3\sin 3x}{\cos 3x}.....-\frac{n\sin nx}{\cos nx}$

$f'(x)=-(\cos x\cos 2x\cos 3x.....\cos nx)(\tan x+2\tan 2x+....+n\tan nx)$

$\lim_{x\to 0}(\cos x\cos 2x\cos 3x....\cos nx)\frac{(\tan x+2\tan 2x+....+n\tan nx)}{2x}$

$=(1)\left(\frac{1}{2}\right)\left(1^2+2^2+.....+n^2\right)=253$

$\frac{n(n+1)(2n+1)}{6}=506$

$n(n+1)(2n+1) = 3036$

$n = 11$

(b) $\lim\limits_{x\to 0} \dfrac{1-\cos^5 x \cos^3 2x \cos^3 3x}{x^2}$

$\lim\limits_{x\to 0} \dfrac{1-f(x)}{x^2}$

$\lim\limits_{x\to 0} \dfrac{-f'(x)}{2x}$

$f(x) = \cos^5 x \cos^3 2x \cos^3 3x$

$\log(f(x)) = 5 \log \cos x + 3 \log \cos 2x + 3 \log \cos 3x$

$$\frac{1}{f(x)} f'(x) = 5 - \frac{\sin x}{\cos x} + 3 \times 2 \frac{(-\sin 2x)}{\cos 2x} + 3 \cdot 3 \left(\frac{-\sin 3x}{\cos 3x} \right)$$

$f'(x) - -f(x)\,[5 \tan x + 6 \tan 2x + 9 \tan 3x]$

$$\Rightarrow \lim_{x\to 0} + \cos^5 x \cos^3 2x \cos^3 3x \frac{(5 \tan x + 6 \tan 2x + 9 \tan 3x)}{2x}$$

$$= (1)\left(\frac{1}{2}\right)(5 + 12 + 27)$$

$$= \frac{44}{2} = 22$$

(c) $$\lim_{x\to 0} \frac{1-\cos 3x \cdot \cos 9x \cdot \cos 27x \ldots\ldots \cos 3^n x}{1-\cos\frac{x}{3}\cos\frac{x}{9}\cos\frac{x}{27}\ldots\ldots\cos\frac{x}{3^n}}$$

$\lim\limits_{x\to 0} \dfrac{1-f(x)}{1-g(x)}$

$\lim\limits_{x\to 0} \dfrac{+f'(x)}{+g'(x)}$

$f(x) = \cos 3x \cos 9x \cos 27x \ldots\ldots \cos 3^n x$

$\log[f(x)] = \log(\cos 3x) + \log(\cos 9x) + \log(\cos 27x) + \ldots\ldots\ldots\ldots + \log(\cos 3^n x)$

$$\frac{f'(x)}{f(x)} = -\frac{3\sin 3x}{\cos 3x} - 3^2\frac{\sin 9x}{\cos 9x} - 3^3\frac{\sin 27x}{\cos 27x} \ldots\ldots\ldots\ldots - 3^n\frac{\sin 3^n x}{\cos 3^n x}$$

$f'(x) = -f(x)\,[3\tan 3x + 3^2\tan 3^2x + 3^3\tan 3^3x + \ldots. + 3^n\tan 3^nx]$

$$g(x) = \cos\frac{x}{3}\cos\frac{x}{9}\cos\frac{x}{27}\ldots\ldots\ldots\cos\frac{x}{3^n}$$

$$\log[g(x)] = \log\left(\cos\frac{x}{3}\right) + \log\left(\cos\frac{x}{9}\right) + \log\left(\cos\frac{x}{27}\right) + \ldots\ldots + \log\left(\frac{x}{3^n}\right)$$

$$\frac{g'(x)}{g(x)} = -\frac{1}{3}\tan\frac{x}{3} - \frac{1}{9}\tan\frac{x}{9} - \frac{1}{27}\tan\frac{x}{27}\ldots\ldots\ldots\frac{1}{3^n}\tan\frac{x}{3^n}$$

$$g'(x) = g(x)\left[\frac{1}{3}\tan\frac{x}{3} + \frac{1}{9}\tan\frac{x}{9} + \frac{1}{27}\tan\frac{x}{27} + \ldots\ldots + \frac{1}{3^n}\tan\frac{x}{3^n}\right]$$

$$= \lim_{x\to 0}\frac{+\left(\cos 3x\cos 9x\ldots\ldots\cos 3^n x\right)\left(3\tan 3x + 3^2\tan 3^2x + 3^3\tan 3^3x\ldots\ldots + 3^n\tan 3^n x\right)}{+\left(\cos\frac{x}{3}\cos\frac{x}{9}\ldots\ldots\cos\frac{x}{3^n}\right)\left(\frac{1}{3}\tan\frac{x}{3} + \frac{1}{3^2}\tan\frac{x}{9} + \ldots\ldots + \frac{1}{3^n}\tan\frac{x}{3^n}\right)}$$

$$\frac{(1)\left(3^2 + 3^4 + 3^6 + \ldots\ldots + 3^{2n}\right)}{(1)\left(\frac{1}{3^2} + \frac{1}{3^4} + \frac{1}{3^6} + \ldots\ldots + \frac{1}{3^{2n}}\right)}$$

$$\Rightarrow \frac{3^2\left(\frac{3^{2n}-1}{3^2-1}\right)}{\frac{1}{3^2}\left(\frac{1-\frac{1}{3^{2n}}}{1-\frac{1}{3^2}}\right)} = \frac{9\left(\frac{3^{2n}-1}{8}\right)}{\frac{1}{9}\left(\frac{3^{2n}-\frac{1}{3^{2n}}}{\frac{3^2-1}{3^2}}\right)}$$

$\Rightarrow 9\cdot 3^{2n} = 3^{10}$

$3^{2n+2} = 3^{10}$

$2n + 2 = 10$

$2n = 8$

$n = 4$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 42 to 43

Let f(x) and g(x) be two differentiable functions, defind as : $f(x)= x^2 + xg'(1) + g''(2)$ and $g(x) = f(1)\, x^2 + x\, f'(x) + f''(x)$

माना f(x) तथा g(x) दो अवकलनीय फलन इस प्रकार परिभाषित है कि $f(x)= x^2+xg'(1)+g''(2)$ तथा $g(x) = f(1)\, x^2 + x\, f'(x) + f''(x)$

42. The value of $f(1) + g(-1)$ is :

$f(1) + g(-1)$ का मान होगा :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol. $f(x) = x^2 + xg'(1) + g''(2)$ $\qquad$ $g(x) = f(1)\, x^2 + xf'(x) + f''(x)$

$f(x) = x^2 + ax + b$ $\qquad$ $g(x) = (x^2 + xf'(x) + f''(x))$

$a = g'(1)$ $\qquad$ $b = g''(2)$ $\qquad$ $c = f(1)$

$$\left.\begin{matrix} f'(x) = 2x + a \\ f''(x) = 2 \end{matrix}\right\} \Rightarrow$$

$g(x) = cx^2 + x\,(2x + a) + 2$

$g(x) = (c + 2)\, x^2 + ax + 2$

$g'(x) = 2\,(c + 2)\, x + a$

$g'(1) = 2c + 4 + a = 0$

$\Rightarrow 2c + 4 = 0$

$c = -2$

$g''(x) = 2\,(c + 2)$

$g''(2) = 2\,(c + 2)$

$b = 2\,(c + 2) \Rightarrow b = 0$

$f(x) = x^2 + x\,(2c + a + 4) + 2(c + 2)$

$x = 1$

$f(1) = c = 1 + 2c + a + 4 + 2c + 4$

$-2 = 1 - 4 + a + 4 - 4 + 4$

$a = -3$

$f(x) = x^2 - 3x$ $\qquad$ $g(x) = -3x + 2$

$f(1) = 1 - 3$ $\quad\quad$ $g(-1) = 3 + 2 = 5$

$= -2$

$f(1) + g(-1) = -2 + 5 = 3$

43. The number of integers in the domain of the function $F(x) = \sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ is :

(A) 0 $\quad$ (B) 1 $\quad$ (C) 2 $\quad$ (D) Infinite

$F(x) = \sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ के प्रान्त में पूर्णांक मानों की संख्या होगी :

(A) 0 $\quad$ (B) 1 $\quad$ (C) 2 $\quad$ (D) अनन्त

Ans. C

Sol. $F(x) = \sqrt{\frac{-(x^2-3x)}{-(3x-2)}} + \sqrt{3-x}$

$= \sqrt{\frac{x^2-3x}{3x-2}} + \sqrt{3-x}$

$\frac{x^2-3x}{3x-2} \geq 0$ $\quad\quad$ $3-x \geq 0$

$x-3 \leq 0$

$x \leq 3$

$\frac{x(x-3)}{3x-2} \geq 0$

$x \in \left[0, \frac{2}{3}\right) \cup [3, \infty)$

$F.A. \Rightarrow x = \left[0, \frac{2}{3}\right) \cup \{3\}$

$I = 0, 3$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

44. Let $f(x) = \dfrac{\sqrt{x - 2\sqrt{x-1}}}{\sqrt{x-1} - 1}$ x. then :

(A) $f'(10) = 1$ (B) $f'(3/2) = -1$

(C) domain of $f(x)$ is $x \geq 1$ (D) None of these

माना $f(x) = \dfrac{\sqrt{x - 2\sqrt{x-1}}}{\sqrt{x-1} - 1}$. x तब :

(A) $f'(10) = 1$ (B) $f'(3/2) = -1$

(C) domain of $f(x)$ is $x \geq 1$ (D) इनमें से कोई नहीं

Ans. AB

Sol. $f(x) = \dfrac{\sqrt{\left(\sqrt{x-1}\right)^2 + 1 - 2\sqrt{x-1}}}{\sqrt{x-1} - 1} \cdot x = \dfrac{\left|\sqrt{x-1} - 1\right|}{\sqrt{x-1} - 1} \cdot x = \begin{bmatrix} -x & \text{if } x \in [1, 2) \\ x & \text{if } x \in (2, \infty) \end{bmatrix}$

45. If $f(x) = x \,.\, |x|$, then its derivative is :

यदि $f(x) = x \,.\, |x|$, तब $f(x)$ का अवकलन है :

(A) $2x$ (B) $-2x$ (C) $2|x|$ (D) $2x \operatorname{sgn} x$

Ans. CD

Sol. $f(x) = x \cdot |x|$

$x \geq 0$ $x < 0$

$f(x) = x^2$ $f(x) = -x^2$

$f'(x) = 2x$ $f'(x) = -2x$

$f'(x) = 2|x|$

46. Let $f(x) = \cos^{-1}(2x^2 - 1)$ then :

(A) $f(x)$ is continuous in $[-1, 1]$ (B) $f(x)$ is derivable in $(-1, 1)$

(C) range of $f(x)$ is $[0, \pi]$ (D) derivative of $f(x)$ w.r.t. $\sin^{-1}x$ at $x = 1/2$ is 2

माना $f(x) = \cos^{-1}(2x^2 - 1)$ तब :

(A) $f(x)$; $[-1, 1]$ में सतत् है (B) $f(x)$; $(-1, 1)$ में अवकलनीय है

(C) $f(x)$ का परिसर $[0, \pi]$ है (D) $f(x)$ का $\sin^{-1}x$ के सापेक्ष $x = 1/2$ पर अवकलन 2 है

Ans. AC

Sol. $f(x) = \cos^{-1}(2x^2 - 1)$

$-1 \le 2x^2 - 1 \le 1$

$0 \le 2x^2 \le 2$

$0 \le x^2 \le 1$

$x \in [-1, 1] \rightarrow$ Continuous

Range $\Rightarrow [0, \pi]$

$x = \cos\theta$

$f(x) = \cos^{-1}(2\cos^2\theta - 1)$

$= 2\theta$

$f(x) = \cos^{-1}x$

$$f'(x) = \frac{-2}{\sqrt{1-x^2}}$$

$g(x) = \sin^{-1}x$

$$g'(x) = \frac{1}{\sqrt{1-x^2}}$$

$= -2$

MATCH THE COLUMN TYPE QUESTION(S)

47. Match the following column :

Column-I	**Column-II**
(A) $f(x) = \begin{cases} \ell n(1+x^3)\cdot\sin\frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \le 0 \end{cases}$	(P) continuous everywhere but not differentiable at $x = 0$
(B) $g(x) = \begin{cases} \ell n^2(1+x)\cdot\sin\frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \le 0 \end{cases}$	(Q) differentiable at $x = 0$ but derivative is discontinuous at $x = 0$
(C) $u(x) = \begin{cases} \ell n\left(1+\frac{\sin x}{2}\right), & \text{if } x > 0 \\ 0, & \text{if } x \le 0 \end{cases}$	(R) differentiable and has continuous derivative

(D) $v(x) = \lim_{t\to 0} \frac{2x}{\pi} \tan^{-1}\left(\frac{2}{t^2}\right)$ (S) continuous and differentiable at $x = 0$

निम्न कॉलम को सुमेलित कीजिए :

Column-I **Column-II**

(A) $f(x) = \begin{cases} \ell n(1+x^3)\cdot \sin\frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \le 0 \end{cases}$ (P) प्रत्येक बिन्दु पर संतत् परन्तु $x = 0$ पर अवकलनीय नहीं

(B) $g(x) = \begin{cases} \ell n^2(1+x)\cdot \sin\frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \le 0 \end{cases}$ (Q) $x = 0$ पर अवकलनीय परन्तु इसका अवकलज $x = 0$ पर असंतत् है

(C) $u(x) = \begin{cases} \ell n\left(1+\frac{\sin x}{2}\right), & \text{if } x > 0 \\ 0, & \text{if } x \le 0 \end{cases}$ (R) अवकलनीय तथा अवकलज भी संतत् है

(D) $v(x) = \lim_{t\to 0} \frac{2x}{\pi} \tan^{-1}\left(\frac{2}{t^2}\right)$ (S) $x = 0$ पर अवकलनीय तथा संतत् है

Ans. (A) R, S; (B) Q, S; (C) P ; (D) R, S

Sol. (A) $f(x)\begin{cases} \ln\left(1+x^3\right)\sin\frac{1}{x} & ; \quad x > 0 \\ 0 & ; \quad x \le 0 \end{cases}$

$f\left(0^+\right) = 0\times[-1,1] \Rightarrow 0$

continuous

LHD = 0

$\text{RHD} \Rightarrow \lim_{x\to 0} \frac{\ln\left(1+x^3\right)\sin\frac{1}{x} - 0}{x\times x^2} \times x^2$

$= 0$

differentiable at 0

$f'(x) = \begin{cases} \frac{3x^2}{1+x^3}\sin\left(\frac{1}{x}\right) - \frac{x}{x^2}\frac{\ln\left(1+x^3\right)}{\times x}\cos\left(\frac{1}{x}\right) & ; \quad x > 0 \\ 0 & ; \quad x \le 0 \end{cases}$

$f'(0^-) = 0 = f'(0) = f'(0^+) \Rightarrow$ Continuous at $x = 0$

(B) $g(x) = \begin{cases} \ln^2(1+x)\cdot\sin\left(\frac{1}{x}\right) & ; \quad x < 0 \\ 0 & ; \quad x \le 0 \end{cases}$

$g(0^+) = 0 = g(0) = g(0^-)$

$\Rightarrow$ Continuous at 0;

LHD = 0

$$\text{RHD} = \lim_{x\to 0} \frac{\ln^2(1+x)\sin\frac{1}{x} - 0}{x \times \frac{n}{x}}$$

$= 0$

Differentiable at $x = 0$

$$g'(x) = \begin{cases} \frac{2\ln(1+x)}{1+x}\sin\frac{1}{x} & ; \quad x > 0 \\ 0 & ; \quad x \le 0 \end{cases}$$

Derivative is not continuous

(C) $u(x) = \begin{cases} \ln\left(1+\frac{\sin u}{2}\right) & ; \quad x > 0 \\ 0 & ; \quad x \le 0 \end{cases}$

$u(0^+) = 0 = u(0) = u(0^-)$

$\Rightarrow$ continuous

LHD = 0

$$\text{RHD} = \lim_{x\to 0}\left[\frac{\ln\left(1+\frac{\sin x}{2}\right) - 0}{x \times \frac{\sin x}{2}}\right]\left(\frac{\sin x}{2}\right)$$

$$= \frac{1}{2}$$

not differentiable at $x = 0$

(D) $v(x)=\lim_{t\to 0}\frac{2x}{11}\tan^{-1}\left(\frac{2}{t^2}\right)$

$=\frac{2x}{\pi}\lim_{t\to 0}\tan^{-1}\left(\frac{2}{t^2}\right)$

$=\frac{2x}{\pi}\times\frac{\pi}{2}$

$=x$

48. Column-I contains function defined on R and Column-II contains their properties. Match them.

Column-I	**Column-II**
(A) $\lim_{n\to\infty}\left(\frac{1+\tan\frac{\pi}{2n}}{1+\sin\frac{\pi}{3n}}\right)^n$ equal	(P) e
(B) $\lim_{x\to 0^+}\frac{1}{(1+\text{cosec}\,x)^{\frac{1}{\ln(\sin x)}}}$ equals	(Q) e^2
(C) $\lim_{x\to 0}\left(\frac{2}{\pi}\cos^{-1}x\right)^{1/x}$ equals	(R) $e^{-2/\pi}$
	(S) $e^{\pi/6}$

कॉलम–I में R में परिभाषित फलन दिये गए है तथा कॉलम–II में उनके गुणधर्म। सुमेलित कीजिये।

Column-I	**Column-II**
(A) $\lim_{n\to\infty}\left(\frac{1+\tan\frac{\pi}{2n}}{1+\sin\frac{\pi}{3n}}\right)^n$ बराबर है	(P) e
(B) $\lim_{x\to 0^+}\frac{1}{(1+\text{cosec}\,x)^{\frac{1}{\ln(\sin x)}}}$ बराबर है	(Q) e^2
(C) $\lim_{x\to 0}\left(\frac{2}{\pi}\cos^{-1}x\right)^{1/x}$ बराबर है	(R) $e^{-2/\pi}$
	(S) $e^{\pi/6}$

Ans. (A) S; (B) P; (C) R

Sol. (A) $\lim\limits_{n\to\infty}\left(\dfrac{1+\tan\dfrac{\pi}{2n}}{1+\sin\dfrac{\pi}{3n}}\right)$ $\qquad\qquad \dfrac{\pi}{6n}=\theta$

$=1^{\infty}$

$\Rightarrow (e)^{\lim\limits_{n\to\infty} n\left(\frac{1+\tan 3\theta}{1+\sin 2\theta}-1\right)}$

$\Rightarrow (e)^{\lim\limits_{n\to\infty} n\left(\frac{1+\frac{3t-t^3}{1-3t^2}-\frac{2t}{1+t^2}-1}{1+\sin 2\theta}\right)}$ $\qquad (t=\tan x)$

$\Rightarrow (e)^{\lim\limits_{n\to\infty} n\frac{\tan\frac{\pi}{6n}}{\frac{\pi}{6n}}\left[\frac{3-t^2}{1-3t^2}-\frac{2}{1+t^2}\right]\cdot\frac{1}{\left(1+\sin\frac{\pi}{3n}\right)}\times\frac{\pi}{6n}}$

$\Rightarrow (e)^{\frac{\pi}{6}(1)(3-2)}=(e)^{\frac{\pi}{6}}$

(B) $\lim\limits_{x\to 0^+}\dfrac{1}{(1+\operatorname{cosec} x)^{\frac{1}{\ln(\sin x)}}}$

$\lim\limits_{x\to 0^+}(e)^{\frac{1}{\ln(1+\operatorname{cosec} x)}}=\lim\limits_{x\to 0^+} e^{\frac{\ln\sin x}{\ln(1+\operatorname{cosec} x)}}$

$\Rightarrow \lim\limits_{x\to 0^+}(e)^{\frac{\frac{\cos x}{\sin x}}{-\frac{\operatorname{cosec} x\cot x}{1+\operatorname{cosec} x}}}$

$\Rightarrow \lim\limits_{x\to 0^+}(e)^{\frac{\cos x}{\sin x}\times\frac{1+\operatorname{cosec} x}{-\operatorname{cosec} x\cot x}}$

$\Rightarrow \lim\limits_{x\to 0^+}(e)^{1+\sin x}$

$=e^1$

(C) $\lim\limits_{x\to 0^+}\left[\dfrac{2}{\pi}\cos^{-1}x\right]^{\frac{1}{x}}$

$$\Rightarrow \lim_{x\to 0^+}\left[\frac{2}{\pi}\left(\frac{\pi}{2}-\sin^{-1}x\right)\right]^{\frac{1}{x}}$$

$$\Rightarrow (e)^{\lim\limits_{x\to 0^+}\frac{1}{x}\left[1-\frac{2}{\pi}\sin^{-1}x-1\right]}$$

$$=(e)^{-\frac{2}{\pi}}$$

Exercise J-M

1. If $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$, $x \in R$, then : **[JEE(Main)Jan.2023]**

यदि $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$, $x \in R$ है, तब : **[JEE(Main)Jan.2023]**

(A) $2f(0)-f(1)+f(3)=f(2)$ (B) $f(1)+f(2)+f(3)=f(0)$

(C) $3f(1)+f(2)=f(3)$ (D) $f(3)-f(2)=f(1)$

Ans. A

Sol. $f(x)=x^3-x^2f'(1)+xf''(2)-f'''(3)$

$f'(x)=3x^2-2xf'(1)+f''(2)$

$f''(x)=6x-2f'(1)$

$f'''(x)=6 \Rightarrow f'''(3)=6$

$\text{from (3)} \rightarrow f''(2)=12-2f'(1)$

$\text{from(2)} \rightarrow f'(1)=3(1)^2-2f'(1)+f''(2)$

$\Rightarrow f''(2)=3f'(1)-3$

$\Rightarrow 12-2f'(1)=3f'(1)-3 \Rightarrow f'(1)=3$

$f''(2)=12-6=6$

$f(x)=x^3-3x^2+6x-6, \quad f(0)=-6$

$f(1)=-2, \quad f(2)=2, \quad f(3)=12$

2. Let $y(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y'-y''$ at $x=-1$ is equal to : **[JEE(Main)Jan.2023]**

माना $y(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$ है। तो $x=-1$ पर $y'-y''$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 464 (B) 944 (C) 496 (D) 976

Ans. C

Sol. $y=(1+x)\left(1+x^2\right)\left(1+x^4\right)\left(1+x^8\right)\left(1+x^{16}\right)$

$\Rightarrow y=\dfrac{(1-x)(1+x)\left(1+x^2\right)\left(1+x^4\right)\left(1+x^8\right)\left(1+x^{16}\right)}{(1-x)}$

$\Rightarrow y=\dfrac{1-x^{32}}{(1-x)}$

$\Rightarrow (1-x)y=1-x^{32}$

$\Rightarrow (1-x)y'-y=-32x^{31}$

$\Rightarrow (1-x)y''-2y'=-32.31x^{30}$

Put $x=-1$

$\Rightarrow y' - y'' = 496$

3. Let $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right) - 2(1 - \sin^2 2\theta)$ and

$S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$. If $4\beta = \sum_{\theta \in S} \theta$, then $f(\beta)$ is equal to : **[JEE(Main)Jan.2023]**

माना $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta)\right) - 2(1 - \sin^2 2\theta)$ एवं

$S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$ है। यदि $4\beta = \sum_{\theta \in S} \theta$, है तब $f(\beta)$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{11}{8}$ (B) $\frac{3}{2}$ (C) $\frac{9}{8}$ (D) $\frac{5}{4}$

Ans. D

Sol. $f(\theta) = 3\left(\sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3x + \theta)\right) - 2\left(1 - \sin^2 2\theta\right)$

$S = \left\{\theta \in [0, \pi] : f'(\theta) = -\frac{\sqrt{3}}{2}\right\}$

$\Rightarrow f(\theta) = 3(\cos^4\theta + \sin^4\theta) - 2\cos^2 2\theta$

$\Rightarrow f(\theta) = 3\left(1 - \frac{1}{2}\sin^2 2\theta\right) - 2\cos^2 2\theta$

$\Rightarrow f(\theta) = 3 - \frac{3}{2}\sin^2 2\theta - 2\cos^2 2\theta$

$= \frac{3}{2} - \frac{1}{2}\cos^2 2\theta = \frac{3}{2} - \frac{1}{2}\left(\frac{1 + \cos 4\theta}{2}\right)$

$f(\theta) = \frac{5}{4} - \frac{\cos 4\theta}{4}$

$f'(\theta) = \sin 4\theta$

$\Rightarrow f'(\theta) = \sin 4\theta = -\frac{\sqrt{3}}{2}$

$\Rightarrow 4\theta = n\pi + (-1)^n \frac{\pi}{3}$

$\Rightarrow \theta = \frac{n\pi}{4} + (-1)^n \frac{\pi}{12}$

$$\Rightarrow \theta = \frac{\pi}{12}, \left(\frac{\pi}{4} - \frac{\pi}{12}\right), \left(\frac{\pi}{2} + \frac{\pi}{12}\right), \left(\frac{3\pi}{4} - \frac{\pi}{12}\right)$$

$$\Rightarrow 4\beta = \frac{\pi}{4} + \frac{\pi}{2} + \frac{3\pi}{4} = \frac{3\pi}{2}$$

$$\Rightarrow \beta = \frac{3\pi}{8} \Rightarrow f(\beta) = \frac{5}{4} - \frac{\cos\frac{3\pi}{2}}{4} = \frac{5}{4}$$

4. Let $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$. Then, at $x = 1$: **[JEE(Main)Jan.2023]**

माना $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$ है। तो $x = 1$ पर **[JEE(Main)Jan.2023]**

(A) $\sqrt{2}y' - 3\pi^2 y = 0$ (B) $y' + 3\pi^2 y = 0$

(C) $2y' + 3\pi^2 y = 0$ (D) $2y' + \sqrt{3}\pi^2 y = 0$

Ans. C

Sol. $Y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{3/2}\right)\right)\right)$

$$\frac{dy}{dx} = 3\sin^2\left(\frac{\pi}{3}\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)\right)^{3/2}\right) \times \frac{\pi}{3}\left(-\sin\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x + 1\right)\right)\right)^{3/2}$$

$$\left(\cos\left(\frac{\pi}{3}\cos\left(\frac{\pi}{3\sqrt{2}}(-4x^3 + 5x^2 + 1)^{3/2}\right) \times \frac{\pi}{3\sqrt{2}}(3/2(-4x^3 + 5x^2 + 1)^{1/2})(-12x^2 + 10x\right)\right)$$

at $x = 1$

$$y = f(1) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}.2\sqrt{2}\right)\right)\right)$$

$$= \sin^3\left(\frac{\pi}{3}(-1/2)\right)$$

$$= \sin^3\left(-\frac{\pi}{6}\right) = \left(-\frac{1}{2}\right)^3 = -\frac{1}{8}$$

$$\left(\frac{dy}{dx}\right)_{x=1} = 3\left(-\frac{1}{2}\right)^2 \cdot \left(-\frac{\pi}{3}\right)\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) \cdot \frac{\pi}{3\sqrt{2}}\left(\frac{3}{2}\sqrt{2}\right)(-2)$$

$$= \frac{3}{4}\cdot\frac{\pi}{3}\cdot\frac{3}{4} = \frac{3\pi^2}{16}$$

Now (3) $$2y' + 3\pi^2 y = 2\left(\frac{3\pi^2}{16}\right) + 3\pi^2\left(-\frac{1}{8}\right) = \frac{3\pi^2}{8} - \frac{3\pi^2}{8} = 0$$

5. If $y(x) = x^x, x > 0$, then y"(2) – 2y'(2) is equal to : **[JEE(Main)Jan.2023]**

यदि $y(x) = x^x, x > 0$ है, तो y"(2) – 2y'(2) बराबर है : **[JEE(Main)Jan.2023]**

(A) $4(\log_e 2)^2 + 2$ (B) $4\log_e 2 + 2$

(C) $8\log_e 2 - 2$ (D) $4(\log_e 2)^2 - 2$

Ans. D

Sol. $y' = x^x$

$y' = x^x + (1 + \ln x)$

$$y'' = x^x(1+\ln x)^2 + x^x.\frac{1}{x}$$

$y''(2) = 4(1 + \ln 2)^2 + 2$

$y'(2) = 4(1 + \ln 2)$

$y''(2) - 2y'(2) = 4(1 + \ln 2)^2 + 2 - 8(1 + \ln 2)$

$= 4(1 + \ln 2)[1 + \ln 2 - 2] + 2$

$= 4(\ln 2)^2 - 1) + 2$

$= 4(\ln 2)^2 - 2$

6. If $f(x) = x^2 + g'(1)x + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$, then the value of f(4) – g(4) is equal to _____. **[JEE(Main)Jan.2023]**

यदि $f(x) = x^2 + g'(1)x + g''(2)$ एवं $g(x) = f(1)x^2 + xf'(x) + f''(x)$ है, तो f(4) – g(4) का मान है। **[JEE(Main)Jan.2023]**

Ans. 14

Sol. $f(x) = x^2 + xg'(1) + g''(2)$ ...(1)

$\therefore f'(x) = 2x + g'(1)$

$f''(x) = 2$

$\therefore g(x) = f(1)x^2 + x(2x + g'(1)) + 2 = f(1)x^2 + 2x^2 + xg'(1) + 2$

$g'(x) = 2f(1)x + 4x + g'(1)$ ...(2)

$g''(x) = 2(1) + 4$ ...(3)

from (2) put x = 1

$g'(1) + 2f(1) + 4 + g'(1)$

$\Rightarrow 2f(1) + 4 = 0$

$\Rightarrow f(1) = -2 \Rightarrow g''(2) = -4 + 4 = 0$ from (3)

from (1)

$f(1) = 1 + g'(1) + 0$

$\Rightarrow g'(1) = -2 - 1 = -3$

$\because f'(x) = 2x - 3$

$f(x) = x^2 - 3x + c$

put $x = 1, f(1) = -2 \quad \therefore c = 0$

$\therefore f(x) = x^2 - 3x$

$\therefore g(x) = -3x + 2$

$\Rightarrow f(4) - g(4) = 4 - (-10) = 14$

7. If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at $(2, 2)$ is equal to : **[JEE(Main)April,2023]**

यदि $2x^y + 3y^x = 20$ है, तब $(2, 2)$ पर $\frac{dy}{dx}$ का मान है : **[JEE(Main)April,2023]**

(A) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$ (B) $-\left(\frac{3+\log_e 4}{2+\log_e 8}\right)$ (C) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$ (D) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$

Ans. D

Sol. $2x^y + 3y^x = 20$

$$2x^y\left[\frac{y}{x} + (\ln x)y'\right] + 3y^x\left[\frac{xy'}{y} + \ln y\right] = 0$$

$$y' = \frac{-(12\ln 2 + 8)}{12 + 8\ln 2} = -\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$$

8. Let $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}, x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to : **[JEE(Main)April,2023]**

यदि $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}, x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$ है, तो $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{2}{9}$ (B) $\frac{-2}{3}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{-1}{3\sqrt{3}}$

Ans. A

Sol. $$f(x) = \sqrt{2}\frac{\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right) - 1}{\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x - \frac{1}{\sqrt{2}}\cos x\right)} = \frac{\sin(x + \pi/4) - 1}{\sin(x - \pi/4)}$$

$$f'(x)=\frac{\cos(x+\pi/4)\sin(x-\pi/4)-\cos(x-\pi/4)(\sin(x+\pi/4)-1)}{\sin^2(x-\pi/4)}$$

$$f'(x)=\frac{\cos(x-\pi/4)-1}{\sin^2(x-\pi/4)}=\frac{-(1-\cos(x-\pi/4))}{1-\cos^2(x-\pi/4)}$$

$$f'(x)=-\frac{1}{1+\cos(x-\pi/4)}$$

$$=f''(x)=-\frac{\sin(x-\pi/4)}{(1+\cos(x-\pi/4))^2}$$

$$f(7\pi/12)=-\frac{1}{\sqrt{3}}$$

$$f''(7\pi/12)=-\frac{2\sqrt{3}}{9}$$

$$f(7\pi/12)f''(7\pi/12)=\frac{2}{9}$$

9. For the differentiable function $f:\mathbb{R}-\{0\}\to\mathbb{R}$, let $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10$, then $\left|f(3)+f'\left(\frac{1}{4}\right)\right|$ is equal to:

[JEE(Main)April,2023]

अवकलनीय फलन $f:\mathbb{R}-\{0\}\to\mathbb{R}$ के लिए माना $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10$, है, तो $\left|f(3)+f'\left(\frac{1}{4}\right)\right|$ बराबर है :

[JEE(Main)April,2023]

(A) $\frac{33}{5}$ (B) 7 (C) $\frac{29}{5}$ (D) 13

Ans. D

Sol. $3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10$ ...(i)

$$x\to\frac{1}{x}$$

$$3f\left(\frac{1}{x}\right)+2f(x)=x-10 \quad \text{...(ii)}$$

From (i) & (ii)

$$5f(x)=\frac{3}{x}-2x-10$$

Now, f(3) = –3

$$5f'(x)=\frac{-3}{x^2}-2$$

$$5f'\left(\frac{1}{4}\right) = -48 - 2$$

$$f'\left(\frac{1}{4}\right) = -10$$

$$\left|f(3) + f'\left(\frac{1}{4}\right)\right| = 13$$

10. Suppose $f(x) = \dfrac{\left(2^x + 2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2 - x + 1\right)}}{\left(7x^2 + 3x + 1\right)^3}$. Then the value of f'(0) is equal to :

[JEE(Main)Jan.2024]

माना $f(x) = \dfrac{\left(2^x + 2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2 - x + 1\right)}}{\left(7x^2 + 3x + 1\right)^3}$ है। तो f'(0) का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $\dfrac{\pi}{2}$ (B) π (C) 0 (D) $\sqrt{\pi}$

Ans. D

Sol. Let f(x) = tan x g(x)

where $g(x) = \dfrac{\left(2^x + 2^{-x}\right)\sqrt{\tan^{-1}\left(x^2 - x + 1\right)}}{\left(7x^2 + 3x + 1\right)^3}$

$$\Rightarrow f'(x) = \sec^2 x g(x) + \tan x g'(x)$$

$$f'(0) = \sec^2(0) g(0) + \tan 0 g'(0)$$

$$= g(0)$$

$$g(0) = \frac{\left(2^0 + 2^{-0}\right)\sqrt{\tan^{-1}\left(0^2 - 0 + 1\right)}}{\left(7(0)^2 + 3(0) + 1\right)^3}$$

$$= 2\sqrt{\tan^{-1} 1} = 2\sqrt{\frac{\pi}{4}} = \sqrt{\pi}$$

11. Let $y = \log_e\left(\dfrac{1 - x^2}{1 + x^2}\right)$, $-1 < x < 1$. Then at $x = \dfrac{1}{2}$, the value of $225(y' - y'')$ is equal to :

[JEE(Main)Jan.2024]

माना $y = \log_e\left(\frac{1-x^2}{1+x^2}\right), -1 < x < 1$ है। तो $x = \frac{1}{2}$ पर $225(y' - y'')$ का मान है : **[JEE(Main)Jan.2024]**

(A) 736 (B) 746 (C) 732 (D) 742

Ans. A

Sol. $y = \log_e\left(\frac{1-x^2}{1+x^2}\right)$

$\frac{dy}{dx} = y' = \frac{-4x}{1-x^4}$

Again $\frac{d^2y}{dx^2} = y'' = \frac{-4\left(1+3x^4\right)}{\left(1-x^4\right)^2}$

Again $y' - y'' = \frac{-4x}{1-x^4} + \frac{4\left(1+3x^4\right)}{\left(1-x^4\right)^2}$

at $x = \frac{1}{2}$

$y' - y'' = \frac{736}{225}$

Thus $225\left(y' - y''\right) = 225 \times \frac{736}{225} = 736$

12. If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ then $\frac{1}{5}f'(0)$ is equal to : **[JEE(Main)Jan.2024]**

यदि $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$ है, तो $\frac{1}{5}f'(0)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 6 (C) 0 (D) 1

Ans. C

Sol. $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$

$$f'(x)=\begin{vmatrix}8\cos^3 x(-\sin x) & 2\sin^4 x & 3+\sin^2 2x\\ 8\cos^3 x(-\sin x) & 2\sin^4 x & \sin^2 2x\\ 8\cos^3 x(-\sin x) & 3+2\sin^4 x & \sin^2 2x\end{vmatrix}+\begin{vmatrix}2\cos^4 x & 8\sin^3 x\cos x & 3+\sin^2 2x\\ 3+2\cos^4 x & 8\sin^3 x\cos x & \sin^2 2x\\ 2\cos^4 x & 8\sin^3 x\cos x & \sin^2 2x\end{vmatrix}+$$

$$\begin{vmatrix}2\cos^4 x & 2\sin^4 x & 4\sin 2x\cos 2x\\ 3+2\cos^4 x & 2\sin^4 x & 4\sin 2x\cos 2x\\ 2\cos^4 x & 3+2\sin^4 x & 4\sin 2x\cos 2x\end{vmatrix}$$

$$f'(0)=\begin{vmatrix}0&0&0\\0&0&0\\0&0&0\end{vmatrix}+\begin{vmatrix}2&0&0\\5&0&0\\2&0&0\end{vmatrix}+\begin{vmatrix}2&0&0\\5&0&0\\2&0&0\end{vmatrix}$$

$f(0)=0$

$\therefore \frac{1}{5}f'(0)=0$

13. If $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$ for all $x\in R$, then $2f(0)+f'(0)$ is equal to :

[JEE(Main)Jan.2024]

यदि सभी $x\in R$ के लिए $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$ हैं, तो $2f(0)+f'(0)$ बराबर है :

[JEE(Main)Jan.2024]

(A) 48 (B) 18 (C) 24 (D) 42

Ans. D

Sol. $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$

$$f'(x)=\begin{vmatrix}3x^2 & 4x & 3\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-x\end{vmatrix}+\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 6x & 2 & 3x^2\\ x^3-x & 4 & x^2-2\end{vmatrix}+\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ 3x^2-1 & 0 & 2x\end{vmatrix}$$

$$2f(0)+f'(0)=2\begin{vmatrix}0&1&1\\2&0&6\\0&4&-2\end{vmatrix}+\begin{vmatrix}0&0&3\\2&0&6\\0&4&-2\end{vmatrix}+\begin{vmatrix}0&1&1\\0&2&0\\0&4&-2\end{vmatrix}+\begin{vmatrix}0&1&1\\2&0&6\\-1&0&0\end{vmatrix}$$

$= 24 + 24 + 0 - 6$

$= 42$

14. Let g(x) be a linear function and $f(x)=\begin{cases} g(x) & , \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , \quad x > 0 \end{cases}$, is continuous at x = 0. If f′(1) = f(–1), then the value of g(3) is : **[JEE(Main)Jan.2024]**

माना g(x) एक रैखिक फलन है तथा $f(x)=\begin{cases} g(x) & , \quad x \le 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , \quad x > 0 \end{cases}$, x = 0 पर संतत है। यदि f′(1) = f(–1) है, तो g(3) का मान है : **[JEE(Main)Jan.2024]**

(A) $\log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$ (B) $\log_e\left(\frac{4}{9}\right)-1$ (C) $\frac{1}{3}\log_e\left(\frac{4}{9}\right)+1$ (D) $\frac{1}{3}\log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$

Ans. A

Sol. Let g(x) = ax + b

f(x) is continuous at x = 0

$$\therefore \lim_{x\to 0^+}\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}=\left(\frac{1}{2}\right)^{\infty}=0$$

$$\lim_{x\to 0^-} g(x)=\lim_{x\to 0^-} ax+b=b$$

So, b = 0

Now, $f(x)=\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}$

$$\ell n f(x)=\frac{1}{x}\left[\ell n(1+x)-\ell n(2+x)\right]$$

$$\frac{1}{f(x)}\cdot f'(x)=\frac{x\left[\frac{1}{1+x}-\frac{1}{2+x}\right]-\ell n\left(\frac{1+x}{2+x}\right)}{x^2}$$

$$f'(1) = f(1)\left[\frac{\left(\frac{1}{2}-\frac{1}{3}\right)-\ell n\left(\frac{2}{3}\right)}{1}\right]$$

$$f'(1) = \frac{2}{3}\left[\frac{1}{6}-\ell n\frac{2}{3}\right] \quad ...(1)$$

Also $f(-1)\ a(-1)+b$

$f(-1) = -a$ ...(2)

$\therefore$ from (1) and (2)

$$-a = \frac{1}{9}-\frac{2}{3}\ell n\frac{2}{3}$$

$$a = \frac{2}{3}\ell n\left(\frac{2}{3}\right)-\frac{1}{9}$$

$\therefore\ g(x) = ax$

$$g(3) = 3\left(\frac{2}{3}\ell n\left(\frac{2}{3}\right)-\frac{1}{9}\right)$$

$$g(3) = 2\ell n\left(\frac{2}{3}\right)-\frac{1}{3}$$

$$g(3) = \ell n\frac{4}{9}-\frac{1}{3}$$

$$g(3) = \ell n\frac{4}{9}-\ell n e^{\frac{1}{3}}$$

$$g(3) = \ell n\left(\frac{4}{9\cdot e^{\frac{1}{3}}}\right)$$

$$g(3) = \log_e\left(\frac{4}{9e^{\frac{1}{3}}}\right)$$

15. Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in R$. Then $f'(10)$ is equal to _____.

[JEE(Main)Jan.2024]

माना $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in R$ है। तो $f'(10)$ बराबर है _____।

[JEE(Main)Jan.2024]

Ans. 202

Sol. $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$

$f'(x) = 3x^2 + 2xf'(1) + f''(2)$

$f''(x) = 6x + 2f'(1)$

$f'''(x) = 6$

$\Rightarrow f'''(3) = 6$

$f''(2) = 12 + 2f'(1)$ _____(1)

$f'(1) = 3 + 2f'(1) + f''(2)$ _____(2)

adding (1) and (2)

$0 = 15 + 3f'(x) \Rightarrow f'(1) = -5$

$f''(2) = 12 - 10 = 2$

Now

$f'(10) = 300 + 20(-5) + 2$

$= 300 - 100 + 2 = 202$

16. If $y = \frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$, then $96\, y'\left(\frac{\pi}{6}\right)$ is equal to _____.

[JEE(Main)Jan.2024]

यदि $y = \frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$ है, तो $96\, y'\left(\frac{\pi}{6}\right)$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 105

Sol. $y = \frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(x\sqrt{x}-1\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2\cdot x - 5\right)\cos^3 x$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(\sqrt{x}\right)^3-1^3\right)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left[3\cos^5 x - 5\cos^3 x\right]$

$y = \frac{\left(\sqrt{x}+1\right)\cdot\sqrt{x}\left(\sqrt{x}-1\right)\left(x+\sqrt{x}+1\right)}{\sqrt{x}(x+\sqrt{x}+1)} + \frac{1}{15}\left[3\cos^5 x - 5\cos^3 x\right]$

$y = (x-1) + \frac{1}{15}\left[3\cos^5 x - 5\cos^3 x\right]$

Now, diff w.r.t x

$y' = 1 - 0 + \frac{1}{15}\left[15\cos^4 x\cdot(-\sin x) - 15\cos^2 x(-\sin x)\right]$

$$y' = 1 - \sin x \cdot \left[\cos^4 x - \cos^2 x\right]$$

$$y'\left(\frac{\pi}{6}\right) = 1 - \sin\frac{\pi}{6}\left[\cos^4\frac{\pi}{6} - \cos^2\frac{\pi}{6}\right]$$

$$= 1 - \frac{1}{2}\left[\frac{9}{16} - \frac{3}{4}\right]$$

$$= 1 - \frac{1}{2}\left[\frac{9-12}{16}\right] \Rightarrow 1 + \frac{3}{32} \Rightarrow \frac{35}{32}$$

$$96.y'\left(\frac{\pi}{6}\right) = 96 \times \frac{35}{32} \Rightarrow 105$$

17. Let $f(x) = x^5 + 2e^{\frac{x}{4}}$ for all $x \in R$. Consider a function g(x) such that (gof) (x) = x for all $x \in R$. Then the value of $8g'(2)$ is : **[JEE(Main)April,2024]**

माना सभी $x \in R$ के लिए $f(x) = x^5 + 2e^{\frac{x}{4}}$ है। एक फलन g(x), जिसके लिए (gof) (x) = x $\forall$ $x \in R$ है, का विचार कीजिए। तो $8g'(2)$ का मान है : **[JEE(Main)April,2024]**

(A) 8 (B) 4 (C) 2 (D) 16

Ans. D

Sol. g(f(n)) = x

Differentiate w.r.t x, we get

$$g'(f(x)) \cdot f'(x) = 1$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

We need to find g'(2), so,
$f(0) = 2 \Rightarrow f(x) = x^5 + 2e^{x/4}$

$$f'(x) = 5x^4 + \frac{1}{2}e^{\frac{x}{4}}$$

$$f'(x) = \frac{1}{2}$$

$$\therefore\ g'(f(0)) = \frac{1}{f'(0)}$$

$$g'(2) = 2$$

$\therefore 8g'(2)=16$

18. Let $f(x)=x^5+2x^3+3x+1$, $x \in \mathbf{R}$, and $g(x)$ be a function such that $g(f(x))=x$ for all $x \in \mathbf{R}$. the $\frac{g(7)}{g'(7)}$ is equal to : **[JEE(Main)April,2024]**

माना $f(x)=x^5+2x^3+3x+1$, $x \in \mathbf{R}$ है, तथा एक फलन $g(x)$ इस प्रकार है कि $g(f(x))=x \ \forall \ x \in R$ है। तो $\frac{g(7)}{g'(7)}$ बराबर है: **[JEE(Main)April,2024]**

(A) 42 (B) 7 (C) 14 (D) 1

Ans. C

Sol. $f(x)=x^5+2x^3+3x+1$

$f'(x)=5x^4+6x^2+3$

$f'(1)=5+6+3=14$

$g(f(x))=x$

$g'(f(x))f'(x)=1$

for $f(x)=7$

$\Rightarrow x^5+2x^3+3x+1=7$

$\Rightarrow x=1$

$g'(7)\ f'(1)=1 \Rightarrow g'(7)=\frac{1}{f'(1)}=\frac{1}{14}$

$x=1, f(x)=7 \Rightarrow g(7)=1$

$\frac{g(7)}{g'(7)}=\frac{1}{1/14}=14$

19. If $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$, then at $\theta=\frac{\pi}{2}$, $y''+y'+y$ is equal to : **[JEE(Main)April,2024]**

यदि $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$ है, तो $\theta=\frac{\pi}{2}$, $y''+y'+y$ बराबर है : **[JEE(Main)April,2024]**

(A) 2 (B) 1/2 (C) 1 (D) 3/2

Ans. A

Sol. $y=\dfrac{2\cos\theta+2\cos^2\theta-1}{4\cos^3\theta-3\cos\theta+8\cos^2\theta-4+5\cos\theta+2}$

$$y=\frac{\left(2\cos^2\theta+2\cos\theta-1\right)}{\left(2\cos^2\theta+2\cos\theta-1\right)\left(2\cos\theta+2\right)}$$

$$y=\frac{1}{2}\left(\frac{1}{1+\cos\theta}\right)$$

$$\Rightarrow \theta=\frac{\pi}{2} \qquad y=\frac{1}{2}$$

$$y'=\frac{1}{2}\left(\frac{-1}{\left(1+\cos\theta\right)^2}\times\left(-\sin\theta\right)\right)$$

$$\Rightarrow \theta=\frac{\pi}{2} \qquad y'=\frac{1}{2}$$

$$y''=\frac{1}{2}\left[\frac{\cos\theta\left(1+\cos\theta\right)^2-\sin\theta\left(2\right)\left(1+\cos\theta\right)\left(-\sin\theta\right)}{\left(1+\cos\theta\right)^4}\right]$$

$$\Rightarrow \theta=\frac{\pi}{2} \qquad y''=1$$

20. Let $f:(-\infty,\infty)-\{0\}\to\mathbb{R}$ be a differentiable function such that $f'(1)=\lim\limits_{a\to\infty}a^2f\left(\frac{1}{a}\right)$. Then $\lim\limits_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ is equal to : **[JEE(Main)April,2024]**

माना $f:(-\infty,\infty)-\{0\}\to\mathbb{R}$ एक अवकलनीय फलन इस प्रकार का है कि $f'(1)=\lim\limits_{a\to\infty}a^2f\left(\frac{1}{a}\right)$, तो $\lim\limits_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{5}{2}+\frac{\pi}{8}$ (B) $\frac{3}{4}+\frac{\pi}{8}$ (C) $\frac{3}{8}+\frac{\pi}{4}$ (D) $\frac{3}{2}+\frac{\pi}{4}$

Ans. A

Sol. $f'(1)=\lim\limits_{a\to\infty}a^2f\left(\frac{1}{a}\right)$

So $f\left(\frac{1}{a}\right)=0$ should be at $x\to\infty$

Now $\lim\limits_{a\to\infty} a^2\left[\left[\frac{1}{2}+\frac{1}{2a}\right]\tan^{-1}\left(\frac{1}{a}\right)+1+\frac{\ell n a^{-2}}{a^2}\right]$

$f\left(\frac{1}{a}\right)=\left[\frac{1}{2}+\frac{1}{2a}\right]\tan^{-1}\left(\frac{1}{a}\right)+1+\frac{\ell n\left(\frac{1}{a^2}\right)}{a^2}$ (according to question $f\left(\frac{1}{a}\right)$ should be zero whenever a tending to infinite but it is 1 so there is a conceptual mistake in the question)

$f(x)=\left(\frac{1}{2}+\frac{x}{2}\right)\tan^{-1}x+1+x^2\ell n\left(x^2\right)$

$f'(x)=\frac{x+1}{2}\cdot\frac{1}{1+x^2}+\frac{\tan^{-1}x}{2}+\frac{x^2}{x^2}2x+\ell nx^2\cdot 2x$

According to NTA $f'(1)=1\times\frac{1}{2}+\frac{\pi}{8}+2=\frac{5}{2}+\frac{\pi}{8}$

21. If $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & X\neq 0\\ 0, & X=0\end{cases}$ then: **[JEE(Main)April,2024]**

यदि $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & X\neq 0\\ 0, & X=0\end{cases}$ तो : **[JEE(Main)April,2024]**

(A) $f''\left(\frac{2}{\pi}\right)=\frac{24-\pi^2}{2\pi}$ (B) $f''\left(\frac{2}{\pi}\right)=\frac{12-\pi^2}{2\pi}$ (C) $f''(0)=1$ (D) $f''(0)=0$

Ans. A

Sol. $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & X\neq 0\\ 0, & X=0\end{cases}$

$\lim\limits_{h\to 0^+} f(0+h)=\lim\limits_{h\to 0}(0+h)^3\cdot\sin\left(\frac{1}{0+h}\right)$

$\Rightarrow \lim\limits_{h\to 0^+} h^3\cdot\sin\left(\frac{1}{h}\right)$

$\Rightarrow 0\times(-1,1]=0$

$\lim\limits_{h\to 0^-} f(0-h)=\lim\limits_{h\to 0}(0-h)^3\sin\left(-\frac{1}{h}\right)$

$\Rightarrow 0 \times [-1,1] = 0$

f(x) is continuous

$$f'(x) = 3 \cdot x^2 \cdot \sin\left(\frac{1}{x}\right) - x \cdot \cos\frac{1}{x}$$

$$f''(x) = 6x \cdot \sin\frac{1}{x} - 3\cos\frac{1}{x} - \cos\left(\frac{1}{x}\right) - \frac{\sin\frac{1}{x}}{x}$$

$$f''\left(\frac{2}{\pi}\right) = \frac{24-\pi^2}{2\pi}$$

22. Suppose for a differentiable function h, h(0)=0, h(1)=1 and $h'(0) = h'(1) = 2$. If $g(x) = h\left(e^x\right)e^{h(x)}$, then $g'(0)$ is equal to : **[JEE(Main)April,2024]**

मान लीजिए कि अवकलनीय फलन h के लिए, h(0)=0, h(1)=1 तथा $h'(0) = h'(1) = 2$ हैं। यदि $g(x) = h\left(e^x\right)e^{h(x)}$, तो $g'(0)$ बराबर है : **[JEE(Main)April,2024]**

(A) 3 (B) 8 (C) 4 (D) 5

Ans. C

Sol. $g(x) = h(e^x).e^{h(x)}$

$g'(x) = h(e^x).e^{h(x)}.h'(x) + e^{h(x)}h'(e^x).e^x$

$g'(0) = h(1)\ ^{eh(0)}h'(0) + e^{h(0)}h'(1)$

$= 2 + 2 = 4$

23. Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to : **[JEE(Main)April,2024]**

माना $f(x) = ax^3 + bx^2 + cx + 41$ इस प्रकार है कि $f(1) = 40$, $f'(1) = 2$ तथा $f''(1) = 4$ है। तो $a^2 + b^2 + c^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 73 (B) 62 (C) 51 (D) 54

Ans. C

Sol. $f(x) = ax^3 + bx^2 + cx + 41$

$f'(x) = 3ax^2 + 2bx + c$

$\Rightarrow f'(1) = 3a + 2b + c = 2$ .....(1)

$f''(n) = 6ax + 2b$

$\Rightarrow f''(1) = 6a + 2b = 4$

$3a + b = 2$ .....(2)

(A) – (2) $\Rightarrow$ b + c = 0 ...(3)

f(1) = 40

a + b + c + 41 = 40

use (3)

a + 41 = 40

by (2)

– 3 + b = 2 $\Rightarrow$ b = 5 & c = – 5

$a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$

24. If $\log_e y = 3\sin^{-1}x$, then $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ is equal to : **[JEE(Main)April,2024]**

यदि $\log_e y = 3\sin^{-1}x$ है, तो $(1-x^2)y'' - xy'$ at $x = \frac{1}{2}$ बराबर है : **[JEE(Main)April,2024]**

(A) $9e^{\frac{\pi}{6}}$ (B) $3e^{\frac{\pi}{2}}$ (C) $3e^{\frac{\pi}{6}}$ (D) $9e^{\frac{\pi}{2}}$

Ans. D

Sol. $\log_e y = 3\sin^{-1}x$

$y = e^{3\sin^{-1}x}$

$\frac{dy}{dx} = e^{3\sin^{-1}x} \cdot \frac{3}{\sqrt{1-x^2}}$

$\sqrt{1-x^2}\frac{dy}{dx} = 3y$

Again differentiate

$\sqrt{1-x^2} \cdot y'' - \frac{2x}{2\sqrt{1-x^2}} y' = 3y'$

$(1-x)^2 y'' - xy' = 3y'\left(\sqrt{1-x^2}\right)$

So value of $3y'\left(\sqrt{1-x^2}\right)$

at $x = \frac{1}{2}$

$3 \cdot \frac{3}{\sqrt{1-x^2}} e^{\sin^{-1}x}\left(\sqrt{1-x^2}\right)$

$= 9e^{3\frac{\pi}{6}} = 9e^{\frac{\pi}{2}}$

25. The value of $\lim\limits_{x\to 0} 2\left(\dfrac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____. **[JEE(Main)April,2024]**

$\lim\limits_{x\to 0} 2\left(\dfrac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}.....\sqrt[10]{\cos 10x}}{x^2}\right)$ का मान होगा _____ **[JEE(Main)April,2024]**

Ans. 55

Sol. $\lim\limits_{x\to 0}\dfrac{2}{x^2}\left(1-\left(1-\dfrac{x^2}{2}\right)\left(1-\dfrac{2^2x^2}{2}\right)^{\frac{1}{2}}\left(1-\dfrac{3^2x^2}{2}\right)^{\frac{1}{3}}.....\left(1-\dfrac{10^2x^2}{2}\right)^{\frac{1}{10}}\right)$

$\lim\limits_{x\to 0}\dfrac{2}{x^2}\left(1-\left(1-\dfrac{x^2}{2}\right)\left(1-\dfrac{2^2x^2}{2.2}\right)\left(1-\dfrac{3^2\cdot x^2}{3.2}\right).....\left(1-\dfrac{10^2x^2}{2.10}\right)\right)$

$= \lim\limits_{x\to 0}\dfrac{2}{x^2}\left(1-\left(1-\dfrac{x^2}{2}-\dfrac{2x^2}{2}-\dfrac{3x^2}{2}.....-\dfrac{10x^2}{2}\right)\right)$

$= \left(\dfrac{2}{2}(1+2+3+.....+10)\right) = \dfrac{10(10+1)}{2} = 55$

Exercise J - A

1. Let f(x) = x sin πx, x > 0. Then for all natural numbers n, f'(x) vanishes at : **[JEE(Adv.)2013]**

(A) a unique point in the interval $\left(n, n+\frac{1}{2}\right)$ (B) a unique point in the interval $\left(n+\frac{1}{2}, n+1\right)$

(C) a unique point in the interval (n, n + 1) (D) two points in the interval (n, n + 1)

मानाकि f(x) = x sin πx, x > 0 तब सभी धन–पूर्णांकों n के लिए f'(x) निम्न पर शून्य होता है : **[JEE(Adv.)2013]**

(A) अंतराल $\left(n, n+\frac{1}{2}\right)$ में एकमात्र एक बिन्दु पर (B) अंतराल $\left(n+\frac{1}{2}, n+1\right)$ में एकमात्र एक बिन्दु पर

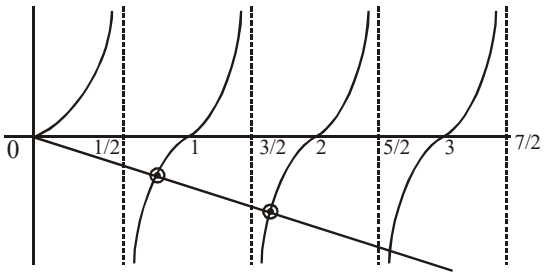
(C) अंतराल (n, n + 1) में एकमात्र एक बिन्दु पर (D) अंतराल (n, n + 1) में दो बिन्दुओं पर

Ans. BC

Sol. $f'(x) = \sin\pi x + \pi x \cos\pi x = 0$

$\tan \pi x = -\pi x$

$y = \tan x\pi$ & $y = -\pi x$



intersection point lies in

$\left(\frac{1}{2},1\right)\cup\left(\frac{3}{2},2\right)\cup\left(\frac{5}{2},3\right)...$

option (B) is correct for $\left(n+\frac{1}{2}, n+1\right)$

as well (n, (n + 1)) because root lies in

$(0,1)\cup(1,2)\cup(2,3)$

2. Let f : R → R, g : R → R and h : R → R be differentiable functions such that $f(x) = x^3 + 3x + 2$, g(f(x)) = x and h(g(g(x))) = x for all x ∈ R. Then : **[JEE(Adv.)2016]**

माना कि f : R → R, g : R → R और h : R → R ऐसे अवकलनीय फलन हैं कि सभी x ∈ R के लिए $f(x) = x^3 + 3x + 2$, g(f(x)) = x और h(g(g(x))) = x हैं। तब : **[JEE(Adv.)2016]**

(A) $g'(2) = \frac{1}{15}$ (B) h'(1) = 666 (C) h(0) = 16 (D) h(g(3)) = 36

Ans. BC

Sol. Here, $g(f(x)) = x \Rightarrow g(x) = f^{-1}(x)$

$h(g(g(x))) = x \Rightarrow h(g(x)) = f(x)$ (i)

$h(g(g(x))) = x \Rightarrow h(x) = f(f(x))$ (ii)

(A) $f(g(x)) = x \Rightarrow f'(g(x))\, g'(x) = 1 \;\therefore\; g'(2) = \dfrac{1}{f'(g(2))} = \dfrac{1}{f'(0)} = \dfrac{1}{3}$

(B) $h'(x) = f'(f(x)) \cdot f'(x)$ ($\because$ using (ii))

Put $x = 1 \Rightarrow h'(1) = 666$

(C) $h(0) = f(f(0)) = 16$

(D) $h(g(3)) = f(3)$ ($\because$ using (i))

Paragraph for Question Nos. 3 to 4

Answer the following by appropriately matching the lists based on the information given in the paragraph. Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order :

$X = \{x : f(x) = 0\}$, $Y = \{x : f'(x) = 0\}$

$Z = \{x : g(x) = 0\}$, $W = \{x : g'(x) = 0\}$

Column-I contains the sets X, Y, Z and W. Column-II contains some information regarding these sets.

Column-I	**Column-II**
(I) X	(P) $\supseteq \left\{\dfrac{\pi}{2}, \dfrac{3\pi}{2}, 4\pi, 7\pi\right\}$
(II) Y	(Q) an arithmetic progression
(III) Z	(R) NOT an arithmetic progression
(IV) W	(S) $\supseteq \left\{\dfrac{\pi}{6}, \dfrac{7\pi}{6}, \dfrac{13\pi}{6}\right\}$
	(T) $\supseteq \left\{\dfrac{\pi}{3}, \dfrac{2\pi}{3}, \pi\right\}$
	(U) $\supseteq \left\{\dfrac{\pi}{6}, \dfrac{3\pi}{4}\right\}$

(Que. 3 से 4) अनुच्छेद में दी गई जानकारी के आधार पर सूचियों का उचित मिलान करके प्रश्न का उत्तर दें। माना कि $f(x) = \sin(\pi \cos x)$ और $g(x) = \cos(2\pi \sin x)$ दो फलन (function) है जो $x > 0$ में परिभाषित हैं।

$X = \{x : f(x) = 0\}$, $Y = \{x : f'(x) = 0\}$

$Z = \{x : g(x) = 0\}$, $W = \{x : g'(x) = 0\}$

सूची-I में X, Y, Z और W समुच्चय है। सूची-II में इन समुच्चयों के बारे में कुछ सूचनाएं हैं।

Column-I	**Column-II**

(I) X

(II) Y

(III) Z

(IV) W

(P) $\supseteq \left\{\frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi\right\}$

(Q) समान्तर श्रेणी (an arithmetic progression)

(R) समान्तर श्रेणी नहीं है (NOT an arithmetic progression)

(S) $\supseteq \left\{\frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}\right\}$

(T) $\supseteq \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$

(U) $\supseteq \left\{\frac{\pi}{6}, \frac{3\pi}{4}\right\}$

3. Which of the following is the only CORRECT combination : **[JEE(Adv.)2019]**

निम्न में से कौन सा एकमात्र संयोजन सही है : **[JEE(Adv.)-2019]**

(A) (I), (Q), (U) (B) (II), (R), (S) (C) (I), (P), (R) (D) (II), (Q), (T)

Ans. D

4. Which of the following is the only CORRECT combination : **[JEE(Adv.)2019]**

निम्न में से कौन सा एकमात्र संयोजन सही है : **[JEE(Adv.)2019]**

(A) (III), (P), (Q), (U) (B) (III), (R), (U) (C) (IV), (P), (R), (S) (D) (IV), (Q), (T)

Ans. C

Sol. X: $f(x) = 0$

$\sin(\pi \cos x) = 0$

$\pi \cos x = n\pi \quad n \in I$

$\cos x = n \quad n \in I$

$\therefore \cos x = 0$ or $\cos x = -1$ or $\cos x = 1$

$x = (2n+1)\frac{\pi}{2}$ $x = n\pi$

$\therefore \in \left\{n\pi, \left(n + \frac{1}{2}\right)\pi\right\}$

$\therefore x = \left\{\frac{\pi}{2}, \pi, \frac{3\pi}{2}, ...\right\}$

Y: $f'(x) = 0$

$\cos(\pi\cos x).\ \pi(-\sin x) = 0$

$\cos(\pi \cos x) = 0$ or $\sin x = 0$

$$\pi \cos x = (2n+1)\frac{\pi}{2} \qquad x = n\pi$$

$$\cos x = \frac{(2n+1)}{2}$$

$$\therefore \cos x = \frac{1}{2}, \frac{-1}{2}$$

$$x = 2n\pi \pm \frac{\pi}{3} = \left(2n + \frac{1}{3}\right)\pi, \left(2n\frac{-1}{3}\right)\pi$$

$$x = 2n\pi \pm \frac{2\pi}{3} = \left(2n + \frac{2}{3}\right)\pi, \left(2n\frac{-2}{3}\right)\pi$$

$$y = \left\{\frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \ldots\right\}$$

Z : $g(x) = 0$

$\cos(2\pi \sin x) = 0$

$2\pi \sin x = (2n+1)\pi/2 \qquad n \in I$

$$\sin x = \frac{(2n+1)}{4}$$

$$\sin x = \frac{1}{4}, \frac{-1}{4}, \frac{3}{4}, \frac{-3}{4}$$

$x = n\pi \pm (-1)^n \sin^{-1}(1/4)$

or $x = n\pi \pm (-1)^n \sin^{-1}(3/4)$

W : $g'(x) = 0$

$-\sin(2\pi \sin x).\ 2\pi \cos x = 0$

$\sin(2\pi \sin x) = 0$ or $\cos x = 0$

$2\pi \sin x = n\pi$ $\qquad x = (2n+1)\frac{\pi}{2}$

$\sin x = n/2$

$\therefore \sin x = 0, ½, \frac{-1}{2}, 1, -1$

$\therefore x = n\pi$

or $x = n\pi \pm (-1)^n \frac{\pi}{6}$

or $x = 2n\pi \pm \frac{\pi}{2}$

5. Let b be a nonzero real number. Suppose $f: R \to R$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$.

for all $x \in R$, then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

(A) If $b > 0$, then f is an increasing function (B) If $b < 0$, then f is an decreasing function

(C) $f(x) f(-x) = 1$ for all $x \in R$ (D) $f(x) - f(-x) = 0$ for all $x \in R$

माना कि b एक शून्येतर वास्तविक संख्या है। मान लिजिये कि $f: R \to R$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 1$ है।

यदि f का अवकलज f' समीकरण $f'(x) = \frac{f(x)}{b^2 + x^2}$ को सभी $x \in R$ के लिये संतुष्ट करता है, तब निम्न में से कौन सा (से) कथन सही है (हैं)? **[JEE(Adv.)2020]**

(A) यदि $b > 0$ है, तब f एक वर्धमान फलन है (B) यदि $b < 0$ है, तब f एक ह्रासमान फलन है

(C) सभी $x \in R$ के लिए $f(x) f(-x) = 1$ है (D) सभी $x \in R$ के लिए $f(x) - f(-x) = 0$ है

Ans. AC

Sol. f is non zero

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{x^2 + b^2} dx$$

$$ln(f(x)) = \frac{1}{b} \tan^{-1} \frac{x}{b} + c$$

$$0 = 0 + c \quad \{\because f(0) = 1\}$$

$$f(x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}}$$

$$f(-x) = e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}}$$

(A) $f(x)$ is positive function for all $x \in R$

$$f'(x) = \frac{f(x)}{x^2 + b^2}$$

$f'(x)$ is also positive function

so $f(x)$ is increasing function

(d) $f(x) - f(-x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}} - e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}} \neq 0$

for all $x \in R$

(c) $f(x) f(-x) = 1$

$e^0 = 1$

$1 = 1$

6. Let α be a positive real number. Let $f : R \to R$ and $g : (\alpha, \infty) \to R$ be the functions defined by $f(x) = \sin\left(\frac{\pi x}{12}\right)$ and $g(x) = \frac{2\log_e\left(\sqrt{x} - \sqrt{\alpha}\right)}{\log_e\left(e^{\sqrt{x}} - e^{\sqrt{\alpha}}\right)}$. Then the value of $\lim\limits_{x \to \alpha^+} f(g(x))$ is _____. **[JEE(Adv.)2022]**

माना कि α एक धनात्मक वास्तविक संख्या (positive real number) है। माना कि फलनों (functions) $f : R \to R$ एवं $g : (\alpha, \infty) \to R$ को $f(x) = \sin\left(\frac{\pi x}{12}\right)$ एवं $g(x) = \frac{2\log_e\left(\sqrt{x} - \sqrt{\alpha}\right)}{\log_e\left(e^{\sqrt{x}} - e^{\sqrt{\alpha}}\right)}$ द्वारा परिभाषित किया जाता है। तब $\lim\limits_{x \to \alpha^+} f(g(x))$ का मान _____ है। **[JEE(Adv.)2022]**

Ans. 00.50

Sol. $\lim\limits_{x \to \alpha^+} g(x) = \lim\limits_{x \to \alpha^+} \frac{\frac{2}{\sqrt{x} - \sqrt{\alpha}}\left(\frac{1}{2\sqrt{x}}\right)}{\frac{1}{e^{\sqrt{x}} - e^{\sqrt{\alpha}}}\left(\frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}}\right)}$ **(Using L'Hospital Rule)**

$= \lim\limits_{x \to \alpha^+} \frac{e^{\sqrt{x}} - e^{\sqrt{\alpha}}}{\sqrt{x} - \sqrt{\alpha}} \cdot \frac{1}{e^{\sqrt{x}}} \cdot 2$

$= \lim\limits_{x \to \alpha^+} \frac{e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{x}}} \cdot \frac{2}{e^{\sqrt{x}}} = 2$

$\lim\limits_{x \to \alpha^+} f(g(x)) = f\left(\lim\limits_{x \to \alpha^+} g(x)\right) = \sin\frac{\pi}{6} = \frac{1}{2} = 00.50$

7. Let R denote the set of all real numbers. Let $f : R \to R$ and $g : R \to (0, 4)$ be functions defined by $f(x) = \log_e(x^2 + 2x + 4)$, and $g(x) = \frac{4}{1 + e^{-2x}}$. Define the composite function $f \circ g^{-1}$ by $(f \circ g^{-1})(x) = f(g^{-1}(x))$, where g^{-1} is the inverse of the function g.

Then the value of the derivative of the composite function $f \circ g^{-1}$ at $x = 2$ is _____. **[JEE(Adv.)2025]**

मान लीजिए कि **R** सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है। मान लीजिए कि फलन (function) $f : R \to R$ और $g : R \to (0, 4)$, $f(x) = \log_e(x^2 + 2x + 4)$, और $g(x) = \frac{4}{1 + e^{-2x}}$ द्वारा परिभाषित हैं। संयुक्त फलन (composite

function) $f \circ g^{-1}$ by $(f \circ g^{-1})(x) = f(g^{-1}(x))$ द्वारा परिभाषित कीजिए, जहाँ g^{-1}, फलन g का प्रतिलोम (inverse) है।

तब x = 2 पर संयुक्त फलन $f \circ g^{-1}$ के अवकलज (derivative) का मान _____ है। **[JEE(Adv.)2025]**

Ans. 0.25

Sol. Let $g^{-1}(x) = h(x)$

$\Rightarrow u(x) = f(h(x)) \quad \rightarrow u'(x) = f'(h(x)).h'(x)$

$u'(2) = f'(h(2))\, h'(2)$

$$h(2) = \alpha \Rightarrow g(\alpha) = 2 \Rightarrow \frac{4}{1+e^{-2\alpha}} = 2 \Rightarrow \alpha = 0$$

$u'(2) = f'(0)\, h'(2)$

$$f'(x) = \frac{1}{x^2+2x+4}(2x+2) \Rightarrow f'(0) = \frac{1}{2}$$

and $h(g(x)) = x$

$\Rightarrow h'(g(x)).g'(x) = 1$ Put $\alpha = 0$

$\Rightarrow h'(g(0))\, g'(0) = 1$

$\Rightarrow h'(2)\, g'(0) = 1$

$g(x) = 4(1 + e^{-2x})^{-1} \quad \Rightarrow g'(x) = -4(1 + e^{-2x})^{-2}\,(e^{-2x})\,(-2)$

$g'(0) = 8(2)^{-2}\,(1) = 2$

$$= u'(2) = \frac{1}{2}\cdot\frac{1}{2} = 1/4 = 0.25$$

8. Let R denote the set of all real numbers. Consider the polynomial function $f : R \to R$ defined by $f(x) = \frac{d^{10}}{dx^{10}}((x^2-1)^{10})$, for all $x \in \mathbb{R}$. Here $\frac{d^{10}}{dx^{10}}((x^2-1)^{10})$ is the 10th order derivative of the function $(x^2-1)^{10}$. Then which of the following statements is (are) **TRUE**: **[JEE(Adv.)2026]**

(A) The coefficient of x^8 in the polynomial f(x) is $(-10)\left(\frac{18!}{8!}\right)$

(B) The value of f(1) + f(–1) is equal to $10!2^{11}$

(C) The degree of the polynomial f(x) is 10

(D) The constant term of the polynomial f(x) is $-\left(\frac{10!}{5!}\right)$

मान लीजिए कि R सभी वास्तविक संख्याओं के समुच्चय को दर्शाता है। बहुपदीय फलन $f: R \to R$ जिसे $f(x) = \frac{d^{10}}{dx^{10}}((x^2-1)^{10})$,

सभी $x \in R$ के लिए, द्वारा परिभाषित किया गया है, पर विचार कीजिए। यहाँ $\frac{d^{10}}{dx^{10}}((x^2-1)^{10})$ फलन $(x^2-1)^{10}$ का 10 वीं कोटि

का अवकलज (10th order derivative) है। तब निम्नलिखित कथनों में से कौन–सा (से) **सत्य** है (हैं) : **[JEE(Adv.)2026]**

(A) बहुपद f(x) में x^8 का गुणांक $(-10)\left(\frac{18!}{8!}\right)$ है

(B) $f(1) + f(-1)$ का मान $10!2^{11}$ के बराबर है

(C) बहुपद f(x) की घात 10 है

(D) बहुपद f(x) का अचर पद $-\left(\frac{10!}{5!}\right)$ है

Ans. ABC

Sol. Let $g(x) = (x^2-1)^{10}$

$g(x) = x^{20} - {}^{10}C_1\, x^{18} + {}^{10}C_2\, x^{16} + ... - {}^{10}C_5\, x^{10} + {}^{10}C_6\, x^8 ..... + 1$

$$f(x) = \frac{20!}{10!}x^{10} - {}^{10}C_1\frac{18!}{8!}x^8 + {}^{10}C_2\frac{16!}{8!}x^6 ..... {}^{10}C_5 \cdot 10!$$

Coefficient of $x^8 = -10\cdot\frac{18!}{8!}$

Degree = 10

Constant term $= -{}^{10}C_5 \cdot 10! = -\left(\frac{10!}{5!}\right)^2$

$g(x) = (x^2-1)^{10}$

$g'(x) = 10(x^2-1)^9 \cdot 2x \Rightarrow g'(1) = 0$

$g''(x) = 2\cdot 10(x^2-1)^9 + 2^2\cdot 10\cdot 9(x^2-1)^8\cdot x \Rightarrow g''(1)$

For f(1) or f(–1) all terms becomes 0 except the last term.

$f(1) = f(-1) = 2^{10}\cdot 10!$

$f(1) + f(-1) = 2^{11}\cdot 10!$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $y = (x + \sqrt{1+x^2})^n$ then $(1+x^2)y_2 + xy_1 =$

(A) ny^2 (B) n^2y (C) n^2y^2 (D) None of these

यदि $y = (x + \sqrt{1+x^2})^n$ है, तो $(1+x^2)y_2 + xy_1 =$

(A) ny^2 (B) n^2y (C) n^2y^2 (D) इनमें से कोई नहीं

Ans. B

Sol. $y = \left(x + \sqrt{1+x^2} \right)^n$

$$y_1 = n\left(x + \sqrt{1+x^2} \right)^{n-1} \left(1 + \frac{2x}{2\sqrt{1+x^2}} \right)$$

$$y_1 = \frac{n\left(x + \sqrt{1+x^2} \right)^n}{\sqrt{1+x^2}}$$

$$y_2 = \frac{n^2\left(x + \sqrt{1+x^2} \right)^{n-1} \left(1 + \frac{x}{\sqrt{1+x^2}} \right)\sqrt{1+x^2} - n\left(x + \sqrt{1+x^2} \right)^n \frac{2x}{2\sqrt{1+x^2}}}{\left(1+x^2\right)}$$

$$\left(1+x^2\right)y_2 = n^2\left(x + \sqrt{1+x^2} \right)^n - \frac{n\left(x + \sqrt{1+x^2} \right)^n}{\sqrt{1+x^2}} \cdot x$$

$(1 + x^2)y_2 = n^2y - y_1x$

$(1 + x^2)y_2 + xy_1 = n^2y$

2. Let $u(x)$ and $v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)} \right)' = q$, then $\frac{p+q}{p-q}$ has the value equal to :

यदि $u(x)$ और $v(x)$ अवकलनीय फलन इस प्रकार है कि $\frac{u(x)}{v(x)} = 7$ यदि $\frac{u'(x)}{v'(x)} = p$ और $\left(\frac{u(x)}{v(x)}\right)' = q$, तब $\frac{p+q}{p-q}$ का मान बराबर :

(A) 1 (B) 0 (C) 7 (D) –7

Ans. A

Sol. $u(x) = 7v(x) \Rightarrow u'(x) = 7v'(x) \Rightarrow p = 7$ (given)

again $\frac{u(x)}{v(x)} = 7 \Rightarrow \left(\frac{u(x)}{v(x)}\right)' = 0 \Rightarrow q = 0$; now $\frac{p+q}{p-q} = \frac{7+0}{7-0} = 1$ **Ans.**

3. If $x = t^3 + t + 5$ and $y = \sin t$ then $\frac{d^2y}{dx^2} =$

यदि $x = t^3 + t + 5$ तथा $y = \sin t$ तब $\frac{d^2y}{dx^2} =$

(A) $-\frac{\left(3t^2+1\right)\sin t + 6t\cos t}{\left(3t^2+1\right)^3}$ (B) $\frac{\left(3t^2+1\right)\sin t + 6t\cos t}{\left(3t^2+1\right)^2}$

(C) $-\frac{\left(3t^2+1\right)\sin t + 6t\cos t}{\left(3t^2+1\right)^2}$ (D) $\frac{\cos t}{3t^2+1}$

Ans. A

Sol. $x = t^3 + t + 5$ $y = \sin t$

$\frac{dx}{dt} = 3t^2 + 1$ $\frac{dy}{dt} = \cos t$

$\frac{dy}{dx} = \frac{\cos t}{3t^2+1}$

$\frac{d^2y}{dx^2} = \frac{-\sin t\frac{dt}{dx}\left(3t^2+1\right) - \cos t\left(6t\frac{dt}{dx}\right)}{\left(3t^2+1\right)^2}$

$$= \frac{-\frac{\sin t}{3t^2+1}\left(3t^2+1\right)-\frac{6t\cos t}{1+3t^2}}{\left(3t^2+1\right)^2}$$

$$= -\frac{\sin t\left(3t^2+1\right)+6t\cos t}{\left(3t^2+1\right)^3}$$

4. If $f(x) = \frac{a+\sqrt{a^2-x^2}+x}{\sqrt{a^2-x^2}+a-x}$ where $a > 0$ and $x < a$, then $f'(0)$ has the value equal to :

यदि $f(x) = \frac{a+\sqrt{a^2-x^2}+x}{\sqrt{a^2-x^2}+a-x}$ जहां $a > 0$ और $x < a$, तब $f'(0)$ का मान होगा :

(A) $\sqrt{a}$ (B) a (C) $\frac{1}{\sqrt{a}}$ (D) $\frac{1}{a}$

Ans. D

Sol. $y = \frac{(a+x)+\sqrt{a-x}\cdot\sqrt{a+x}}{(a-x)+\sqrt{a-x}\cdot\sqrt{a+x}}$

$$y = \frac{\sqrt{a+x}\left(\sqrt{a+x}+\sqrt{a-x}\right)}{\sqrt{a-x}\left(\sqrt{a+x}+\sqrt{a-x}\right)} = \left(\frac{a+x}{a-x}\right)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2}\sqrt{\frac{a-x}{a+x}}\left(\frac{(a-x)+(a+x)}{(a-x)^2}\right) = \frac{1}{2}\frac{\sqrt{a-x}}{\sqrt{a+x}}\times\frac{2a}{(a-x)^2}; \quad \therefore \quad \left.\frac{dy}{dx}\right|_{x=0} = \frac{1}{a} \text{ Ans.}$$

5. Suppose that $f(0) = 0$ and $f'(0) = 2$, and let $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ is equal to :

माना कि $f(0) = 0$ और $f'(0) = 2$, तथा $g(x) = f(-x + f(f(x)))$. $g'(0)$ का मान बराबर :

(A) 0 (B) 1 (C) 6 (D) 8

Ans. C

Sol. $g(x) = f\left(-x+f\left(f(x)\right)\right); \quad f(0) = 0; \quad f'(0) = 2$

$$g'(x) = f'\left(-x+f\left(f(x)\right)\right)\cdot\left[-1+f'\left(f(x)\right)\cdot f'(x)\right]$$

$$g'(0) = f'\left(f(0)\right)\cdot\left[-1+f'(0)\cdot f'(0)\right]$$

$= f'(0)\,[-1 + (2)(2)]$

$= (2)(3) = 6$ Ans.

6. If ln$(3\sin x - 4\cos x + 7 + 5y) = (\sin^2 x)y$, then $y'(\pi)$ is equal to :

यदि ln$(3\sin x - 4\cos x + 7 + 5y) = (\sin^2 x)y$ हो, तब $y'(\pi)$ बराबर होगा :

(A) $\frac{5}{3}$ (B) 0 (C) $\frac{3}{5}$ (D) $\frac{-3}{5}$

Ans. C

Sol. We have ln $(3\sin x - 4\cos x + 7 + 5y) = \left(\sin^2 x\right)y$ ..... (1)

Put $x = \pi$ in equation (1), we get ln $(11 + 5y) = 0 \Rightarrow 11 + 5y = 1 \Rightarrow 5y = -10$

$\therefore$ $y = -2$. So, $(\pi, -2)$ lies on the given curve.

Now on differentiating both the sides of equation (1) w.r.t. x, we get

$$\frac{1}{(3\sin x - 4\cos x + 7 + 5y)} \times \left(3\cos x + 4\sin x + 5\frac{dy}{dx}\right) = \left(\sin^2 x\right)\frac{dy}{dx} + (\sin 2x)y$$

As $(\pi, -2)$ satisfy it, we get $1 \times \left(-3 + 0 + 5\frac{dy}{dx}\right) = 0$

$$\therefore \left.\frac{dy}{dx}\right]_{(\pi,-2)} = \frac{3}{5} \Rightarrow y'(\pi) = \frac{3}{5}.$$ Ans.

7. If $f''(x) = -f(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and given that $F(5) = 5$, then $F(10)$ is equal to :

[JEE(Adv.)-2006]

यदि $f''(x) = -f(x)$ एवं $g(x) = f'(x)$ तथा $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ एवं $F(5) = 5$ हो तब $F(10)$ का मान है :

[JEE(Adv.)-2006]

(A) 5 (B) 10 (C) 0 (D) 15

Ans. A

Sol. $\because$ $g(x) = f''(x)$ ..... (i)

$\Rightarrow$ $g'(x) = f''(x) = -f(x)$ ..... (ii)

$$F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$$

$\Rightarrow \quad F'(x) = 2f(x/2) \cdot f'(x/2) \cdot \frac{1}{2} + 2\, g(x/2) \cdot g'(x/2) \cdot \frac{1}{2}$

$\Rightarrow \quad F'(x) = f\left(\frac{x}{2}\right) f'\left(\frac{x}{2}\right) + g\left(\frac{x}{2}\right) g'\left(\frac{x}{2}\right)$

$\Rightarrow \quad F'(x) = f\left(\frac{x}{2}\right) \cdot f'\left(\frac{x}{2}\right) - f'\left(\frac{x}{2}\right) f\left(\frac{x}{2}\right)$ {Using (i) & (ii)}

$\Rightarrow \quad F'(x) = 0$

$\Rightarrow \quad F(x) = \text{constant} \Rightarrow \quad F(5) = F(10) = 5$

8. If $y = (\tan^{-1}x)^2$ then $(x^2 + 1)^2 D^2y + 2x(x^2 + 1) Dy$ has the value equal to :

(A) 0 (B) 2 (C) 4 (D) None of these

यदि $y = (\tan^{-1}x)^2$ तब $(x^2 + 1)^2 D^2y + 2x(x^2 + 1) Dy$ बराबर :

(A) 0 (B) 2 (C) 4 (D) इनमें से कोई नहीं

Ans. B

Sol. $(1 + x^2)\, y_1 = 2 \tan^{-1}x$

$(1 + x^2)^2 \cdot y_1^2 = 4(\tan^{-1}x)^2 = 4y$

$(1 + x^2)^2 \cdot 2y_1y_2 + y_1^2 \cdot 2(1 + x^2) \cdot 2x = 4y_1$

$(1 + x^2)^2\, y_2 + 2x(1 + x^2)y_1 = 2$ Ans.]

Alternatively: $y = (\tan^{-1}x)^2$

$$y_1 = \frac{2\tan^{-1}x}{1+x^2}; \qquad (1 + x^2)y_1 = 2 \tan^{-1}x; \quad (1 + x^2)y_2 + 2xy_1 = \frac{2}{1+x^2}$$

$(1 + x^2)^2y_2 + 2x(1 + x^2)y_1 = 2$

Hence $\boxed{k = 2}$ Ans.

9. Let $g(x)$ be an antiderivative for $f(x)$. Then $ln(1+(g(x))^2)$ is an antiderivative for :

(A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) None of these

यदि $g(x)$, $f(x)$ का समाकलन है तब $ln(1+(g(x))^2)$ समाकलन निम्न में से किसका समाकलन होगा :

(A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given $\int f(x)\,dx = g(x) \quad \Rightarrow \quad g'(x) = f(x)$

now $\frac{d}{dx}\left(ln(1+g^2(x)\right) = \frac{2g(x)g'(x)}{1+g^2(x)} = \frac{2f(x)g(x)}{1+g^2(x)}$

10. If $y = (A + Bx)\,e^{mx} + (m-1)^{-2}\,e^x$ then $\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ is equal to :

यदि $y = (A + Bx)\,e^{mx} + (m-1)^{-2}\,e^x$ है, तो $\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ बराबर होगा :

(A) e^x (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$

Ans. A

Sol. $y = (A + Bx)e^{mx} + (m-1)^{-2} \cdot e^x$

$y \cdot e^{-mx} = (A + Bx) + (m-1)^{-2} \cdot e^{(1-m)x}$

$e^{-mx} \cdot y_1 - mye^{-mx} = B - (m-1)^{-1} \cdot e^{-(m-1)x}$

$e^{-mx} \cdot y_2 - y_1\,e^{-mx} \cdot m - m[e^{-mx} \cdot y_1 - y\,e^{-mx} \cdot m] = e^{-(m-1)x}$

$e^{-mx} \cdot y_1 - m_2y_1\,e^{-mx} + my \cdot e^{-mx} = e^{-(m-1)x}$

$^{y}{}_2 - 2my_1 + my = e^x$

11. Let $f(x) = x^n$, n being a non-negative integer. The number of values of n for which $f'(p+q) = f'(p) + f'(q)$ is valid for all $p, q > 0$ is :

(A) 0 (B) 1 (C) 2 (D) None of these

माना $f(x) = x^n$, n अऋणात्मक पूर्णांक है तब n के मानों की संख्या जिसके लिए $f'(p+q) = f'(p) + f'(q)$ सभी $p, q > 0$ के लिए सत्य है :

(A) 0 (B) 1 (C) 2 (D) इनमें से कोई नहीं

Ans. C

Sol.

12. The ends A and B of a rod of length $\sqrt{5}$ are sliding along the curve $y = 2x^2$. Let x_A and x_B be the x-coordinate of the ends. At the moment when A is at (0, 0) and B is at (1, 2) the derivative $\frac{dx_B}{dx_A}$ has the value equal to :

एक $\sqrt{5}$ लम्बाई की छड़ के अन्तिम सिरे A व B वक्र $y = 2x^2$ के अनुदिश फिसल रहे हैं यदि x_A व x_B सिरों के x-निर्देशांक हो तो अवकलन $\frac{dx_B}{dx_A}$ का मान क्या होगा जबकि A (0, 0) व B (1, 2) पर है :

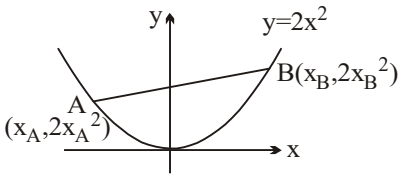
(A) 1/3 (B) 1/5 (C) 1/8 (D) 1/9

Ans. D

Sol. We have $y = 2x^2$

$(AB)^2 = (x_B - x_A)^2 + (2x_B^2 - 2x_A^2)^2 = 5$

or $(x_B - x_A)^2 + 4(x_B^2 - x_A^2)^2 = 5$



differentiating w.r.t. x_A and denoting $\frac{dx_B}{dx_A} = D$

$2(x_B - x_A)(D - 1) + 8(x_B^2 - x_A^2)(2x_BD - 2x_A) = 0$

put $x_A = 0$; $x_B = 1$

$2(1 - 0)(D - 1) + 8(1 - 0)(2D - 0) = 0$

$2D - 2 + 16D = 0 \quad \Rightarrow \quad D = 1/9$ Ans.

13. The derivative of the function, $f(x)=\cos^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x - 3\sin x)\right\} + \sin^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x + 3\sin x)\right\}$ w.r.t. $\sqrt{1 + x^2}$ at $x = \frac{3}{4}$ is :

दिये गये फलन $f(x)=\cos^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x - 3\sin x)\right\} + \sin^{-1}\left\{\frac{1}{\sqrt{13}}(2\cos x + 3\sin x)\right\}$ का अवकलन $\sqrt{1 + x^2}$ के सापेक्ष $x = \frac{3}{4}$ पर होगा :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{10}{3}$ (D) 0

Ans. C

Sol. Put $\cos\phi = \frac{2}{\sqrt{13}}$; $\sin\phi = \frac{3}{\sqrt{13}}$; $\tan\phi = \frac{3}{2}$

$y = \cos^{-1}\{\cos(x+\phi)\} + \sin^{-1}\{\cos(x-\phi)\}$

$= \cos^{-1}\{\cos(x+\phi) + \frac{\pi}{2} - \cos^{-1}\{\cos(\phi - x)\}$ (think !)

$= x + \phi + \frac{\pi}{2} - \phi + x$

$y = 2x + \frac{\pi}{2}$; $z = \sqrt{1+x^2}$

now compute $\frac{dy}{dz}$

14. Let $f(x) = \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right)$ then the primitive of $f(x)$ w.r.t. x is (where C is an arbitrary constant) :

यदि $f(x) = \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right)$ तब $f(x)$ का x के सापेक्ष समाकल है (जहां C स्वेच्छ अचर है) :

(A) $-\frac{3\sin 3x}{4} + C$ (B) $-\frac{3\cos 3x}{4} + C$ (C) $\frac{\sin 3x}{4} + C$ (D) $\frac{\cos 3x}{4} + C$

Ans. D

Sol. Note that $\sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right) = 0$

$\Rightarrow \quad \sin^3 x + \sin^3\left(x + \frac{2\pi}{3}\right) + \sin^3\left(x + \frac{4\pi}{3}\right) = -\frac{3}{4}\sin 3x$ $\quad (a+b+c = 0 \Rightarrow a^3+b^3+c^3 = 3abc)$

$\therefore \quad -\frac{3}{4}\int \sin 3x\, dx = \frac{\cos 3x}{4} + C$ Ans.

15. People living at Mars, instead of the usual definition of derivative D f(x), define a new kind of derivative, Df(x) by the formula $Df(x) = \underset{h\to 0}{\text{Limit}} \frac{f^2(x+h) - f^2(x)}{h}$ where $f^2(x)$ means $[f(x)]^2$. If $f(x) = x\,\ell nx$ then $D^*f(x)\big|_{x=e}$ has the value :

यदि अवकलन की सामान्य परिभाषा D f(x) के स्थान पर अवकलन की नई परिभाषा Df(x) सूत्र $Df(x) = \underset{h\to 0}{\text{Limit}} \frac{f^2(x+h)-f^2(x)}{h}$

से दी जाती है। जहां $f^2(x)$ का अर्थ $[f(x)]^2$ है। यदि $f(x) = x\,\ell nx$ तब $D^*f(x)\big|_{x=e}$ पर होगा :

(A) e (B) 2e (C) 4e (D) 8e

Ans. C

Sol. $D^*f(x) = 2f(x).f'(x)$

$D^*(x\,\ell nx) = 2x\,\ell nx\,(1+\ell nx)$

16. Let P(x) be the polynomial $x^3 + ax^2 + bx + c$, where $a, b, c \in R$. If $P(-3) = P(+2) = 0$ and $P'(-3) < 0$, which of the following is a possible value of 'c'?

माना $P(x) = x^3 + ax^2 + bx + c$, जहां $a, b, c \in R$. यदि $P(-3) = P(+2) = 0$ और $P'(-3) < 0$, तब 'c' के सम्भव मान है :

(A) – 27 (B) – 18 (C) – 6 (D) – 3

Ans. A

Sol. $P(x) = x^3 + ax^2 + bx + c$

$P'(x) = 3x^2 + 2ax + b$

$P(-3) = 0 \quad \Rightarrow \quad -27 + 9a - 3b + c = 0 \quad ....(1)$

$P(2) = 0 \quad \Rightarrow \quad 8 + 4a + 2b + c = 0 \quad ....(2)$

hence (1) – (2) gives

$5a - 5b - 35 = 0$

$a - b = 7 \quad ....(3)$

$P'(-3) = 27 - 6a + b < 0 \quad \Rightarrow \quad 27 - 6(a-b) - 5b < 0 \Rightarrow \quad 27 - 42 - 5b < 0$

$\Rightarrow \quad -15 - 5b < 0 \Rightarrow \quad 3 + b > 0 \quad \Rightarrow \quad b > -3$, hence $a - 7 = b > -3$

$a - 7 > -3 \quad$ [from (3)]

$a > 4 \quad ....(4)$

$\therefore$ from (2)

$8 + 16 - 6 + c_{max} = 0$

$c_{max} = -18 \quad \Rightarrow \quad c < -18$

Aliter: $P(x) \equiv (x+3)(x-2)(x-\alpha)$

where α is the third root i.e. $P(x) = x^3 + (1-\alpha)x^2 + (-6-\alpha)x + 6\alpha$ now proceed.

17. f is a differentiable function such that $x = f(t^2)$, $y = f(t^3)$ and $f'(1) \neq 0$ if $\left(\frac{d^2y}{dx^2}\right)_{t=1} =$

f एक अवकलनीय फलन इस प्रकार है कि $x = f(t^2)$, $y = f(t^3)$ तथा $f'(1) \neq 0$ यदि $\left(\frac{d^2y}{dx^2}\right)_{t=1} =$

(A) $\frac{3}{4}\left(\frac{f''(1)+f'(1)}{(f'(1))^2}\right)$ (B) $\frac{3}{4}\left(\frac{f'(1).f''(1)-f''(1)}{(f'(1))^2}\right)$

(C) $\frac{4}{3}\frac{f''(1)}{(f'(1))^2}$ (D) $\frac{4}{3}\left(\frac{f'(1)f''(1)-f''(1)}{(f'(1))^2}\right)$

Ans. A

Sol. $x = f(t^2)$, $y = f(t^3)$

$$\frac{dx}{dt} = 2t\ f'\left(t^2\right) \qquad \frac{dy}{dt} = 3t^2\ f'\left(t^3\right)$$

$$\frac{dy}{dx} = \frac{3t^2 f'\left(t^3\right)}{2tf'\left(t^2\right)}$$

$$= \frac{\left[3\frac{dt}{dx}f'\left(t^3\right)+3t\ f''\left(t^3\right)\left(3t^2\right)\frac{dt}{dx}\right]2f'\left(t^2\right)-3t\ f'\left(t^3\right)\left[2f''\left(t^2\right)\times 2t\frac{dt}{dx}\right]}{\left[2f'\left(t^2\right)\right]^2}$$

$$\Rightarrow \frac{3\frac{dt}{dx}\left[2f'\left(t^2\right)f'\left(t^3\right)+3t^3f''\left(t^3\right)\times 2f'\left(t^2\right)-4t^2f'\left(t^3\right)f''\left(t^2\right)\right]}{\left[2f'\left(t^2\right)\right]^2}$$

$$= \frac{3}{2f'(1)}\frac{\left[2\left(f'(1)\right)\right]^2+6f'(1)f''(1)-4f'(1)f''(1)}{4\left[f'(1)\right]^2}$$

$$\Rightarrow \frac{3}{8f'(1)}\left[\frac{\left[2f'(1)\right]^2+2f'(1)f''(1)}{f'(1)^2}\right]$$

$$\Rightarrow \frac{3}{4}\left[\frac{f'(1)+f''(1)}{\left[f'(1)\right]^2}\right]$$

18. If P(x) is a polynomial of degree less than or equal to 2 and S is the set of all such polynomials so that P(1) = 1, P(0) = 0 and P'(x) > 0 $\forall\ x \in [0, 1]$, then :

यदि P(x) एक दो या दो से छोटी घात का बहुपद है एवं S उन सभी बहुपदों के समुच्चय है जिनके लिये P(1) = 1, P(0) = 0 तथा P'(x) > 0 $\forall$ x $\in$ [0, 1] हो तो :

(A) S = ϕ (B) S = $\{(1-a)x^2 + ax,\ 0 < a < 2\}$

(C) S=$\{(1-a)x^2 + ax, a \in (0, \infty)\}$ (D) S = $\{(1-a)x^2 + ax,\ 0 < a < 1\}$

Ans. B

Sol. Let $P(x) = bx^2 + ax + c \qquad b \neq 0$

$P(0) = 0 \quad \Rightarrow \quad c = 0$

$P(1) = 1 \quad \Rightarrow \quad a + b = 1$

$\Rightarrow P(x) = (1-a)x^2 + ax$

$\Rightarrow P'(x) = 2(1-a)x + a$

Now $P'(x) > 0$ for $x \in [0, 1]$

$P'(0) > 0$ & $P'(1) > 0$

$\Rightarrow \quad a > 0 \quad \& \quad 2(1-a) + a > 0$

$\Rightarrow \quad 0 < a < 2$

$\therefore \quad S = \{(1-a)x^2 + ax;\ 0 < a < 2\}$

19. If $f(x)$ is a continuous and differentiable function and $f(1/n) = 0$, $\forall$ n $\geq$ 1 and n $\in$ I, then :

(A) f(x) = 0, x $\in$ (0, 1] (B) f(0) = 0, f ' (0) = 0

(C) f '(x) = 0 = f "(x), x $\in$ (0, 1] (D) f(0) = 0 and f '(0) need not to be zero

यदि $f(x)$ एक सतत् एवं अवकलनीय फलन हो एवं $f(1/n) = 0$, $\forall$ n $\geq$ 1 तथा n $\in$ I तो :

(A) f(x) = 0, x $\in$ (0, 1] (B) f(0) = 0, f ' (0) = 0

(C) f '(x) = 0 = f "(x), x $\in$ (0, 1] (D) f(0) = 0 तथा f '(0) शून्य होना आवश्यक नही है

Ans. B

Sol. $f\left(\frac{1}{n}\right) = 0 \qquad \forall n \geq 1, n \in \text{integer}$

$f\left(\frac{1}{1}\right) = f\left(\frac{1}{2}\right) = f\left(\frac{1}{3}\right) = f\left(\frac{1}{4}\right) = \ldots\ldots = 0$

$\lim_{n\to\infty} f\left(\frac{1}{n}\right) = 0$

$f(0) = 0$

$f'(0) = \lim_{n\to 0} \frac{f(n) - f(0)}{n}$

$$= \lim_{\frac{1}{n}\to\infty} \frac{f\left(\frac{1}{n}\right)-0}{\frac{1}{n}}$$

$f'(0) = 0$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

20. For the function $y = f(x) = (x^2 + bx + c)e^x$, which of the following holds?

(A) if $f(x) > 0$ for all real $x \not\Rightarrow f'(x) > 0$ (B) if $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$

(C) if $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$ (D) if $f'(x) > 0$ for all real $x \not\Rightarrow f(x) > 0$

फलन $y = f(x) = (x^2 + bx + c)e^x$, के लिए निम्न में से सत्य है?

(A) यदि $f(x) > 0$ for all real $x \not\Rightarrow f'(x) > 0$ (B) यदि $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$

(C) यदि $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$ (D) यदि $f'(x) > 0$ for all real $x \not\Rightarrow f(x) > 0$

Ans. AC

Sol. $f(x) = (x^2 + bx + c)e^x$

$\therefore$ $f'(x) = (x^2 + (b + 2)x + (b + c))e^x$

$f(x) > 0$ iff $D = b^2 - 4c < 0$

now $f'(x) > 0$ iff $D' = (b + 2)^2 - 4(b + c) = D + 4 < 0$

Thus for $f'(x) > 0$ $D + 4 < 0$ holds. $\Rightarrow$ $D < 0$ $\Rightarrow$ $f(x) > 0$

Note that the converse need not be true, e.g. $b = c = 1$, $f(x) > 0$ but $f'(-1) = 0$

21. If $y = \tan x \tan 2x \tan 3x$ then $\frac{dy}{dx}$ has the value equal to :

यदि $y = \tan x \tan 2x \tan 3x$ तब $\frac{dy}{dx}$ बराबर :

(A) $3\sec^2 3x \tan x \tan 2x + \sec^2 x \tan 2x \tan 3x + 2\sec^2 2x \tan 3x \tan x$

(B) $2y(\operatorname{cosec} 2x + 2\operatorname{cosec} 4x + 3\operatorname{cosec} 6x)$

(C) $3\sec^2 3x - 2\sec^2 2x - \sec^2 x$

(D) $\sec^2 x + 2\sec^2 2x + 3\sec^2 3x$

Ans. ABC

Sol. $y = \tan x \tan 2x \tan 3x$

$$\frac{dy}{dx} = \sec^2 x \tan 2x \tan 3x + 2\tan x \sec^2 2x \tan 3x + 3\tan x \tan 2x \sec^2 3x$$

$$\frac{dy}{dx} = y\left(\frac{\sec^2 x}{\tan x} + \frac{2\sec^2 2x}{\tan 2x} + \frac{3\sec^2 3x}{\tan 3x}\right)$$

$$= y\left(\frac{2}{\sin 2x} + \frac{2\times 2}{\sin 4x} + \frac{2\times 3}{\sin 6x}\right)$$

$= 2y\ (\text{cosec } 2x + 2 \text{ cosec } 4x + 3 \text{ cosec } 6x)$

$y = \tan x \tan 2x \tan 3x$

$\because\ \tan x \tan 2x \tan 3x = \tan 3x - \tan 2x - \tan x$

$y = \tan 3x - \tan 2x - \tan x$

$$\frac{dy}{dx} = 3\sec^2 3x - 2\sec^2 2x - \sec^2 x$$

22. Let $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$ and α, β, γ satisfy $\alpha < \beta < \gamma$ are the roots of $f'(x) = 0$ then which of the following is/are correct ([.] denotes greatest integer function)?

माना $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$ तथा $f'(x) = 0$ के मूल α, β, γ है, जो कि $\alpha < \beta < \gamma$ को सन्तुष्ट करता है तो निम्न में से कौनसा/कौनसे कथन सत्य है? ([.] महत्तम पूर्णांक फलन को दर्शाता है)

(A) $[\alpha] = -2$ (B) $[\beta] = -1$ (C) $[\beta] = 0$ (D) $[\alpha] = 1$

Ans. AC

Sol. $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$

$= (x^2 + 2 - 3x)(x^2 + 2 + 3x)$

$= (x^2 + 2)^2 - 9x^2$

$= x^4 + 4 + 4x^2 - 9x^2$

$= x^4 - 5x^2 + 4$

$f'(x) = 4x^3 - 10x = 0$

$2x(2x^2 - 5) = 0$

$$x = 0, \pm\sqrt{\frac{5}{2}}$$

$$\alpha = -\sqrt{\frac{5}{2}}, \qquad \beta = 0, \qquad \gamma = \sqrt{\frac{5}{2}}$$

$$[\alpha] = -2 \qquad [\beta] = 0 \qquad [\gamma] = 1$$

23. If $y = e^{x\sin(x^3)} + (\tan x)^x$ then $\frac{dy}{dx}$ may be equal to :

यदि $y = e^{x\sin(x^3)} + (\tan x)^x$ है तो $\frac{dy}{dx}$ का मान हो सकता है :

(A) $e^{x\sin(x^3)}[3x^3\cos(x^3)+\sin(x^3)] + (\tan x)^x[l\,n\tan x + 2x\cos ec2x]$

(B) $e^{x\sin(x^3)}[x^3\cos(x^3)+\sin(x^3)] + (\tan x)^x[l\,n\tan x + 2x\cos ec2x]$

(C) $e^{x\sin(x^3)}[x^3\sin(x^3)+\cos(x^3)] + (\tan x)^x[l\,n\tan x + 2\cos ec2x]$

(D) $e^{x\sin(x^3)}[3x^3\cos(x^3)+\sin(x^3)] + (\tan x)^x\left[l\,n\tan x + \frac{x\sec^2 x}{\tan x}\right]$

Ans. AD

Sol. $y = \underbrace{e^{x\sin x^3}}_{y_1} + \underbrace{(\tan x)^x}_{y_2}$

$y_1 = e^{x\sin x^3}$

$\log y_1 = x\sin x^3$

$\frac{1}{y_1}\frac{dy_1}{dx} = \sin x^3 + 3x^3\cos x^3$

$\frac{dy_1}{dx} = e^{x\sin x^3}\left[\sin x^3 + 3x^3\cos x^3\right]$

$y_2 = (\tan x)^x$

$\log y_2 = x\log\tan x$

$\frac{1}{y_2}\frac{dy_2}{dx} = \log(\tan x) + \frac{x}{\tan x}\times\sec^2 x$

$\frac{dy_2}{dx} = (\tan x)^x\left[\log(\tan x) + 2x\ \text{cosec}\ 2x\right]$

$\frac{dy}{dx} = e^{x\sin(x)^3}\left[\sin x^3 + 3x^3\cos x^3\right] + (\tan x)^x\left[\log(\tan x) + 2x\ \text{cosec}\ 2x\right]$

24. Let $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + \ldots\ldots + (1-x)(1-x^2)\ldots\ldots(1-x^{n-1})x^n$; $(n \geq 4)$ then :

माना $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + \ldots\ldots + (1-x)(1-x^2)\ldots\ldots(1-x^{n-1})x^n$; $(n \geq 4)$ है तो :

(A) $f(x) = -\prod_{r=1}^{n}(1-x^r)$ (B) $f(x) = 1-\prod_{r=1}^{n}(1-x^r)$

(C) $f'(x) = (1-f(x))\left(\sum_{r=1}^{n}\frac{r\,x^{r-1}}{(1-x^r)}\right)$ (D) $f'(x) = f(x)\left(\sum_{r=1}^{n}\frac{r\,x^{r-1}}{(1-x^r)}\right)$

Ans. BC

Sol. $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + \ldots\ldots + (1-x)(1-x^2)\ldots\ldots(1-x^{n-1})x^n$

$1 - f(x) = 1 - x - (1-x)x^2 - (1-x)(1-x^2)x^3 \ldots\ldots - (1-x)(1-x^2)\ldots\ldots(1-x^{n-1})x^n$

$1 - f(x) = (1-x)\ [(1-x^2) - (1-x^2)x^3 \ldots\ldots - (1-x^2)(1-x^3)\ldots\ldots(1-x^{n-1})x^2]$

$1 - f(x) = (1-x)(1-x^2)\ [(1-x^3) \ldots\ldots - (1-x^3)\ldots\ldots(1-x^{n-1})x^n]$

$1 - f(x) = (1-x)(1-x^2)(1-x^3)\ldots\ldots(1-x^n)$

$$f(x) = 1 - \prod_{r=1}^{n}(1-x^r)$$

$$\log(1-f(x)) = \log(1-x) + \log(1-x^2) + \log(1-x^3)\ldots\ldots + \log(1-x^n)$$

$$\frac{1}{1-f(x)}\left[-f'(x)\right] = \frac{-1}{1-x} + \frac{-2x}{1-x^2} + \frac{-3x^2}{1-x^3}\ldots\ldots + \frac{-nx^{n-1}}{1-x^n}$$

$$f'(x) = \left[1-f(x)\right]\left[\sum_{r=1}^{n}\frac{rx^{r-1}}{1-x^r}\right]$$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. If $\sqrt{x^2+y^2} = e^{\arcsin\frac{y}{\sqrt{x^2+y^2}}}$. Prove that $\frac{d^2y}{dx^2} = \frac{2(x^2+y^2)}{(x-y)^3}$, $x > 0$.

यदि $\sqrt{x^2+y^2} = e^{\arcsin\frac{y}{\sqrt{x^2+y^2}}}$. तो सिद्ध कीजिये $\frac{d^2y}{dx^2} = \frac{2(x^2+y^2)}{(x-y)^3}$, $x > 0$.

Ans.

Sol. $\sqrt{x^2+y^2} = e^{\sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)}$

$$\Rightarrow \frac{1}{2}\ell n\left(x^2+y^2\right) = \sin^{-1}\left(\frac{\frac{y}{x}}{\sqrt{1+\left(\frac{y}{x}\right)^2}}\right)$$

$$\Rightarrow \frac{1}{2}\ell n\left(x^2+y^2\right) = \tan^{-1}\left(\frac{y}{x}\right)$$

Differentiating with respect to 'x'.

$$\Rightarrow \frac{x+y\frac{dy}{dx}}{x^2+y^2} = \frac{x\frac{dy}{dx}-y}{x^2}\cdot\frac{x^2}{y^2+x^2}$$

$$\Rightarrow x+y\frac{dy}{dx} = x\frac{dy}{dx}-y$$

$$\Rightarrow (y-x)\frac{dy}{dx} = -(y+x)$$

$$\frac{dy}{dx} = \frac{y+x}{x-y}$$

Again differentiating with respect to x

$$\frac{d^2y}{dx^2}=\frac{(x-y)2-2x\left(1-\frac{dy}{dx}\right)}{(x-y)^2}$$

$$=\frac{2}{(x-y)^2}\left[X-y-x\left(\frac{X-y-x-y}{X-y}\right)\right]$$

$$=\frac{2}{(X-y)^3}\left[(X-y)^2+2xy\right]$$

$$=\frac{2\left(X^2+y^2\right)}{(X-y)^3}$$

2. Find a polynomial function $f(x)$ such that $f(2x)=f'(x)f''(x)$.

एक ऐसा बहुपद $f(x)$ बताइए जिसके लिये $f(2x)=f'(x)f''(x)$ है।

Ans. $\frac{4x^3}{9}$

Sol. Let $f(x)=a_0x^n+a_1x^{n-1}+...+a_n$

Given f(2x) = f′(x) f′′(x) _____(1)

↓ ↓ ↓

degree degree degree

n n – 1 n – 2

both side highest power

$n=n-1+n-2$

$n=3$

$\Rightarrow f(x)=a_0x^3+a_1x^2+a_2x+a_3$

$\Rightarrow$ from (2)

$8a_0x^3+4a_1x^2+2a_2x+a^3$

$=(3a_0x^2+2a_1x+a_2)(6a_0x+2a_1)$

comparing coefficients of equal power

$8a_0 = 18a_0^2 \Rightarrow a_0 = \frac{4}{9}$

$a_1 = 0 \qquad a_2 = 0 \qquad a_3 = 0$

$f(x) = \frac{4}{9}x^3$

3. If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ then $(x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = ky$, then find the value of 'k'.

यदि $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ तो $(x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = ky$ है, तो 'k' का मान ज्ञात कीजिए।

Ans. 25

Sol. $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$

Differentiating with respect to y

$$\Rightarrow 2\frac{dx}{dy} = \frac{1}{5}y^{\frac{1}{5}-1} - \frac{1}{5}y^{-\frac{1}{5}-1}$$

$$= \frac{1}{5y}\left(y^{\frac{1}{5}} - y^{-\frac{1}{5}}\right)$$

$$= \frac{1}{5y}\sqrt{\left(y^{\frac{1}{5}} + y^{-\frac{1}{5}}\right)^2 - 4}$$

$$\Rightarrow 10y\frac{dx}{dy} = \sqrt{4x^2-4}$$

$$\Rightarrow 5y = \sqrt{x^2-1}\frac{dy}{dx} \qquad \text{_____(1)}$$

differentiating with respect to x

$$5\frac{dy}{dx} = \sqrt{x^2-1}\frac{d^2y}{dx^2} + \frac{x}{\sqrt{x^2-1}}\frac{dy}{dx}$$

$$\left(x^2-1\right)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=5\sqrt{x^2-1}\frac{dy}{dx}=25y$$

$k = 25$

4. The function $f: R \to R$ satisfies $f(x^2) \cdot f''(x) = f'(x) \cdot f'(x^2)$ for all real x. Given that $f(1) = 1$ and $f'''(1) = 8$, compute the value of $f'(1) + f''(1)$.

फलन $f: R \to R$, सम्बन्ध $f(x^2) \cdot f''(x) = f'(x) \cdot f'(x^2)$ को सभी वास्तविक x के लिये संतुष्ट करता है। दिया गया है कि $f(1) = 1$ तथा $f'''(1) = 8$ है, तो $f'(1) + f''(1)$ का मान ज्ञात कीजिए।

Ans. 6

Sol. $f\left(x^2\right)\cdot f''(x) = f'(x)\cdot f'\left(x^2\right)$ _____(1)

put $x = 1$ $f(1) = 1$ $f'''(1) = 8$

$f(1)\, f''(1) = f'(1)^2$

$f'(1)^2 = f''(1)$ _____(2)

Differentiating with respect to x of (1)

$f(x^2)\, f'''(x) + f'(x^2)\cdot 2x\cdot f''(x) = f''(x)\cdot f'(x^2) + f'(x)\cdot 2x\cdot f''(x^2)$

Put $x = 1$

$f(1)\cdot f'''(1) + f'(1)\, 2\cdot f''(1) = f''(1)\cdot f'(1) + f'(1)\cdot 2\cdot f''(1)$

$8 = f''(1)\cdot f'(1)$ _____(3)

from (1) and (3) $f'(1) = 2$

$f''(1) = 4$

$f'(1) + f''(1) = 6$

5. (a) Show that the substitution $z = \ell n\left(\tan\frac{x}{2}\right)$ changes the equation $\frac{d^2y}{dx^2}+\cot x\frac{dy}{dx}+4y\cos ec^2x=0$ to $(d^2y/dz^2) + 4y = 0$.

सिद्ध कीजिये कि प्रतिस्थापन $z = \ell n\left(\tan\frac{x}{2}\right)$ समीकरण $\frac{d^2y}{dx^2}+\cot x\frac{dy}{dx}+4y\cos ec^2x=0$ को $(d^2y/dz^2) + 4y = 0$ में परिवर्तित कर देगा।

Ans.

Sol. $z = \ell n\left(\tan\frac{x}{2}\right)$

$$\frac{dz}{dx} = \frac{1}{\sin x} \qquad \text{Now } \frac{dy}{dz} = \frac{dy}{dx}.\frac{dx}{dz} = \frac{dy}{dx}\sin x$$

$$\frac{d^2y}{dz^2} = \frac{d}{dx}\left(\frac{dy}{dx}\sin x\right).\frac{dx}{dz}$$

$$\frac{d^2y}{dz^2} = \left(\sin x\frac{d^2y}{dx^2} + \cos x\frac{dy}{dx}\right)\sin x \quad .....(i)$$

$$\frac{d^2y}{dz^2} = \sin^2 x\frac{d^2y}{dx^2} + \sin x\cos x\frac{dy}{dx}$$

$$\text{cosec}^2 x\frac{d^2y}{dz^2} = \frac{d^2y}{dx^2} + \cot x\frac{dy}{dx}$$

Now put value from given equation

$$\text{cosec}^2x\frac{d^2y}{dz^2} + 4y\text{cosec}^2x = 0$$

$$\frac{d^2y}{dz^2} + 4y = 0$$

(b) If the dependent variable y is changed to 'z' by the substitution $y = \tan z$ then the differential equation

$\frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$ is changed to $\frac{d^2z}{dx^2} = \cos^2 z + k\left(\frac{dz}{dx}\right)^2$, then find the value of *k*.

यदि परतंत्र चर y को प्रतिस्थापन $y = \tan z$ द्वारा 'z' में परिवर्तित कर दिया जाए तो अवकल समीकरण $\frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$,

$\frac{d^2z}{dx^2} = \cos^2 z + k\left(\frac{dz}{dx}\right)^2$ में बदल जाएगी, तो k का मान बताइए।

Ans. $k = 2$

Sol. $y = \tan z$ _____(1)

$$\frac{dy}{dx} = \sec^2 z.\frac{dz}{dx} \qquad \text{_____(2)}$$

Diff^n w.r.t. 'x'

$$\frac{d^2y}{dx^2} = \sec^2 z \cdot \frac{d^2z}{dx^2} + 2\sec^2 z \cdot \tan z \left(\frac{dz}{dx}\right)^2 \qquad \text{_____(3)}$$

put values from (1), (2) and (3)

in given eq^n $\quad \frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$

$$\Rightarrow \sec^2 z \cdot \frac{d^2z}{dx^2} + 2\sec^2 z \cdot \tan z \left(\frac{dz}{dx}\right)^2$$

$$= 1 + \frac{2(1+\tan z)}{\sec^2 z}\left(\sec^2 z \frac{dz}{dx}\right)^2$$

$$\Rightarrow \frac{d^2z}{dx^2} + 2\tan z\left(\frac{dz}{dx}\right)^2 = \cos^2 z + 2(1+\tan z)\left(\frac{dz}{dx}\right)^2$$

$$\Rightarrow \frac{d^2z}{dx^2} = \cos^2 z + 2\left(\frac{dz}{dx}\right)^2$$

Hence k = 2

6. Show that $R = \frac{\left[1+(dy/dx)^2\right]^{\frac{3}{2}}}{d^2y/dx^2}$ can be reduced to the form $R^{2/3} = \frac{1}{\left(d^2y/dx^2\right)^{2/3}} + \frac{1}{\left(d^2x/dy^2\right)^{2/3}}$.

सिद्ध कीजिये की $R = \frac{\left[1+(dy/dx)^2\right]^{\frac{3}{2}}}{d^2y/dx^2}$ को $R^{2/3} = \frac{1}{\left(d^2y/dx^2\right)^{2/3}} + \frac{1}{\left(d^2x/dy^2\right)^{2/3}}$ में विघटित किया जा सकता है।

Ans.

Sol. $R = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left(\frac{d^2y}{dx^2}\right)}$ _____(1)

Let $\frac{dy}{dx} = \tan\theta \qquad \Rightarrow \frac{d^2y}{dx^2} = \sec^2\theta.\frac{d\theta}{dx}$

$$\Rightarrow \frac{1}{\frac{d^2y}{dx^2}} = \cos^2\theta.\frac{dx}{d\theta}$$

Now from $R = \sec^3\theta\cos^2\theta\cdot\frac{dx}{d\theta}$

$R = \sec\theta\frac{dx}{d\theta}$ $\Rightarrow \frac{d\theta}{dx} = \frac{\sec\theta}{R}$

$\frac{d^2y}{dx^2} = \frac{\sec^3\theta}{R}$ _____(2)

$\frac{dx}{dy} = \cot\theta \Rightarrow \frac{d^2x}{dy^2} = -\cos ec^2\theta\cdot\frac{d\theta}{dy}$

$= -\cos\alpha^2\theta\cdot\frac{d\theta}{dx}\cdot\frac{dx}{dy}$

$\frac{d^2x}{dy^2} = -\cos ec^2\theta\cdot\frac{\sec\theta}{R}\cdot\cot\theta = -\frac{\cos ec^3\theta}{R}$ _____(3)

Now $\frac{1}{\left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}} + \frac{1}{\left(\frac{d^2x}{dy^2}\right)^{\frac{2}{3}}} = R^{\frac{2}{3}}\left[\sin^2\theta + \cos^2\theta\right]$

$= R^{\frac{2}{3}}$

7. Suppose $p(x) = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_n x^n$. If $|p(x)| \le |e^{x-1} - 1|$ for all $x \ge 0$ prove that $|a_1 + 2a_2 + \ldots\ldots + n a_n| \le 1$. **[JEE-2000]**

माना $p(x) = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_n x^n$ है। यदि $x \ge 0$ के लिए $|p(x)| \le |e^{x-1} - 1|$ हो, तो सिद्ध कीजिए कि $|a_1 + 2a_2 + \ldots\ldots + n a_n| \le 1$. **[JEE-2000]**

Ans.

Sol. $p(x) = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_nx^n$

$\Rightarrow \quad p'(x) = a_1 + 2a_2x + \ldots\ldots + na_nx^{n-1}$

$\Rightarrow \quad p'(1) = a_1 + 2a_2 + \ldots\ldots + na_n \quad \ldots\ldots(i)$

Now $|p(x)| \le |e^{x-1} - 1| \ \forall\ x \ge 0$ (given)

$\Rightarrow \quad |p(1)| \le |e^0 - 1| = |1 - 1| = 0$

But $\quad |p(1)| \ge 0 \quad \Rightarrow \quad p(1) = 0$

Now for $\quad -1 < h < \infty$, $h \ne 0$, $1 + h > 0$

and $\quad |p(x)| \le |e^{x-1} - 1| \ \forall \ x \ge 0$

$\Rightarrow \quad |p(1 + h)| \le |e^h - 1| \ \forall \ h > -1, h \ne 0$

$\Rightarrow \quad |p(1 + h) - p(1)| \le |e^h - 1| \quad \because \quad p(1) = 0$

$$\Rightarrow \quad \left|\frac{p(1+h)-p(1)}{h}\right| \le \left|\frac{e^h - 1}{h}\right|$$

Taking limit as $h \to 0$, we get

$$\lim_{h\to 0}\left|\frac{p(1+h)-p(1)}{h}\right| \le \lim_{h\to 0}\left|\frac{e^h - 1}{h}\right| \quad \Rightarrow \quad |p'(1)| \le 1$$

$\Rightarrow \quad |a_1 + 2a_2 + 3a_3 + \ldots\ldots\ldots na_n| \le 1 \qquad \{\text{from (i)}\}$

Hence proved

8. Suppose f and g are two functions such that $f, g : R \to R, f(x) = \ell n\left(1 + \sqrt{1 + x^2}\right)$ and $g(x) = \ell n\left(x + \sqrt{1 + x^2}\right)$

then find the value of $x\, e^{g(x)}\left(f\left(\frac{1}{x}\right)\right)' + g'(x)$ at $x = 1$.

माना f तथा g दो फलन इस प्रकार है कि $f, g : R \to R$, $f(x) = \ell n\left(1 + \sqrt{1 + x^2}\right)$ तथा $g(x) = \ell n\left(x + \sqrt{1 + x^2}\right)$ तो

$x\, e^{g(x)}\left(f\left(\frac{1}{x}\right)\right)' + g'(x)$ का $x = 1$ पर मान निकालिए।

Ans. Zero

Sol. $$\left[f\left(\frac{1}{x}\right)\right]' = f'\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right) \qquad \text{_____(1)}$$

$$f(x) = \ell n\left(1 + \sqrt{1 + x^2}\right)$$

$$f'(x) = \frac{1}{1 + \sqrt{1 + x^2}} \cdot \frac{x}{\sqrt{1 + x^2}}$$

$$f'\left(\frac{1}{x}\right)=\frac{x}{x+\sqrt{1+x^2}}\cdot\frac{1}{\sqrt{1+x^2}}$$

$$e^{g(x)}f'\left(\frac{1}{x}\right)=\frac{x}{\sqrt{1+x^2}}$$

$$x.e^{g(x)}\cdot f'\left(\frac{1}{x}\right)=\frac{x^2}{\sqrt{1+x^2}}$$

$$g(x)=\ell n\left(x+\sqrt{1+x^2}\right)$$

$$g'(x)=\frac{1}{\sqrt{1+x^2}}$$

Now $x.e^{g(x)}\left[f\left(\frac{1}{x}\right)\right]^1+g'(x)$

$$=x.e^{g(x)}\cdot f'\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right)+g'(x)$$

$$=-\frac{1}{\sqrt{1+x^2}}+\frac{1}{\sqrt{1+x^2}}=0$$

$$\Rightarrow x.e^{g(x)}\left[f\left(\frac{1}{k}\right)\right]^1+g'(x)=0$$

then $\left[x.e^{g(x)}\left[f\left(\frac{1}{x}\right)\right]^1+g'(x)\right]_{x=1}=0$

9. Let $f(x)=\begin{bmatrix} xe^x & x\le 0 \\ x+x^2-x^3 & x>0 \end{bmatrix}$ then prove that :

(a) f is continuous and differentiable for all x.

(b) f' is continuous and differentiable for all x.

माना $f(x)=\begin{bmatrix} xe^x & x\le 0 \\ x+x^2-x^3 & x>0 \end{bmatrix}$ तो सिद्ध कीजिये। :

(a) f सभी x के मानों के लिये सतत् एवं अवकलनीय है।

(b) f' सभी x के मानों के लिये सतत् एवं अवकलनीय है।

Ans.

Sol. $f(x)=\begin{cases} x\cdot e^{x} & x\le 0 \\ x+x^{2}-x^{3} & x\ge 0 \end{cases}$

$\because x.e^{x}$ cont^{n} and diff^{n} $\forall x\in R$

$\therefore xe^{x}$ cont^{n} and diff^{n} $\forall x>0$

it means

$f(x)=\begin{cases} x.e^{x} & x\le 0 \\ x+x^{2}-x^{3} & x>0 \end{cases}$ contn and diff $\forall x\in R-\{0\}$

Now we check cont^{n} and diff^{n} at $x=0$

$\left.\begin{array}{l} f(0-0)=0 \\ f(0)=0 \\ f(0+0)=0 \end{array}\right\}$ f is cont^{n} at $x=0$

$f'(0-0)=1+0=1$ = L.H.D

$f'(0+0)=1+2(0)-3(0)=1=$ R.H.D

f is diff at $x=0$

Hence t is cont^{n} and diff^{n} at $x=0$

$g(x)=f'(x)\begin{cases} xe^{x}+e^{x} & x\le 0 \\ 1+2x-3x^{2} & x>0 \end{cases}$

$f'(x)$ cont^{n} and diff $\forall x\in R-\{0\}$

Now we check cont^{n} and diff^{n} at $x=0$

$\left.\begin{array}{l} g(0-0)=1=\text{L.H.L} \\ g(0)=1 \\ g(0+0)=1=\text{R.H.L} \end{array}\right\}$ $g=f'$ cont^{n} at $x=0$

hence f' cont^{n} $\forall x\in R$

$\left.\begin{array}{l} g'(0-0)=2=\text{L.H.D} \\ g'(0+0)=2=\text{R.H.D} \end{array}\right\}$ $g=f'$ diff^{n} at $x=0$

Hence f ' diff $\forall x \in R$

10. Let $f(x)=\begin{vmatrix} a+x & b+x & c+x \\ \ell+x & m+x & n+x \\ p+x & q+x & r+x \end{vmatrix}$. Show that $f''(x)=0$ and that $f(x)=f(0)+kx$ where k denotes the sum of all the co-factors of the elements in $f(0)$.

माना $f(x)=\begin{vmatrix} a+x & b+x & c+x \\ \ell+x & m+x & n+x \\ p+x & q+x & r+x \end{vmatrix}$ तो सिद्ध कीजिये की $f''(x)=0$ तथा $f(x)=f(0)+kx$ जहाँ k, सारणिक $f(0)$ के सभी अवयवों के सह खण्डजो का योग है।

Ans.

Sol. $f(x)=\begin{vmatrix} a+x & b+x & c+x \\ \ell+x & m+x & n+x \\ p+x & q+x & r+x \end{vmatrix}$

$c_3 \to c_3 - c_1 \quad c_2 \to c_2 - c_1$

$$f(x)=\begin{vmatrix} a+x & b-a & c-a \\ \ell+x & m-\ell & n-\ell \\ p+x & q-p & r-p \end{vmatrix}$$

$$f'(x)=\begin{vmatrix} 1 & b-a & c-a \\ 1 & m-\ell & n-\ell \\ 1 & q-p & r-p \end{vmatrix}+0+0=k$$

$\Rightarrow f'(x)=k$ _____(1)

Differentiate w.r.t. x

$\Rightarrow f''(x)=0$ Hence proved

$$\Rightarrow k=\begin{vmatrix} 1 & b+x & c+x \\ 1 & m+x & n+x \\ 1 & q+x & r+x \end{vmatrix}+\begin{vmatrix} a+x & 1 & c+x \\ \ell+x & 1 & n+x \\ p+x & 1 & r+x \end{vmatrix}+\begin{vmatrix} a+x & b+x & 1 \\ \ell+x & m+x & 1 \\ p+x & q+x & 1 \end{vmatrix}$$

$c_2 \to c_2 - xc_1$ $\quad$ $c_3 + c_3 - x_4$

Now $\begin{vmatrix} 1 & b+x & c+x \\ 1 & m+x & n+x \\ 1 & q-x & r+x \end{vmatrix} = \begin{vmatrix} 1 & b & c \\ 1 & m & n \\ 1 & q & r \end{vmatrix}$ $\quad$ $\left\{\begin{matrix} \text{apply} & c_2 \to c_2 - xc_1 \\ & c_3 \to c_3 - xc_1 \end{matrix}\right\}$

$= (mr - qn) - (br - qc) + (bn + cm)$

$=$ sum of cofactor of a_{11}, a_{21}, a_{31} in f(0)

Illy

$$\begin{vmatrix} a+x & 1 & c+x \\ \ell+x & 1 & n+x \\ p+x & 1 & r+x \end{vmatrix} = \text{sum of cofactor of } a_{12}, a_{22}, a_{32} \text{ of f(0)}$$

And

$$\begin{vmatrix} a+x & b+x & 1 \\ \ell+x & m+x & 1 \\ p+x & q+x & 1 \end{vmatrix} = \text{sum of cofactor of } a_{13}, a_{23}, a_{33} \text{ of f(0)}$$

Hence $\quad k =$ sum of cofactor of all elements of s(0)

$\Rightarrow$ from eq. (1) $\quad f'(x) = k$

Integrate

$\Rightarrow f(x) = kx + c$ $\quad$ put $x = 0$ $\quad$ $c = f(0)$

then $f(x) = kx + f(0)$ $\quad$ H.P.

11. Let $a_1 > a_2 > a_3 \ldots\ldots\ldots a_n > 1$; $p_1 > p_2 > p_3 \ldots\ldots\ldots > p_n > 0$; such that $p_1 + p_2 + p_3 + \ldots\ldots + p_n = 1$

Also $F(x) = \left(p_1 a_1^x + p_2 a_2^x + \ldots\ldots + p_n a_n^x\right)^{\frac{1}{x}}$. Compute

माना $a_1 > a_2 > a_3 \ldots\ldots\ldots a_n > 1$; $p_1 > p_2 > p_3 \ldots\ldots\ldots > p_n > 0$; जैसे कि $p_1 + p_2 + p_3 + \ldots\ldots + p_n = 1$ है, साथ ही $F(x) = \left(p_1 a_1^x + p_2 a_2^x + \ldots\ldots + p_n a_n^x\right)^{1/x}$ है तो ज्ञात कीजिये।

(a) $\lim\limits_{x \to 0^+} F(x)$ $\quad$ (b) $\lim\limits_{x \to \infty} F(x)$ $\quad$ (c) $\lim\limits_{x \to -\infty} F(x)$

Ans. (a) $a_1^{p_1} \cdot a_2^{p_2} .....a_n^{p_n}$; (b) a_1 ; (c) a_n

Sol. (a) $F(x) = \left(p_1 a_1^x + p_2 a_2^x + ... + p_n a_n^x\right)^{\frac{1}{x}}$

$$\lim_{x \to 0^+} F(x) = \lim_{x \to 0^+} \left(p_1 a_1^x + p_2 a_2^x + ... + p_n a_n^x\right)^{\frac{1}{x}} \qquad \left[1^\infty\right]$$

$$\lim_{x \to 0^+} F(x) = e^{\lim_{x \to 0} \frac{\left(p_1 a_1^x + p_2 a_2^x + ... + p_n a_n^x - 1\right)}{x}} \qquad [\%]$$

$$= e^{\lim_{x \to 0} p_1 a_1^x . \ell n a_1 + p_2 a_a^x \ell n a_2 + ... + p_n a_n^x . \ell n a_2} \qquad [\text{By L} - \text{hospital rule}]$$

$$= e^{\ell n\left(a_1^{p_1} \cdot a_2^{p_2} .....a_n^{p_n}\right)} = a_1^{p_1} \cdot a_2^{p_2} .....a_n^{p_n}$$

(b) $a_1 > a_2 > a_3 > .....a_n > 1$

$$\Rightarrow \frac{a_2}{a_1} < 1, \frac{a_3}{a_1} < 1, ..., \frac{a_n}{a_1} < 1$$

$$\Rightarrow \left(\frac{a_i}{a_1}\right)^\infty = 0 \quad \forall i \in \{2, 3, ...n\}$$

$$\lim_{x \to \infty} F(x) = \lim_{x \to \infty} a_1 \left[p_1 + p_2 \left(\frac{a_2}{a_1}\right)^x + ... + p_n \left(\frac{a_n}{a_1}\right)^x\right]^{\frac{1}{x}}$$

$$= a_1 \left[p_1 + 0 + 0 + ... + 0\right]^0 = a_1$$

(c) $\frac{a_1}{a_n} > 1, \frac{a_2}{a_n} > 1, \frac{a_3}{a_n} > 1, ... \frac{a_{n-1}}{a_n} > 1$

$$\left(\frac{a_i}{a_n}\right)^{-\infty} = 0 \qquad \forall i \in \{1, 2, 3, ..., n-1\}$$

$$\lim_{x \to -\infty} F(x) = \lim_{x \to -\infty} a_n \left[p_1 \left(\frac{a_1}{a_n}\right)^x + p_2 \left(\frac{a_2}{a_n}\right)^n + ... + p_{n-1} \left(\frac{a_n - 1}{a_n}\right)^n + p_n\right]^{\frac{1}{x}}$$

$$= a_n \left[0 + 0 + ... + 0 + p_n\right]^\circ$$

$= a_n$

12. If $x_1, x_1, x_2, x_3, \ldots\ldots x_{n-1}$ be n zero's of the polynomial $P(x) = x^n + \alpha x + \beta$, where $x_i \neq x_j \ \forall$ i and $j = 1, 2, 3, \ldots (n-1)$. Prove that the value of $Q(x) = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) \ldots\ldots (x_1 - x_{n-1})$, is equal to ${}^nC_2\ x_1^{n-2}$.

यदि $x_1, x_1, x_2, x_3, \ldots\ldots x_{n-1}$ बहुपद $P(x) = x^n + \alpha x + \beta$ के 'n' मूल है जहाँ $x_i \neq x_j \ \forall$ i तथा $j = 1, 2, 3, \ldots (n-1)$ तो सिद्ध कीजिये $Q(x) = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) \ldots\ldots (x_1 - x_{n-1})$ का मान ${}^nC_2\ x_1^{n-2}$ के बराबर है।

Ans.

Sol. $P(x) = x^n + \alpha x + \beta = (x - x_1)^2 (x - x_3) \ldots\ldots (x - x_{n-1})$

$$\Rightarrow \frac{x^n + \alpha x + \beta}{(x - x_1)^2} = (x - x_2)(x - x_3) \ldots\ldots (x - x_{n-1})$$

$$\Rightarrow \lim_{x \to x_1} \frac{x^n + \alpha x + \beta}{(x - x_n)^2} = (x_1 - x_2)(x_1 - x_3) \ldots\ldots (x_1 - x_{n-1})$$

$$\Rightarrow \lim_{x \to x_1} \frac{nx^{n-1} + \alpha}{2(x - x_1)} = Q(x) \qquad \text{[By L – Hospital rule]}$$

$$\Rightarrow \frac{n(n-1)}{2} x_1^{n-2} = Q(x) \qquad \text{[Again by L – Hospital rule]}$$

$$\Rightarrow {}^nC_2 \cdot x_1^{n-2} = Q(x) \quad \text{H.P.}$$

13. If $\frac{d^2}{dx^2}\left(\frac{\sin^4 x + \sin^2 x + 1}{\sin^2 x + \sin x + 1}\right) = a\sin^2 x + b\sin x + c$ then the value of $b + c - a$ is.

यदि $\frac{d^2}{dx^2}\left(\frac{\sin^4 x + \sin^2 x + 1}{\sin^2 x + \sin x + 1}\right) = a\sin^2 x + b\sin x + c$ है तो $b + c - a$ का मान होगा?

Ans. 7

Sol. $P^4 + P^2 + 1 = (P^2 - P + 1)(P^2 + P + 1)$

$$\Rightarrow \frac{P^4 + P^2 + 1}{P^2 + P + 1} = P^2 - P + 1$$

$$\frac{\sin^4 x+\sin^2 x+1}{\sin^2 x+\sin x+1}=\sin^2-\sin x+1$$

Now $\frac{d^2}{dx^2}\left[\frac{\sin^4 x+\sin^2 x+1}{\sin^2 x+\sin x+1}\right]=2\cos 2x+\sin x$

$$=2\left(1-2\sin^2 x\right)+\sin x$$

$$=-4\sin^2 x+\sin x+2$$

Compairing with $a\sin^2x + b\sin x + c$

$a = -4 \qquad b = 1 \quad c = 2$

$b + c - a = 7$

14. Let $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5}$ and let $g(x) = f^{-1}(x)$. Find $g'''(0)$.

माना $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5}$ तथा $g(x) = f^{-1}(x)$ है तो $g'''(0)$ का मान ज्ञात कीजिये ?

Ans. 1

Sol. $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5} \qquad \Rightarrow f(0)=0$

$$f'(x)=1+x+x^2+x^3+x^4=\frac{x^5-1}{x-1}$$

$$f''(x)=1+2x+3x^2+4x^3$$

$$f'''(x)=2+6x+12x^2$$

$$f'(0)=1, \qquad f''(0)=1, \qquad f'''(0)=2$$

$g(x)=f^{-1}(x) \Rightarrow f[g(x)]=x$ and $g[f(x)]=x$

$\Rightarrow g[f(x)]=x$

Diff^n w.r.t. x

$\Rightarrow g'\left[f(x)\right]\ f'(x)=1$ _____(1) $\rightarrow g'(0)=1$

Again Diff^n w.r.t. x

$\Rightarrow g''\left(f(x)\right)\cdot\left\{f'(x)\right\}^2+g'\left(f(x)\right)\cdot f''(x)=0$ _____(2)

put $x=0$

$\Rightarrow g''\left(f(0)\right)\cdot\left\{f'(0)\right\}^2+g'\left(f(0)\right)\cdot f''(0)=0$

$\Rightarrow g''(0)+g'(0)=0$

$g''(0)=-1$

Again diff^n w.r.t. x of eq^n (1)

$\Rightarrow g''\left(f(x)\right)\cdot\left\{f'(x)\right\}^3+g''\left(f(x)\right)\cdot 2f'(x)\cdot f''(x)+g''\left(f(x)\right)\cdot f'(x)\cdot f''(x)+g'\left(f(x)\right)\cdot f''(x)=0$

put $x=0$

$\Rightarrow g'''(0)+g''(0)2+g''(0)+g'(0)\cdot(2)=0$

$g'''(0)+3(-1)+2=0$

$g'''(0)=1$

15. The value of x, $x\in(2,\infty)$ where $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$ is not differentiable is.

$x\in(2,\infty)$ में x का मान ज्ञात कीजिये जहां $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$ अवकलनीय नहीं है ?

Ans. 4

Sol. $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$

Let $8x-16=p^2$ $\Rightarrow x=\dfrac{p^2+16}{8},$ $p>0$

$f(x)=\dfrac{\sqrt{p^2+16+8p}+\sqrt{p^2+16-8p}}{2\sqrt{2}}$

$$= \frac{|p+4|+|p-4|}{2\sqrt{2}} = \frac{p+4+|p-4|}{2\sqrt{2}}$$

$$f(x) = \begin{cases} \frac{2p}{2\sqrt{2}} & p \geq 4 \quad \Rightarrow 8x \geq 32 \quad \Rightarrow x \geq 4 \\ 2\sqrt{2} & P < 4 \quad \Rightarrow 2 < X < 4 \end{cases}$$

$$f'(4+0) = \frac{1}{2\sqrt{16}} \cdot \frac{8}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$f'(4-0) = 0$$

f is not differential at x = 4

16. If $y = \left[\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \right]^2 + k\, \ell n \left(x+\sqrt{x^2-a^2} \right)$, then find the value of $(x^2-a^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$:

यदि $y = \left[\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \right]^2 + k\, \ell n \left(x+\sqrt{x^2-a^2} \right)$ तो $(x^2-a^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ का मान ज्ञात कीजिए :

Ans. 2

Sol. $$y = \left[\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \right]^2 + k\, \ell n \left(x+\sqrt{x^2-a^2} \right)$$

Differentiating both sides w. r. t.

$$\frac{dy}{dx} = 2\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \frac{1}{\sqrt{x^2-a^2}} + \frac{k}{\sqrt{x^2-a^2}}$$

$$\sqrt{x^2-a^2} \cdot \frac{dy}{dx} = 2\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) + k$$

Differentiating w.r.t. x

$$\sqrt{x^2-a^2}\,\frac{d^2y}{dx^2} + \frac{x}{\sqrt{x^2-a^2}} \cdot \frac{dy}{dx} = \frac{2}{\sqrt{x^2-a^2}}$$

$$\Rightarrow \left(x^2 - a^2\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 2$$

17. If $(a + bx)\, e^{y/x} = x$, show that $x^3 y'' = (xy' - y)^2$ _____.

यदि $(a + bx)\, e^{y/x} = x$ तो प्रदर्शित कीजिए कि $x^3 y'' = (xy' - y)^2$ _____.

Ans.

Sol. $e^{\frac{y}{x}} = \dfrac{x}{a+bx}$ $\Rightarrow$ taking log on both side

$$\Rightarrow \frac{y}{x} = \ell n\left(\frac{x}{a+bx}\right)$$

$$\Rightarrow y = x\cdot\ell n\left(\frac{x}{a+bx}\right)$$

Differentiating w.r.t. "x"

$$\Rightarrow \frac{dy}{dx} = \ell n\left(\frac{x}{a+bx}\right) + x\left[\frac{1}{x} - \frac{b}{a+bx}\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} + \frac{x[a]}{x(a+bx)} \qquad \text{_____(1)}$$

$$\Rightarrow x\cdot\frac{dy}{dx} - y = \frac{ax}{a+bx} = ae^{\frac{y}{x}} \qquad \text{_____(2)}$$

from (1)

$$\frac{d^2y}{dx^2} = \frac{xy' - y}{x^2} - \frac{ab}{(a+bx)^2}$$

$$\Rightarrow x^3\frac{d^2y}{dx^2} = (xy' - y)\cdot x - \frac{b}{a}x\left(\frac{ax}{a+bx}\right)^2$$

$$\Rightarrow (xy' - y)x - \frac{b}{a}x(xy' - y)^2$$

$$\Rightarrow x(xy' - y)\left[1 - \frac{b}{a}\frac{ax}{a+bx}\right]$$

$$\Rightarrow (xy'-y)\frac{ax}{a+bx}$$

$$\Rightarrow x^3y''' = (xy'-y)^2$$

18. If $\frac{x+b}{2} = a\tan^{-1}(a\,\ell n\,y)$, $a>0$ prove that $yy'' - yy'\ \ell n\,y = (y')^2$ _____.

यदि $\frac{x+b}{2} = a\tan^{-1}(a\,\ell n\,y)$, $a>0$ तो सिद्ध करो कि $y\,y'' - yy'\ \ell n\,y = (y')^2$ _____.

Ans.

Sol. $\frac{x+b}{2} = a\tan^{-1}(a\ell ny) \qquad a>0$

$$\Rightarrow \tan\left(\frac{x+b}{2a}\right) - a\ell ny = 0$$

$$\because \frac{dy}{dx} = -\frac{\delta f/dx}{\delta f/dy} = -\frac{\frac{1}{2a}\cdot\sec^2\left(\frac{x+b}{2a}\right)}{-\frac{a}{y}}$$

$$\Rightarrow \frac{2a^2y'}{y} = 1+\tan^2\left(\frac{x+b}{2a}\right)$$

$$\Rightarrow \frac{2a^2\cdot y'}{y} = 1+a^2\ell n^2 y$$

Differentiating w.r.t. x

$$\Rightarrow 2a^2\cdot\frac{yy''-(y')^2}{y^2} = 2a^2\cdot\ell ny\cdot\frac{1}{y}\cdot y'$$

$$\Rightarrow yy''-(y')^2 = y\cdot y'\ell ny$$

$$\Rightarrow yy''-yy'\ell ny = (y')^2$$

19. If $y^3 - y = 2x$, then prove that $\frac{d^2y}{dx^2} = -\frac{24y}{(3y^2-1)^3}$. Hence, show that $\left(x^2 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = \frac{y}{9}$ _____.

यदि $y^3 - y = 2x$ तो सिद्ध करो कि $\frac{d^2y}{dx^2} = -\frac{24y}{(3y^2-1)^3}$ तथा इस प्रकार प्रदर्शित कीजिए कि $\left(x^2 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = \frac{y}{9}$ _____.

Ans.

Sol. $y^3 - y = 2x \qquad y^2 - 1 = \frac{2x}{y}$

$$\Rightarrow \left(3y^2 - 1\right)\frac{dy}{dx} = 2$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{3y^2 - 1} = \frac{2}{\alpha}(\text{say})$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{12y\cdot\frac{dy}{dx}}{\left(3y^2-1\right)} = -\frac{12y}{\left(3y^2-1\right)^2}\cdot\frac{2}{\left(3y^2-1\right)}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{24y}{\left(3y^2-1\right)^3} \qquad 3y^2 - \alpha \qquad y^2 = \frac{\alpha+1}{3} \qquad y^2 - 1 = \frac{\alpha-2}{3}$$

Now $\text{L.H.S.} = \left(x^2 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$

$$= \left[\frac{y^2\left(y^2-1\right)^2}{4} - \frac{1}{27}\right]\left[\frac{-24y}{\left(3y^2-1\right)^3}\right] \times \frac{y\left(y^2-1\right)}{2}\cdot\frac{2}{3y^2-1}$$

$$= y\left[\left[\frac{93y^2\left(y^2-1\right)^2 - 4}{108}\right]\left[\frac{-24}{\left(3y^2-1\right)^3}\right] + \frac{y^2-1}{3y^2-1}\right]$$

$$= y\left[\frac{\left(\frac{9(\alpha+1)(\alpha-2)^2}{9}-4\right)}{108}\left(-\frac{24}{\alpha^3}\right)+\frac{\alpha-2}{3\alpha}\right]$$

$$= \frac{y}{9}\left[\left((\alpha+1)(\alpha-2)^2-4\right)\left(-\frac{2}{\alpha^3}\right)+\frac{3\alpha-6}{\alpha}\right]$$

$$= \frac{y}{9}\left[\frac{8-2(\alpha+1)\left(\alpha^2-4\alpha+4\right)+3\alpha^3-6\alpha^2}{\alpha^3}\right]$$

$$= \frac{y}{9}\left[\frac{8-2\alpha^3+6\alpha^2-8+3\alpha^3-6\alpha^2}{\alpha^3}\right]$$

$$= \frac{y}{9} = \text{R.H.S.}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 20 to 22

In certain problem the differentiation of $\{f(x) \cdot g(x)\}$ appears. One student commits mistake and differentiates as $\frac{df}{dx}\cdot\frac{dg}{dx}$ but he gets correct result if $f(x) = x^3$ and $g(x)$ is a decreasing function for which $g(0) = \frac{1}{3}$.

किसी प्रश्न में $\{f(x) \cdot g(x)\}$ का अवकलन पूछा जाता है। एक विद्यार्थी गलती करके इसका अवकलन $\frac{df}{dx}\cdot\frac{dg}{dx}$ कर देता है परन्तु उसे सही परिणाम प्राप्त होता है यदि $f(x) = x^3$ तथा $g(x)$ एक ह्रासमान फलन है जिसके लिये $g(0) = \frac{1}{3}$

20. The function $g(x)$ is $(x > 3)$:

फलन $g(x)$ होगा $(x > 3)$:

(A) $\frac{3}{(x-3)^3}$ (B) $\frac{4}{(x-3)^3}$ (C) $\frac{9}{(x-3)^3}$ (D) $\frac{27}{(x-3)^3}$

Ans. C

Sol. $f(x)=x^3$ ATQ $\frac{d}{dx}\left[f(x)\cdot g(x)\right]=\frac{df}{dx}\cdot\frac{dg}{dx}$

$\Rightarrow 3x^2\cdot g(x)+g'(x)\cdot x^3=3x^2\cdot g'(x)$

$\Rightarrow 3x^2 g(x)=\left(3x^2-x^3\right)g'(x)$

$\Rightarrow \int\frac{3}{3-x}dx=\int\frac{dg}{g}$

$\Rightarrow -3\ell n|3-x|=\ell ng+\ell nc$

$\Rightarrow c.g(x)=\frac{1}{(3-x)^3}$

$x=0$ $\quad C\cdot\left(\frac{1}{3}\right)=\frac{1}{27}\quad\Rightarrow C=\frac{1}{9}$

$x>3$ $\quad g(x)=\frac{9}{|3-x|^3}=\frac{9}{(x-3)^3}$

21. Derivative of $\{f(x-3)\,.\,g(x)\}$ with respect to x at $x=100$ is :

$x=100$ पर $\{f(x-3)\,.\,g(x)\}$ का x के सापेक्ष अवकलज होगा :

(A) 0 (B) 1 (C) – 1 (D) 2

Ans. A

Sol. $\frac{d}{dx}\{f(x-3)\cdot g(x)\}=g'(x)f(x-3)+f'(x-3)g(x)$

$\frac{d}{dx}\{f(x-3)\cdot g(x)\}=-\frac{27}{(x-3)^4}\cdot(x-3)^3+3(x-3)^2\frac{9}{(x-3)^3}$

$=0$

$\Rightarrow \frac{d}{dx}\{f(x-3)\cdot g(x)\}_{x=100}=0$

22. $\lim\limits_{x\to 0}\frac{f(x).g(x)}{x(1+g(x))}$ will be :

$\lim_{x\to 0}\frac{f(x).g(x)}{x(1+g(x))}$ होगा :

(A) 0 (B) – 1 (C) 1 (D) 2

Ans. A

Sol. $\lim_{x\to 0}\frac{f(x)\cdot g(x)}{x\left[1+g(x)\right]}=\lim_{x\to 0}\frac{x^3\frac{9}{(x-3)^3}}{x\left[1+\frac{9}{(x-3)^3}\right]}$

$\lim_{x\to 0}\frac{9x^2}{(x-3)^3+9}$

$= 0$

Paragraph for Question Nos. 23 to 24

Let $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}$ and $g(x) = \lambda \tan x + (1 - \lambda) \sin x - x$, where $\lambda \in R$ and $x \in [0, \pi/2)$.

माना $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}$ तथा $g(x) = \lambda \tan x + (1 - \lambda) \sin x - x$ है जहां $\lambda \in R$ तथा $x \in [0, \pi/2)$ है।

23. g'(x) equals :

g'(x) का मान होगा :

(A) $\frac{(1-\cos x)(f(x)-\lambda)}{\cos x}$ (B) $\frac{(1-\cos x)(\lambda-f(x))}{\cos x}$

(C) $\frac{(1-\cos x)(\lambda-f(x))}{f(x)}$ (D) $\frac{(1-\cos x)(\lambda-f(x))}{(f(x))^2}$

Ans. C

Sol. $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}$ $\quad x\in\left[0,\frac{\pi}{2}\right)\Rightarrow f(x)>0$

$f(x)=\frac{1}{\sec^2 x+\sec x+1}$

$$g(x)=\lambda\tan x+(1-\lambda)\sin x-x$$

$$g'(x)=\lambda\sec^2 x-\lambda\cos x+\cos x-1$$

$$=\frac{\lambda(1-\cos^3 x)}{\cos^2 x}-(1-\cos x)$$

$$=(1-\cos x)\left[\frac{\lambda(1+\cos x+\cos^2 x)}{\cos^2 x}-1\right]$$

$$=(1-\cos x)\frac{(\lambda-f(x))}{f(x)}$$

24. The exhaustive set of values of 'λ' such that $g'(x)\geq 0$ for any $x\in[0,\pi/2)$:

'λ' के सम्पूर्ण मानों का समुच्चय जिसके लिए $x\in[0,\pi/2)$ में $g'(x)\geq 0$ है :

(A) $[1,\infty)$ (B) $[0,\infty)$ (C) $\left[\frac{1}{2},\infty\right)$ (D) $\left[\frac{1}{3},\infty\right)$

Ans. D

Sol. $g'(x)=\frac{(1-\cos x)(\lambda-f(x))}{f(x)}\geq 0$

$1-\cos x=+ve \qquad \Rightarrow \lambda-f(x)\geq 0$

$f(x)=+ve \qquad \lambda\geq f(x) \qquad$ _____(1)

Now $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}=\frac{1}{\sec^2 x+\sec x+1}\ x\in\left[0,\frac{\pi}{2}\right)$

$\sec^2 x=t\in[1,\infty)$

$f(t)=\frac{1}{t^2+t+1}\ t\in[1,\infty)$

$f(t)\in\left(0,\frac{1}{3}\right] \qquad \because t^2+t+1\in[3,\infty)$

$t\in[1,\infty)$

Range of f(x) is $\left(0, \frac{1}{3}\right]$

$O < f(x) \le \frac{1}{3}$ _____(2)

from (1) and (2) $\lambda \ge \frac{1}{3}$ $\Rightarrow \lambda \in \left[\frac{1}{3}, \infty\right)$

MATCH THE COLUMN TYPE QUESTION(S)

25. Match the following columns :

Column-I	**Column-II**
(A) If $f'(x) = \sqrt{3x^2+6}$ and $y = f(x^3)$ then at $x = 1$, $\frac{dy}{dx}$	(P) – 2
(B) If f be a diffr. fun. such that $f(xy) = f(x) + f(y)$, $\forall\ x, y \in R$ then $f(e) + f(1/e) =$	(Q) 4
(C) If f be a twice diffr. fun such that $f''(x) = -f(x)$ and $f'(x) = g(x)$, If $h(x) = (f(x))^2 + (g(x))^2$ and $h(5) = 9$ then $h(10) = ?$	(R) 0
(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, then $\frac{dy}{dx}$	(S) –1
	(T) 9

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) यदि $f'(x) = \sqrt{3x^2+6}$ तथा $y = f(x^3)$ तो $x = 1$ पर $\frac{dy}{dx}$ होगा	(P) – 2
(B) यदि f एक अवकलनीय फलन है इस प्रकार कि $f(xy) = f(x) + f(y)$, $\forall\ x, y \in R$ तो $f(e) + f(1/e) =$	(Q) 4
(C) यदि f एक दो बार अवकलनीय फलन है इस प्रकार कि(R) 0 $f''(x) = -f(x)$ तथा तथा $f'(x) = g(x)$, तथा $h(x) = (f(x))^2 + (g(x))^2$ तथा $h(5) = 9$ तो $h(10) = ?$	(R) 0
(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, तो $\frac{dy}{dx}$	(S) –1
	(T) 9

Ans. A – T, B – R, C – T, D – P

Sol. (A) $f'(x) = \sqrt{3x^2+6}$ $y = f(x^3)$

$\frac{dy}{dx} = f'(x^3).3x^2$

$$\left.\frac{dy}{dx}\right)_{x=1} = f'(1)\cdot 3 = \sqrt{3+6}\cdot 3 = 9$$

$A \to T$

(B) $f(xy) = f(x) + f(y)$ and $x, y \in R$ and f differentiable

$\Rightarrow f(x) = \ell nx$

$$f(x) + f\left(\frac{1}{x}\right) = 0 \qquad \Rightarrow f(e) + f\left(\frac{1}{e}\right) = 0$$

$B \to R$

(C) $f''(x) = -f(x)$ $\qquad f'(x) = g(x)$

$h(x) = \left[f(x)\right]^2 + \left[g(x)\right]^2 \qquad h(5) = 9$

$h'(x) = 2f(x)\cdot f'(x) + 2g(x)\cdot g'(x)$

$= 2f(x)\cdot f'(x) + 2f'(x).f''(x)$

$= 2f(x)\cdot f'(x) - 2f'(x)\cdot f(x) = 0$

$h'(x) = 0$

$h(x) = \text{const} \quad \Rightarrow h(5) = 9 \quad \Rightarrow h(x) = 9$

$\Rightarrow h(10) = 9$

$C \to T$

(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$

$$y = \frac{\pi}{2} - \cot^{-1}(\cot x) + \frac{\pi}{2} - \tan^{-1}(\tan x)$$

$$y = \frac{\pi}{2} - x + \frac{\pi}{2} - x$$

$y = \pi - 2x$

$$\frac{dy}{dx} = -2 \qquad \Rightarrow D \to P$$